

Why Did Our Models Score

$R^2 \approx 0.92\text{--}0.97$?

A scientific investigation of overfitting and generalization
in synthetic-data cheese shelf-life prediction

This report documents an investigation prompted by a legitimate methodological concern: the internal test scores of the production shelf-life models were unusually high, and the evaluation protocol that produced them had not been examined at the level of the synthetic data-generation structure.

The investigation reproduces the original training and evaluation procedure from source, executes the diagnostic experiments that were requested, identifies the specific property of the evaluation design responsible for the optimistic scores, and re-evaluates seventeen model configurations under a corrected protocol.

Every number reported here is generated from result files in the project repository. Where the repository does not contain the evidence needed to answer a question, that is stated explicitly rather than filled in.

Scope

- Six production specialists
- Four production model families
- Seventeen evaluated configurations
- Six diagnostic experiments
- Two evaluation protocols

Dataset

- V7 specialist corpus
- 34 000 synthetic rows
- 7–12 generation rules per specialist

Prepared for

- Two-hour technical review

Summary of the investigation

The concern

The production shelf-life specialists reported internal test scores in the range $R^2 \approx 0.92$ – 0.97 for the general-shelf-life task. A score of that magnitude on a biological prediction problem is unusual enough to warrant an explanation, and the question that was put to this project was whether those scores reflected genuine predictive skill, classical overfitting, or an evaluation procedure that was not sufficiently independent of the training data.

That question is legitimate and this report treats it as such. The investigation was not conducted to defend the original numbers. It was conducted to determine what they measure.

What was done

Six diagnostic experiments were executed against the production dataset, covering model complexity reduction, a classification reformulation, K -fold cross-validation at two values of K , category aggregation, validation-set removal, and a model zoo spanning seventeen configurations from a constant-output baseline to the production gradient ensembles. The original production training and evaluation code was reproduced from source rather than described from documentation, and the structure of the synthetic corpus was audited directly.

What was found

The central result

Under the original evaluation protocol, **100% of test rows in every one of the six specialists originate from a synthetic generation rule that is also present in that specialist's training set.** The protocol correctly prevents a *context* from appearing on both sides of the split — context overlap between train and test is exactly zero in all six specialists — but it places no constraint at all on the generation rule. The held-out rows are therefore new samples of relationships the model has already been trained on.

This single structural fact explains the pattern of results better than any model-side explanation, and the diagnostic experiments confirm it by elimination:

- Reducing model capacity does not reduce the score. A ridge regression — a linear model with no interaction terms — reaches $R^2 = 0.910$ on the soft general-shelf-life specialist where the production LightGBM reaches 0.971 . A problem that a linear model solves to within six points of a 2000-tree gradient ensemble is not a problem whose high scores are produced by model complexity.
- Increasing K in cross-validation does not reduce the score, because K -fold with the same grouping variable reshuffles the same already-seen rules.
- Removing the validation set changes test R^2 by a mean of $+0.0003$ across all model–specialist pairs: no material effect in either direction.
- Aggregating all three cheese categories into a single training set does not reduce the score either.

The corrected evaluation

The protocol was replaced with *leave-one-generation-rule-out* cross-validation: each fold removes every row belonging to one `source_rule_id` from training and predicts that rule's rows with a model that has never seen it. All seventeen configurations were re-evaluated under this protocol across all six specialists.

Effect of the protocol change

The three production tree ensembles retain a median fold R^2 in the region of 0.70–0.87, substantially below their context-holdout scores but clearly above a no-skill baseline. Several simpler models that looked competitive under the original protocol collapse to strongly negative R^2 once the generation rule is withheld — ridge regression falls from a mean context-holdout R^2 near 0.90 to a strongly negative leave-rule-out mean. The evaluation change, not any change to the models, produces the entire difference.

What this does and does not establish

The revised numbers are an honest measure of one specific ability: the ability to predict rows generated by a *synthetic rule* the model has not seen, using other synthetic rules as training data. They are not a measure of real-world predictive accuracy on physical cheese, and this report is explicit throughout about the difference. Chapter 13 enumerates the dependencies that survive the corrected protocol, and Chapter 14 examines the external literature evaluation that would be required to make a real-world claim.

Principal evidence limitation

The synthetic data-generation source code is not present in this repository; only its output is. Every statement in this report about generation-rule behaviour is therefore inferred statistically from the generated data, and is labelled as such. Separately, no dataset file in this repository at any point in its recoverable history contains a non-synthetic `data_origin` tag, so the original experimental observations that must have calibrated the generation rules cannot be counted, characterised, or audited from here. Both limitations are documented in full in Section 2.1.

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The prediction problem and the system under investigation

What to say at the start of this section

Open by stating the problem in one sentence — predicting how many days a cheese formulation remains acceptable under given storage conditions — then explain that the system is not one model but six, and that this structure matters for everything that follows, because it determines how little data each individual model actually sees.

1.1 The prediction task

The system predicts *shelf life in days* for a cheese formulation under specified storage conditions. The target variable, `shelf_life_days`, is continuous and strictly positive, so the problem is posed as regression. Given a feature vector $\mathbf{x}$ describing the cheese matrix, the treatment applied to it, the storage environment, and the spoilage or safety indicator being tracked, the model estimates

$$\hat{y} = f(\mathbf{x}), \quad \hat{y} \in \mathbb{R}_{>0} \text{ (days)}. \quad (1.1)$$

Two distinct questions are asked of the same physical system, and the project treats them as two separate prediction tasks:

General shelf life

the time until a quality indicator crosses an acceptability threshold — yeasts and moulds, total viable count, sensory acceptability, pH drift, peroxide value, oxidation index, visible moulds, or clostridial late blowing. This is a *quality* endpoint.

Safety endpoint

the time until a pathogen-related criterion is reached — *Listeria monocytogenes*, *Staphylococcus aureus*, *Salmonella*. This is a *safety* endpoint and, as Section 13.4 discusses, its target semantics are not identical to those of the quality endpoint.

1.2 Six specialists, not one model

The production architecture routes each prediction to one of six specialist models, indexed by the Cartesian product of cheese category and prediction task:

$$\mathcal{S} = \{\text{soft, semi-hard, hard}\} \times \{\text{general_shelf_life, safety_endpoint}\}, \quad |\mathcal{S}| = 6. \quad (1.2)$$

This was verified against the implementation, not assumed: the training driver `train_specialists.py` iterates over exactly these three categories and two tasks, loading one category file per category and filtering it on the `model_task` column. Each specialist is trained independently, with its own feature schema detection, its own preprocessing fit, and its own set of four candidate model families.

The practical consequence of this architecture is that the effective training set for any one model is much smaller than the headline corpus size. The hard/safety-endpoint specialist, for example, trains on 2 240 rows, not on 34 000. This becomes important in Chapter 2, where the motivation for synthetic expansion is examined quantitatively.

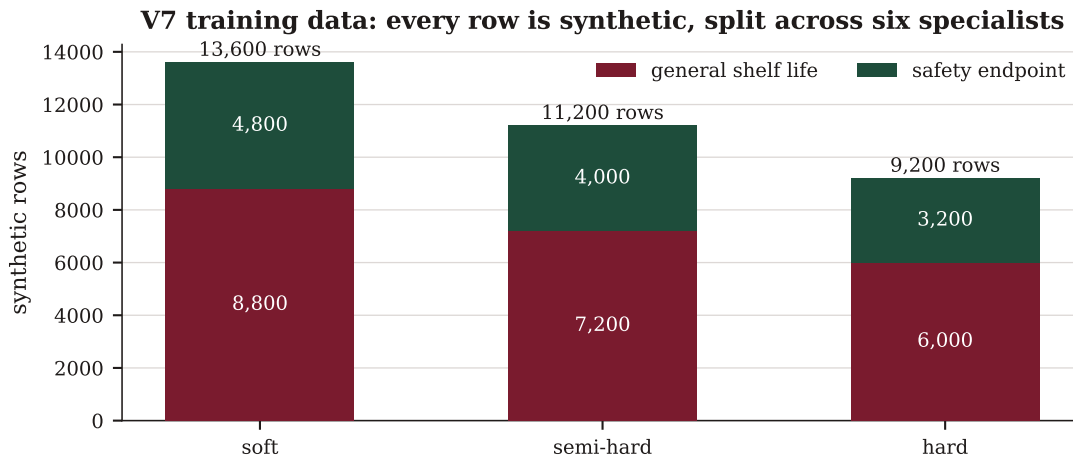


Figure 1.1: Composition of the V7 training corpus. The three category files contain 34 000 rows in total, partitioned into six specialist training sets. Every row carries `data_origin = synthetic-literature-constrained_v7`; there is no real-observation subset in this corpus. Generated from the three raw specialist CSV files.

1.3 The four candidate model families

For each of the six specialists, four model families are trained and compared:

Random Forest

a bagged ensemble of unpruned regression trees; variance reduction through bootstrap resampling and feature subsampling.

LightGBM

gradient boosting with leaf-wise (best-first) tree growth and histogram-binned split finding.

XGBoost

gradient boosting with depth-wise tree growth and explicit L_1/L_2 leaf-weight regularization in the objective.

Explainable Boosting Machine (EBM)

a glass-box generalized additive model with a bounded number of pairwise interaction terms, cyclically boosted one feature at a time.

Their exact production hyperparameters are recovered from source and reported in Table 1.1; the mathematics of each is developed in Chapters 8, 10 and 11.

Note on capacity

Two of the four production models use *two thousand* boosting rounds with early stopping, and the Random Forest grows 400 fully-developed trees with a minimum leaf size of two. These are high-capacity configurations. If the high scores were produced by capacity, reducing capacity should reduce them. Chapter 5 tests exactly that, and it does not.

1.4 Feature space

Schema detection is performed per specialist at training time by `detect_schema()`: any column that is not the target and not on the exclusion list is classified as numeric, binary, or categorical by inspecting its dtype and its set of unique values. The V7 files carry 63 columns, of which the identifier, provenance, and zero-variance columns are excluded, leaving the physically meaningful inputs.

Table 1.1: Production hyperparameters, read directly from `train_specialists.py`. The seed is global (SEED = 42) and shared by the splitter and every model.

Model	Configuration as implemented
Random Forest	<code>n_estimators=400, max_depth=None, min_samples_leaf=2, random_state=42, n_jobs=-1</code>
LightGBM	<code>n_estimators=2000, learning_rate=0.03, num_leaves=31, min_child_samples=20, subsample=0.9, colsample_bytree=0.9; early stopping after 50 rounds on the validation set</code>
XGBoost	<code>n_estimators=2000, learning_rate=0.03, max_depth=6, subsample=0.9, colsample_bytree=0.9, eval_metric="rmse"; early stopping after 50 rounds</code>
EBM	<code>interactions=10, max_bins=256, learning_rate=0.02, max_rounds=3000, min_samples_leaf=4, outer_bags=8</code>

The features that reach the model describe four groups:

- **Matrix composition:** `matrix_ph`, `matrix_water_activity`, `matrix_moisture_pct`, `matrix_fat_pct`, `matrix_protein_pct`, `matrix_salt_pct`, `matrix_ripening_days`, `pasteurization_applied`, `food_matrix`, `physical_form`. (`base_cheese_name` is *excluded* as redundant with `food_matrix`; it is retained in the data and used throughout this report to label results.)
- **Storage environment:** `storage_temperature_c`, `packaging_type`, `headspace_oxygen_pct`, `headspace_co2_pct`, `headspace_n2_pct`.
- **Treatment:** `treatment_type`, `application_method`, `ingredient_count`, `primary_ingredient_name`, `primary_ingredient_family`, `canonical_concentration_value`, `is_control`. (The raw `primary_concentration` and its unit column are excluded in favour of the unit-harmonised canonical value.)
- **Indicator definition:** `indicator_group`, `indicator_type`, `indicator_threshold`, `initial_indicator_value`, and, for safety specialists only, `initial_inoculum_log_cfu_g` and `growth_support`.

1.4.1 What is deliberately excluded, and why it matters

The exclusion list is not a formality. Reading it is one of the more informative things one can do in this codebase, because the comments record checks that were actually performed. Eleven V7-specific columns are excluded, each for a stated and verifiable reason: `routing_category` is 1:1 with `cheese_category`; `physical_form_observed` is constant zero; `endpoint_threshold_basis` and `is_challenge_test` are bijective with `model_task` and therefore constant within any one specialist; `evidence_class`, `observed_fields`, `source_url`, `source_note`, `generation_rule_version`, and `matrix_profile_confidence` are provenance rather than physics.

Critically for this investigation, `source_rule_id` itself is excluded from the feature set. The models never see the rule label. The problem identified in Chapter 6 is therefore *not* that the rule identifier leaks into the features — it does not — but that the *evaluation split* ignores it.

One exclusion worth discussing with the panel

`target_is_lower_bound` is excluded with the comment that it “describes the TARGET’s censoring semantics, not a live input a user could supply.” That reasoning is sound for a deployed predictor: the flag would not be known at prediction time. But it means the regression objective treats censored and uncensored targets identically during fitting. Section 13.4 returns to this.

Why synthetic data, and what it costs

What to say at the start of this section

This is the section most likely to be interrupted with a direct question. Do not begin with a justification. Begin with the honest statement of what the repository can and cannot show, then give the scientific argument for why synthetic expansion was an intelligible choice, then state plainly what it costs. The panel will trust the argument more if the limitation is volunteered rather than extracted.

2.1 The evidence limitation, stated first

Before any argument about motivation, the following must be stated, because it constrains everything that can honestly be claimed afterwards.

Limitation 1 – the original experimental observations are not in this repository

Every dataset file used for training at any recoverable point in this project’s history was loaded and its `data_origin` column value-counted. The result appears in Table 2.1: across five successive dataset versions spanning 30 500 to 48 240 rows, every single `data_origin` value is a synthetic-generation tag. No file contains a real, experimental, or `real_paper_derived` row.

Consequently the original experimental or literature-extracted observations that must have calibrated the generation process cannot be counted, characterised, or audited from this repository. Any figure for their number – including the figure of roughly 400 that has been quoted informally – is unverifiable here. This report does not assert one.

Limitation 2 – the generation code is not in this repository

Every Python file in the repository was searched for code that *assigns* `source_rule_id`, as opposed to reading it for grouping or splitting. There is none. Every script that touches the column – the training driver, the serving layer, the diagnostic experiments – consumes it as a pre-existing column of an already-generated file. The parameter distributions, sampling ranges, target equations, noise model and censoring mechanism of the generator are therefore not inspectable here.

Everything this report says about generation-rule behaviour is inferred *statistically from the generated data* and is labelled as such. No generation equation is reconstructed or guessed.

A direct consequence for the meeting: the honest answer to “how many real observations did you start with” is that the number is not recoverable from this repository and should be requested from whoever maintains the generation pipeline. Stating that is stronger than quoting a number that cannot be substantiated.

Table 2.1: Every training dataset version in the repository, with the `data_origin` values actually present in it. Produced by loading each file and value-counting the column.

Dataset file	rows	<code>data_origin</code> values present
CHEESE_SHELF_LIFE_REVIS	30,500	<code>synthetic_literature_constrained</code> (20,000); <code>synthetic_cheese_database_anchored</code> (10,500)
CHEESE_SHELF_LIFE_CORRECTED_- GENERALIZED	41,285	<code>synthetic_literature_constrained</code> (20,000); <code>synthetic_literature_recalibrated</code> (10,785); <code>synthetic_cheese_database_anchored</code> (10,500)
CHEESE_SHELF_LIFE_RECALIBRATED_- V3	48,240	<code>synthetic_literature_recalibrated_v3</code> (31,800); <code>retained_soft_rebalanced_v3</code> (16,440)
CHEESE_SHELF_LIFE_TARGETED_V4	35,650	<code>synthetic_literature_constrained</code> (16,185); <code>synthetic_cheese_database_anchored</code> (9,865); <code>synthetic_independent_literature_calibrated_v4</code> (5,150); <code>synthetic_recalibrated_v4</code> (4,450)
V7 specialists (soft + semi-hard + hard)	34,000	<code>synthetic_literature_constrained_v7</code> (34,000)
Verified by loading each file and value-counting <code>data_origin</code> ; no non-synthetic tag appears anywhere.		

2.2 Why a small experimental corpus is hard to model, stated precisely

The claim “there was not enough data” is too loose to defend in front of a panel, and it is also not quite the right claim. There is no universal minimum row count for a Random Forest. What matters is the relationship between the diversity of the relationships to be learned and the number of *independent* observations supporting them. Four specific structural difficulties apply here, and each can be quantified from the data that does exist.

2.2.1 Fragmentation across specialists

The architecture divides whatever data exists by six. Figure 1.1 shows the division for the synthetic corpus, and the same division would apply to any real corpus. If a real corpus of a few hundred observations were distributed across six specialists in the proportions the synthetic corpus uses, the smallest specialist — hard/safety-endpoint, holding 6.6% of the corpus — would receive on the order of tens of rows. Tens of rows cannot support the estimation of a nonlinear function over twenty-plus heterogeneous inputs, and this conclusion does not depend on knowing the exact original count.

2.2.2 Dimensionality and heterogeneity of the input space

The feature space is not merely high-dimensional; it is *combinatorial*. A single prediction is conditioned on the cheese product, its physical form, its composition, the treatment and its concentration, the application method, the storage temperature, the packaging and its headspace gas composition, and the specific indicator being tracked with its threshold. In the V7 corpus the hard/general specialist alone spans six base cheese products, four treatment ingredients including the untreated control, and eight distinct indicator types — and that is after synthetic expansion. The product of the observed level counts is far larger than any realistic experimental programme could populate.

2.2.3 Sparse and uneven coverage

Coverage is not uniform across that space, and the unevenness is severe even in the expanded corpus. Figures 2.2 and 3.1 show it directly: in the soft/general specialist, one generation rule contributes 5 654 of 8 800 rows while two others contribute 17 and 8 rows respectively — a ratio of roughly 700:1 between the largest and smallest. Regions of the space represented by single-digit row counts cannot support reliable estimation, and Chapter 9 shows exactly what happens when the model is asked to predict them.

2.2.4 Statistical estimability

The relevant formal statement is the bias–variance decomposition. For a model $\hat{f}$ trained on a random sample $\mathcal{D}$, the expected squared error at a point $\mathbf{x}_0$ decomposes as

$$\mathbb{E}_{\mathcal{D}} \left[\left(y_0 - \hat{f}(\mathbf{x}_0) \right)^2 \right] = \underbrace{\sigma_{\varepsilon}^2}_{\text{irreducible}} + \underbrace{\left(\mathbb{E}_{\mathcal{D}}[\hat{f}(\mathbf{x}_0)] - f(\mathbf{x}_0) \right)^2}_{\text{bias}^2} + \underbrace{\mathbb{E}_{\mathcal{D}} \left[\left(\hat{f}(\mathbf{x}_0) - \mathbb{E}_{\mathcal{D}}[\hat{f}(\mathbf{x}_0)] \right)^2 \right]}_{\text{variance}}. \quad (2.1)$$

A flexible learner drives the bias term down but leaves the variance term governed by how much independent information $\mathcal{D}$ carries near $\mathbf{x}_0$. The key word is *independent*: the variance term does not shrink when the same underlying relationship is resampled, only when genuinely new information is added. This is the precise sense in which synthetic expansion can fail to help, and it is the theoretical statement of the empirical result in Chapter 6.

2.3 The trade-off that synthetic expansion actually makes

Synthetic generation from literature-constrained rules addresses coverage: it populates combinations of temperature, concentration, packaging and indicator that no experimental programme enumerated, in a way that is consistent with the relationships the source literature reports. That is a legitimate and common strategy, and it makes the six-specialist architecture trainable at all.

What it does not do is add independent scientific information. This distinction deserves a formal statement, because it is the intellectual core of the entire investigation.

Sample size is not information

Let n be the number of rows in a training set and let G be the number of distinct generative mechanisms that produced them. Classical learning theory bounds depend on the number of *independent* draws. When each mechanism g contributes n_g rows that are deterministic or near-deterministic functions of g 's parameters, the effective number of independent units is closer to G than to $n = \sum_g n_g$.

In the V7 corpus, $n = 34\,000$ but $G \in [7, 12]$ per specialist. The ratio n/G exceeds 1 000 for the largest rules. A model that learns G relationships well will score highly on any test set drawn from those same G relationships, at any value of n , and *regardless of its own capacity*.

Effective information: thousands of rows, but only 7–11 generation rules per specialist

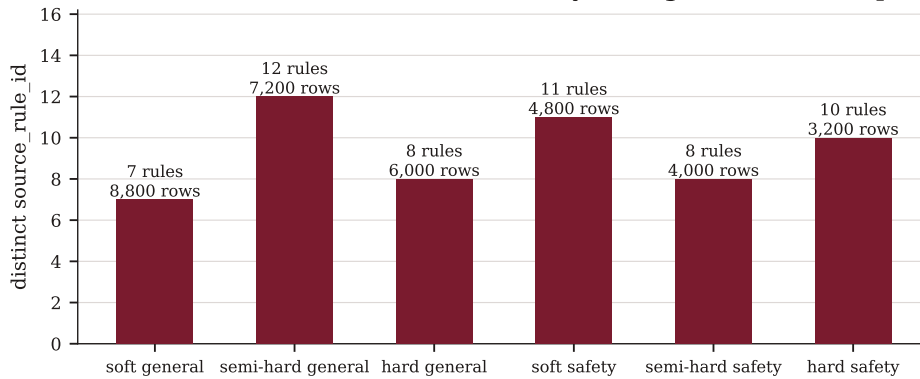


Figure 2.1: The quantity that governs generalization is not the row count but the rule count. Each specialist is built from between seven and twelve distinct `source_rule_id` values. Counted directly from the three V7 specialist files.

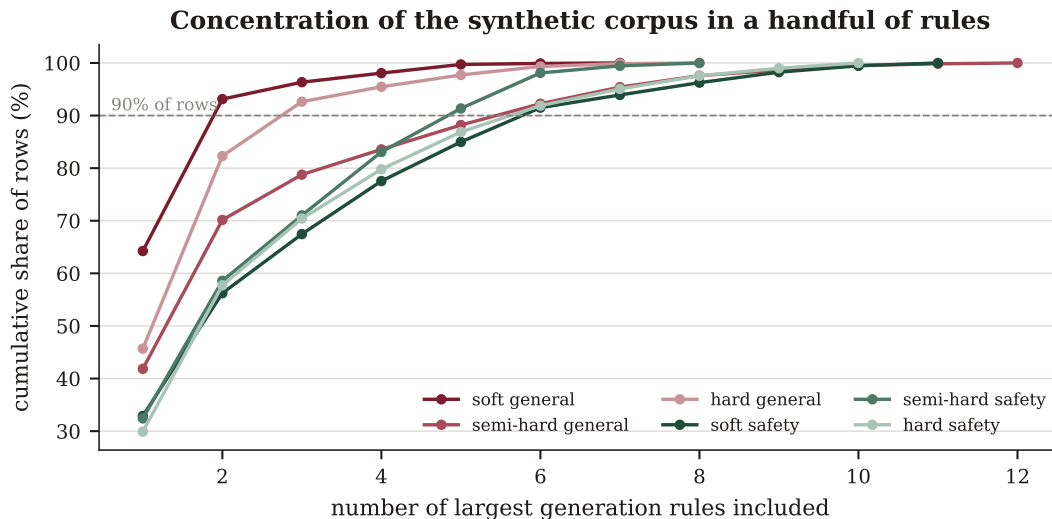


Figure 2.2: Cumulative share of each specialist’s rows as generation rules are added largest-first. In four of the six specialists, two rules account for more than 90% of all rows.

2.4 Alternatives that existed, and why they are not free

The panel may reasonably ask whether synthetic expansion was the only option. It was not, and it is worth being explicit about the alternatives and their own costs:

Train simpler models on the real data only

Fewer parameters would reduce variance, but the six-specialist architecture would still divide a few hundred observations by six, and the categorical inputs (product, indicator, packaging) would still be sparsely populated. This option trades one failure mode for another rather than removing it.

Hierarchical or multi-task modelling

Fitting one model with category-level random effects, rather than six independent models, would pool statistical strength across categories while still allowing category-specific behaviour. This is a genuine and largely untried alternative in this project, and Chapter 15 lists it as a recommendation.

Transfer from a mechanistic baseline

Predictive microbiology supplies parametric growth models whose parameters could be fitted to the small real corpus, with machine learning used only for the residual. This constrains the model with physics instead of with synthetic rows.

Collecting more experimental data

The scientifically strongest option and the slowest: shelf-life studies run for the duration of the shelf life being measured, across multiple temperatures, with microbiological enumeration at each time point.

Likely question – “Why did you use synthetic data at all?”

Suggested answer. Because the prediction problem is split across six specialists and a combinatorial input space, and the experimental corpus available at the time could not populate that space — some specialists would have received only tens of rows. Synthetic generation under literature-constrained rules was used to make the architecture trainable, not because synthetic rows were believed equivalent to measurements.

What we have since established, and what this report documents, is that expansion of this

kind increases sample size without increasing independent information: the corpus has 34 000 rows but only seven to twelve generation rules per specialist. That is exactly why the original evaluation was optimistic, and it is why we now evaluate by holding out whole generation rules. I would also say plainly that the original experimental observations are not present in this repository, so I cannot give you a verified count for them.

The synthetic generation system, reconstructed from its output

What to say at the start of this section

Be explicit that this chapter reconstructs the generator's *behaviour* from the data it produced, because the generator's code is not in this repository. Then show the rule inventory — it is the single most informative table in the report, and it makes the rest of the argument almost self-evident.

3.1 What a `source_rule_id` actually is

Each row of the V7 corpus carries a `source_rule_id`. Reading the actual values shows immediately that a rule identifier is *not* a single publication and *not* a single physical relationship. It is a pipe-delimited composite of source components. Examples taken verbatim from the soft specialist:

<code>source_rule_id</code>	rows
V7 PH_CHEESE	5 654
V7 PLANT_REVIEW	2 541
V7 RICOTTA_MAP	283
V7 NATAMYCIN_REVIEW	152
V7 RICOTTA_MAP PLANT_REVIEW	145
V7 HMMC_NAT NATAMYCIN_REVIEW	17
V7 RICOTTA_MAP NATAMYCIN_REVIEW	8

Three structural properties follow directly, and all three matter for the evaluation design:

1. **Rules share components.** RICOTTA_MAP appears alone, with PLANT_REVIEW, and with NATAMYCIN_REVIEW. NATAMYCIN_REVIEW appears in three different composites. Rules are therefore not mutually independent even as labels — they are intersections drawn from a smaller set of underlying source components.
2. **Composite rules are small.** The single-component rules are large (5 654 and 2 541 rows); the three-way composites are tiny (17 and 8 rows). The composites appear to represent the specific intersections where two sources both constrain the same formulation.
3. **The distribution is extremely unbalanced.** The largest rule in soft/general is 707 times the size of the smallest.

What cannot be said

Because the generator is absent, this report cannot state the mathematical form of any rule's target equation, its parameter distributions, its noise model, or whether two rules that share a component also share calibration constants. The composite naming makes shared structure *likely*, but that is an inference from identifiers, not a verified property of the generating code.

Chapter 13 treats this as a live limitation of the revised evaluation, not a closed question.

3.2 Complete rule inventory for all six specialists

Table 3.1 enumerates every generation rule in the production corpus with the statistics that can be computed from the data: row count, number of distinct contexts, number of distinct cheese products and indicator types, the storage-temperature range spanned, and the mean and standard deviation of the target. Table 3.3 gives the products and treatment ingredients associated with each rule.

Table 3.1: Generation-rule inventory for all six specialists. Every column is computed from the V7 specialist files; the target equations behind these rules are not recoverable here.

Rule (source_rule - id)	rows	ctx	prod.	ind.	temp range (°C)	$\bar{y}$ (d)	s_y (d)
soft / general shelf life							
PH_CHEESE	5,654	2091	18	10	2–12	24.4	25.4
PLANT_REVIEW	2,541	1604	18	9	2–12	27.6	28.4
RICOTTA_MAP	283	109	1	10	2–12	6.3	2.0
NATAMYCIN_REVIEW	152	152	16	1	2–12	31.4	28.7
RICOTTA_MAP PLANT_REVIEW	145	88	1	9	2–12	7.2	2.2
HMMC_NAT NATAMYCIN_REVIEW	17	17	2	1	2–8	8.1	1.8
RICOTTA_MAP NATAMYCIN_REVIEW	8	8	1	1	2–4	9.5	1.1
semi-hard / general shelf life							
HARD_SEMI_REVIEW	3,013	1302	9	9	2–12	60.1	24.3
HARD_SEMI_REVIEW PLANT_REVIEW	2,038	1024	9	8	2–12	67.0	27.7
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW	620	155	9	1	8–15	185.6	54.3
HARD_SEMI_REVIEW SALOIO_COATING	346	165	1	9	2–12	29.8	6.3
HARD_SEMI_REVIEW LMM_MAP	333	143	1	9	2–12	55.2	16.4
HARD_SEMI_REVIEW SALOIO_COATING PLANT_REVIEW	292	136	1	8	2–12	34.0	7.5
HARD_SEMI_REVIEW LMM_MAP PLANT_REVIEW	227	112	1	8	2–12	60.9	18.7
HARD_SEMI_REVIEW NATAMYCIN_REVIEW	157	157	9	2	2–12	66.9	29.2
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW SALOIO_COATING	76	19	1	1	8–15	187.3	48.2
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW LMM_MAP	64	16	1	1	8–15	149.9	52.6
HARD_SEMI_REVIEW SALOIO_COATING NATAMYCIN_REVIEW	22	22	1	2	2–12	34.4	7.6
HARD_SEMI_REVIEW LMM_MAP NATAMYCIN_REVIEW	12	12	1	2	2–10	54.6	13.0
hard / general shelf life							
HARD_SEMI_REVIEW	2,741	1277	6	7	2–12	74.6	23.5

Rule (source_rule - id)	rows	ctx	prod.	ind.	temp range (°C)	$\bar{y}$ (d)	s_y (d)
HARD_SEMI_REVIEW PLANT_REVIEW	2,197	1020	6	6	2–12	82.0	25.7
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW	620	155	6	1	8–15	189.0	37.7
HARD_SEMI_REVIEW NATAMYCIN_REVIEW	170	170	6	2	2–12	79.7	24.0
HARD_SEMI_REVIEW SHRED_NAT	135	60	1	7	2–12	38.4	8.3
HARD_SEMI_REVIEW SHRED_NAT PLANT_REVIEW	97	45	1	6	2–12	42.5	7.6
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW SHRED_NAT	32	8	1	1	8–15	184.3	33.1
HARD_SEMI_REVIEW SHRED_NAT NATAMYCIN_REVIEW	8	8	1	2	4–12	50.2	9.4
soft / safety endpoint							
FDA_LM LM_CHALLENGE MARKET_LM	1,579	484	16	1	4–12	27.2	17.1
RISK_REVIEW PH_CHEESE	1,120	401	19	2	4–12	36.4	14.1
S_AUREUS S_AUREUS_REVIEW PH_CHEESE	539	213	19	1	4–12	36.5	13.8
RISK_REVIEW PH_CHEESE PLANT_REVIEW	484	341	19	2	4–12	37.4	14.0
FDA_LM LM_CHALLENGE MARKET_LM PLANT_REVIEW	357	314	16	1	4–12	27.3	16.6
S_AUREUS S_AUREUS_REVIEW PH_CHEESE PLANT_REVIEW	313	205	19	1	4–12	36.6	13.6
FDA_LM LM_CHALLENGE MARKET_LM RICOTTA_NISIN	116	29	1	1	4–12	10.4	6.5
FDA_LM LM_CHALLENGE MARKET_LM FETA_OREGANO	112	28	1	1	4–12	34.8	13.8
FDA_LM LM_CHALLENGE MARKET_LM QF_MARKET	96	24	1	1	4–12	10.5	4.8
FETA_OREGANO RISK_REVIEW PH_CHEESE	58	21	1	1	4–12	41.1	11.2
FETA_OREGANO RISK_REVIEW PH_CHEESE PLANT_REVIEW	26	17	1	1	4–12	39.8	12.6
semi-hard / safety endpoint							
FDA_LM LM_CHALLENGE MARKET_LM	1,297	407	11	1	4–15	56.8	21.0
RISK_REVIEW PH_CHEESE	1,046	382	11	2	4–15	60.1	17.5
S_AUREUS S_AUREUS_REVIEW PH_CHEESE	498	192	11	1	4–15	61.2	21.4
RISK_REVIEW PH_CHEESE PLANT_REVIEW	482	340	11	2	4–15	61.4	16.9
FDA_LM LM_CHALLENGE MARKET_LM PLANT_REVIEW	331	286	11	1	4–15	55.6	20.8
S_AUREUS S_AUREUS_REVIEW PH_CHEESE PLANT_REVIEW	270	180	11	1	4–15	60.7	20.7
KASHAR_E0 RISK_REVIEW PH_CHEESE	54	19	1	1	4–15	62.1	17.0
KASHAR_E0 RISK_REVIEW PH_CHEESE PLANT_REVIEW	22	16	1	1	4–15	65.1	16.3

Rule (source_rule_id)	rows	ctx	prod.	ind.	temp range (°C)	$\bar{y}$ (d)	s_y (d)
hard / safety endpoint							
FDA_LM LM_CHALLENGE MARKET_LM	957	296	6	1	4–15	82.3	20.9
RISK_REVIEW PH_CHEESE	890	324	5	2	4–15	81.7	20.2
RISK_REVIEW PH_CHEESE PLANT_REVIEW	406	282	5	2	4–15	81.7	20.2
S_AUREUS S_AUREUS_REVIEW PH_CHEESE	300	115	6	1	4–15	83.0	22.1
FDA_LM LM_CHALLENGE MARKET_LM PLANT_REVIEW	227	190	6	1	4–15	83.2	21.9
S_AUREUS S_AUREUS_REVIEW PH_CHEESE PLANT_REVIEW	160	108	6	1	4–15	82.2	21.7
CHEDDAR_SALMONELLA RISK_REVIEW PH_CHEESE	102	37	1	1	4–15	89.3	21.4
CHEDDAR_ECOLI RISK_REVIEW PH_CHEESE	80	28	1	1	4–15	82.9	20.9
CHEDDAR_SALMONELLA RISK_REVIEW PH_CHEESE PLANT_REVIEW	46	29	1	1	4–15	88.7	20.8
CHEDDAR_ECOLI RISK_REVIEW PH_CHEESE PLANT_REVIEW	32	24	1	1	4–15	86.2	19.0

Table 3.3: Cheese products and primary treatment ingredients associated with each generation rule, as they appear in the data.

Rule	cheese products (up to 4)	primary ingredients (up to 3)
soft / general shelf life		
PH_CHEESE	brie; burrata; camembert; cottage cheese ...	acetic acid; calcium propionate; chitosan ...
PLANT_REVIEW	brie; burrata; camembert; cottage cheese ...	clove essential oil; grape seed extract; green tea extract ...
RICOTTA_MAP	ricotta	acetic acid; calcium propionate; chitosan ...
NATAMYCIN_REVIEW	brie; burrata; camembert; cottage cheese ...	natamycin
RICOTTA_MAP PLANT_REVIEW	ricotta	clove essential oil; grape seed extract; green tea extract ...
HMMC_NAT NATAMYCIN_REVIEW	fior di latte; fresh mozzarella	natamycin
RICOTTA_MAP NATAMYCIN_REVIEW	ricotta	natamycin
semi-hard / general shelf life		
HARD_SEMI_REVIEW	colby; edam; gouda; halloumi ...	acetic acid; calcium propionate; chitosan ...
HARD_SEMI_REVIEW PLANT_REVIEW	colby; edam; gouda; halloumi ...	clove essential oil; grape seed extract; green tea extract ...
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW	colby; edam; gouda; halloumi ...	lysozyme; none; potassium nitrate ...
HARD_SEMI_REVIEW SALOIO_COATING	saloio cheese	acetic acid; calcium propionate; chitosan ...

Rule	cheese products (up to 4)	primary ingredients (up to 3)
HARD_SEMI_REVIEW LMM_MAP	low-moisture mozzarella	acetic acid; calcium propionate; chitosan ...
HARD_SEMI_REVIEW SALOIO_COATING PLANT_REVIEW	saloio cheese	clove essential oil; grape seed extract; green tea extract ...
HARD_SEMI_REVIEW LMM_MAP PLANT_REVIEW	low-moisture mozzarella	clove essential oil; grape seed extract; green tea extract ...
HARD_SEMI_REVIEW NATAMYCIN_REVIEW	colby; edam; gouda; halloumi ...	natamycin
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW SALOIO_COATING	saloio cheese	lysozyme; none; potassium nitrate ...
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW LMM_MAP	low-moisture mozzarella	lysozyme; none; potassium nitrate ...
HARD_SEMI_REVIEW SALOIO_COATING NATAMYCIN_REVIEW	saloio cheese	natamycin
HARD_SEMI_REVIEW LMM_MAP NATAMYCIN_REVIEW	low-moisture mozzarella	natamycin
hard / general shelf life		
HARD_SEMI_REVIEW	cheddar; emmental; gruyère; manchego ...	acetic acid; calcium propionate; chitosan ...
HARD_SEMI_REVIEW PLANT_REVIEW	cheddar; emmental; gruyère; manchego ...	clove essential oil; grape seed extract; green tea extract ...
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW	cheddar; emmental; gruyère; manchego ...	lysozyme; none; potassium nitrate ...
HARD_SEMI_REVIEW NATAMYCIN_REVIEW	cheddar; emmental; gruyère; manchego ...	natamycin
HARD_SEMI_REVIEW SHRED_NAT	cheddar	acetic acid; calcium propionate; chitosan ...
HARD_SEMI_REVIEW SHRED_NAT PLANT_REVIEW	cheddar	clove essential oil; grape seed extract; green tea extract ...
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW SHRED_NAT	cheddar	lysozyme; none; potassium nitrate ...
HARD_SEMI_REVIEW SHRED_NAT NATAMYCIN_REVIEW	cheddar	natamycin
soft / safety endpoint		
FDA_LM LM_CHALLENGE MARKET_LM	brie; burrata; camembert; cottage cheese ...	caprylic acid; chitosan; lactic acid ...
RISK_REVIEW PH_CHEESE	brie; burrata; camembert; cottage cheese ...	acetic acid; caprylic acid; chitosan ...
S_AUREUS S_AUREUS_REVIEW PH_CHEESE	brie; burrata; camembert; cottage cheese ...	lactic acid; nisin; none ...
RISK_REVIEW PH_CHEESE PLANT_REVIEW	brie; burrata; camembert; cottage cheese ...	clove essential oil; oregano essential oil; thyme essential oil
FDA_LM LM_CHALLENGE MARKET_LM PLANT_REVIEW	brie; burrata; camembert; cottage cheese ...	oregano essential oil; thyme essential oil
S_AUREUS S_AUREUS_REVIEW PH_CHEESE PLANT_REVIEW	brie; burrata; camembert; cottage cheese ...	clove essential oil; oregano essential oil; thyme essential oil

Rule	cheese products (up to 4)	primary ingredients (up to 3)
FDA_LM LM_CHALLENGE MARKET_LM RICOTTA_ NISIN	ricotta	caprylic acid; chitosan; lactic acid ...
FDA_LM LM_CHALLENGE MARKET_LM FETA_ OREGANO	feta	caprylic acid; chitosan; lactic acid ...
FDA_LM LM_CHALLENGE MARKET_LM QF_MARKET	queso fresco	caprylic acid; chitosan; lactic acid ...
FETA_OREGANO RISK_ REVIEW PH_CHEESE	feta	acetic acid; caprylic acid; chitosan ...
FETA_OREGANO RISK_ REVIEW PH_CHEESE PLANT_REVIEW	feta	clove essential oil; oregano essential oil; thyme essential oil
semi-hard / safety endpoint		
FDA_LM LM_CHALLENGE MARKET_LM	colby; edam; gouda; halloumi ...	caprylic acid; chitosan; lactic acid ...
RISK_REVIEW PH_CHEESE	colby; edam; gouda; halloumi ...	acetic acid; caprylic acid; chitosan ...
S_AUREUS S_AUREUS_ REVIEW PH_CHEESE	colby; edam; gouda; halloumi ...	lactic acid; nisin; none ...
RISK_REVIEW PH_ CHEESE PLANT_REVIEW	colby; edam; gouda; halloumi ...	clove essential oil; oregano essential oil; thyme essential oil
FDA_LM LM_CHALLENGE MARKET_LM PLANT_ REVIEW	colby; edam; gouda; halloumi ...	oregano essential oil; thyme essential oil
S_AUREUS S_AUREUS_ REVIEW PH_CHEESE PLANT_REVIEW	colby; edam; gouda; halloumi ...	clove essential oil; oregano essential oil; thyme essential oil
KASHAR_EO RISK_ REVIEW PH_CHEESE	kashar	acetic acid; caprylic acid; chitosan ...
KASHAR_EO RISK_ REVIEW PH_CHEESE PLANT_REVIEW	kashar	clove essential oil; oregano essential oil; thyme essential oil
hard / safety endpoint		
FDA_LM LM_CHALLENGE MARKET_LM	cheddar; emmental; gruyère; manchego ...	caprylic acid; chitosan; lactic acid ...
RISK_REVIEW PH_CHEESE	emmental; gruyère; manchego; parmigiano reg- giano ...	acetic acid; caprylic acid; chitosan ...
RISK_REVIEW PH_ CHEESE PLANT_REVIEW	emmental; gruyère; manchego; parmigiano reg- giano ...	clove essential oil; oregano essential oil; thyme essential oil
S_AUREUS S_AUREUS_ REVIEW PH_CHEESE	cheddar; emmental; gruyère; manchego ...	lactic acid; nisin; none ...
FDA_LM LM_CHALLENGE MARKET_LM PLANT_ REVIEW	cheddar; emmental; gruyère; manchego ...	oregano essential oil; thyme essential oil
S_AUREUS S_AUREUS_ REVIEW PH_CHEESE PLANT_REVIEW	cheddar; emmental; gruyère; manchego ...	clove essential oil; oregano essential oil; thyme essential oil
CHEDDAR_SALMONELLA RISK_REVIEW PH_CHEESE	cheddar	acetic acid; caprylic acid; chitosan ...
CHEDDAR_ECOLI RISK_ REVIEW PH_CHEESE	cheddar	acetic acid; caprylic acid; chitosan ...
CHEDDAR_SALMONELLA RISK_REVIEW PH_ CHEESE PLANT_REVIEW	cheddar	clove essential oil; oregano essential oil; thyme essential oil
CHEDDAR_ECOLI RISK_ REVIEW PH_CHEESE PLANT_REVIEW	cheddar	clove essential oil; oregano essential oil; thyme essential oil

Rows per generation rule (log scale): every specialist is dominated by one or two rules

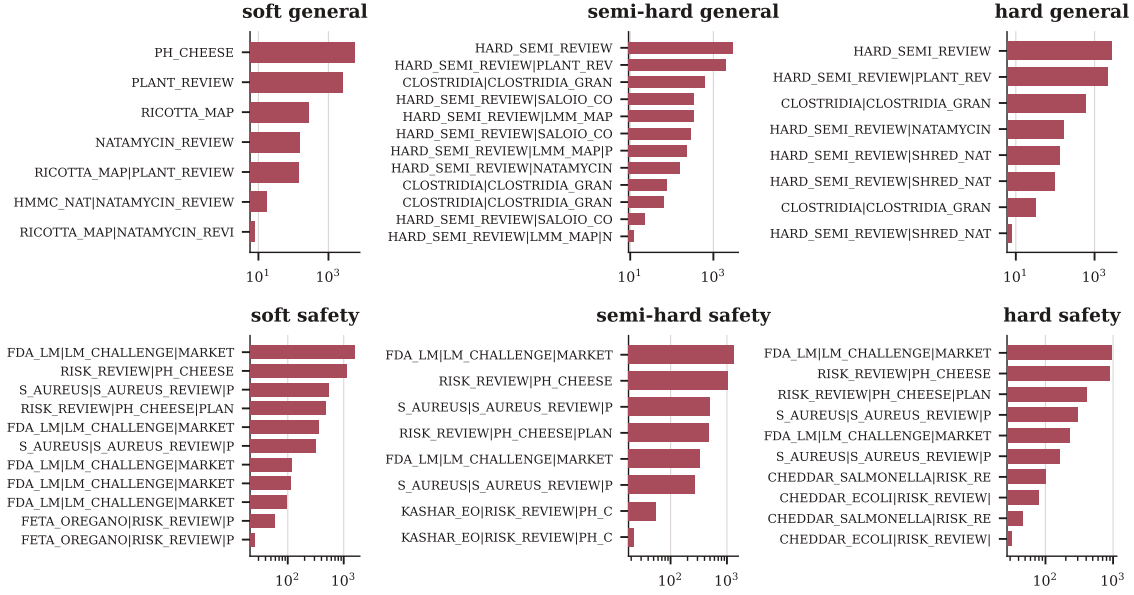


Figure 3.1: Rows per generation rule, logarithmic scale, for each of the six specialists. The imbalance is a property of the corpus, not of the evaluation, and it is the reason the mean and median of per-fold scores diverge so sharply in Chapter 7.

3.3 The dependency between rows generated by one rule

Consider two rows drawn from the same rule. They differ in their feature values — a different storage temperature, a different concentration, a different indicator threshold — so they are distinct points in feature space and no row-level duplication exists. What they share is the *mapping* from those features to the target.

Write the generator for rule g as a function ϕ_g with parameters θ_g fixed by that rule’s source constraints, applied to a sampled input $\mathbf{x}$ and perturbed by noise:

$$y = \phi_g(\mathbf{x}; \theta_g) + \varepsilon, \quad \mathbf{x} \sim \mathcal{X}_g, \quad \varepsilon \sim \mathcal{E}_g. \quad (3.1)$$

The form of ϕ_g is unknown to this report. But regardless of its form, the following holds: if the training set contains many draws from $\mathcal{X}_g$ and the test set contains further draws from the same $\mathcal{X}_g$ under the same ϕ_g and θ_g , then a sufficiently flexible learner can recover ϕ_g to within $\mathcal{E}_g$ and will score on the test rows as well as the noise floor allows.

The learning problem in that situation is *interpolation within a known mechanism*. It is a real learning problem — the model still has to recover ϕ_g from finitely many samples — but it is a fundamentally easier one than predicting a mechanism never seen, and it is not the problem a deployed shelf-life predictor faces when it meets a new formulation.

3.4 Contexts are not mechanisms

The protocol’s grouping variable, `context_id`, identifies a formulation-plus-condition scenario; one context contributes several rows (different indicators, different time points). The V7 soft/general specialist has 8 800 rows spread over its contexts, and each generation rule spans many contexts.

That nesting is the entire problem. Holding out a context removes one sample of a mechanism; it does not remove the mechanism. The next chapter quantifies exactly how much mechanism-level information survives the original split, and the answer turns out to be: all of it.

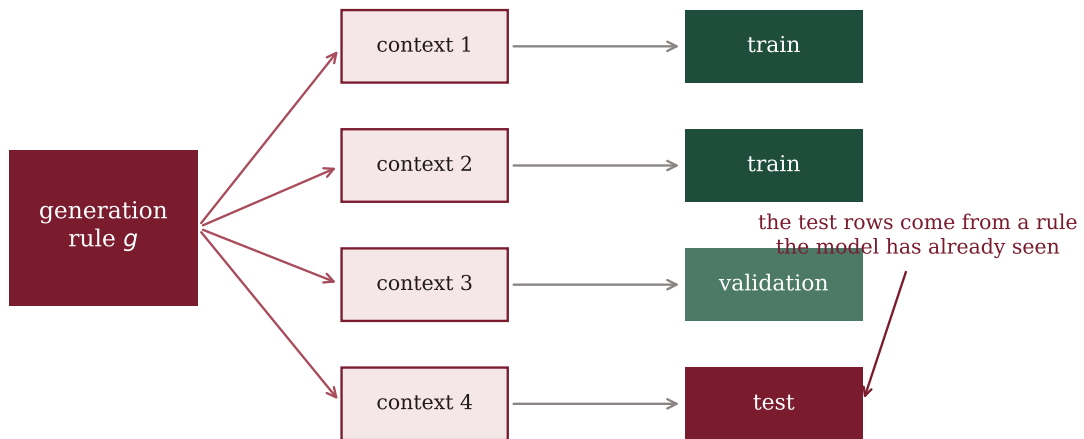
Original protocol: grouping by context_id, not by generation rule

Figure 3.2: The original evaluation protocol in schematic form. Grouping by context_id guarantees that no single context contributes rows to two partitions. It places no constraint whatever on the generation rule, so rows from rule g reach training, validation and test simultaneously.

The original evaluation, reproduced from source

What to say at the start of this section

State that the original protocol was not merely read but re-executed, with the production seed, so the numbers discussed are the actual production numbers rather than a description of them. Then present the protocol’s design intent honestly: it was written to prevent a specific, real form of leakage, and it succeeds at that.

4.1 The splitting procedure as implemented

The production split is performed by `make_context_splits()` in `model_service.py`, called from the training driver with the global seed. Its logic, read from source:

1. Compute, for each `context_id`, whether that context contains any row with `data_origin == "real_paper_derived"`.
2. Partition the context identifiers into two strata by that boolean.
3. Within each stratum, shuffle the context identifiers with `numpy.random.default_rng(42)` and allocate them 70/15/15 to train, validation and test according to `SPLIT_FRACTIONS = {train: 0.70, validation: 0.15, test: 0.15}`.
4. Return the three row subsets selected by context membership.

The design intent is explicit in the function’s own docstring: it is a “leakage-safe split” that groups by context “so a context is never split across train/val/test,” stratified so that a small real-data subset would be represented in every partition rather than left to chance.

The protocol does what it claims — and that is the point

Re-executing the production splitter with the production seed and checking context membership directly confirms **zero context overlap between train and test in all six specialists**. The protocol is not broken. It correctly enforces the constraint it was designed to enforce. The investigation’s finding is not that the code is wrong, but that the constraint it enforces is not the constraint this dataset requires.

4.1.1 An observation about the stratification

The stratification variable is `data_origin == "real_paper_derived"`. In the V7 corpus, every row has `data_origin == "synthetic_literature_constrained_v7"`. The predicate is therefore uniformly false, one stratum is empty, and the stratification is a no-op on this data.

This is not an error — it is a piece of machinery written when real rows were expected to be present, and it would function if they were. It is worth mentioning to the panel because it is independent corroboration of Section 2.1: the pipeline was built anticipating real observations that the corpus does not currently contain.

4.2 The original numerical results

Table 4.1 gives the complete production results for the V7 dataset: all six specialists $\times$ four model families, with the training, validation and test R^2 and the test RMSE, recovered from the stored `metrics.json` and `training_manifest.json` artefacts rather than from any summary document. Table 4.2 gives the corresponding V6 results.

Table 4.1: Original production results, V7 dataset, context-holdout protocol. Recovered from `artifacts_v7/{category}/{task}/metrics.json`.

Specialist	Model	n_{train}	n_{test}	train R^2	val R^2	test R^2	test RMSE
soft/general shelf life	random_forest	6,160	1,320	0.997	0.955	0.956	5.99
	lightgbm	6,160	1,320	0.997	0.972	0.971	4.89
	xgboost	6,160	1,320	0.999	0.969	0.968	5.11
	ebm	6,160	1,320	0.985	0.944	0.944	6.79
semi-hard/general shelf life	random_forest	5,040	1,080	0.995	0.925	0.905	14.57
	lightgbm	5,040	1,080	0.990	0.931	0.920	13.39
	xgboost	5,040	1,080	0.995	0.930	0.927	12.74
	ebm	5,040	1,080	0.983	0.900	0.909	14.30
hard/general shelf life	random_forest	4,200	900	0.994	0.910	0.927	11.68
	lightgbm	4,200	900	0.991	0.916	0.932	11.30
	xgboost	4,200	900	0.995	0.911	0.933	11.26
	ebm	4,200	900	0.977	0.880	0.917	12.47
soft/safety endpoint	random_forest	3,360	720	0.997	0.548	0.572	10.62
	lightgbm	3,360	720	0.876	0.583	0.573	10.61
	xgboost	3,360	720	0.832	0.592	0.594	10.34
	ebm	3,360	720	0.944	0.393	0.360	12.98
semi-hard/safety endpoint	random_forest	2,800	600	0.997	0.015	0.342	15.76
	lightgbm	2,800	600	0.449	0.089	0.303	16.21
	xgboost	2,800	600	0.468	0.115	0.317	16.05
	ebm	2,800	600	0.936	-0.280	-0.127	20.63
hard/safety endpoint	random_forest	2,240	480	0.996	-0.131	-0.097	21.86
	lightgbm	2,240	480	0.036	-0.003	-0.024	21.12
	xgboost	2,240	480	0.046	0.008	-0.011	20.98
	ebm	2,240	480	0.953	-0.731	-0.528	25.79

4.2.1 Reading the table honestly

Three distinct patterns are present, and conflating them would be a mistake.

The general-shelf-life specialists score very high. Soft/general reaches test $R^2 = 0.971$ (LightGBM), hard/general 0.932, semi-hard/general 0.927. These are the numbers that prompted the concern.

The safety-endpoint specialists do not. On V7, soft/safety reaches only 0.594, semi-hard/safety 0.342, and hard/safety is *negative* at -0.097 for Random Forest and -0.528 for EBM. A negative R^2 means the model predicts worse than the test-set mean. Any explanation of the high scores must also be consistent with these low ones, which rules out “the models memorise everything” as a general account.

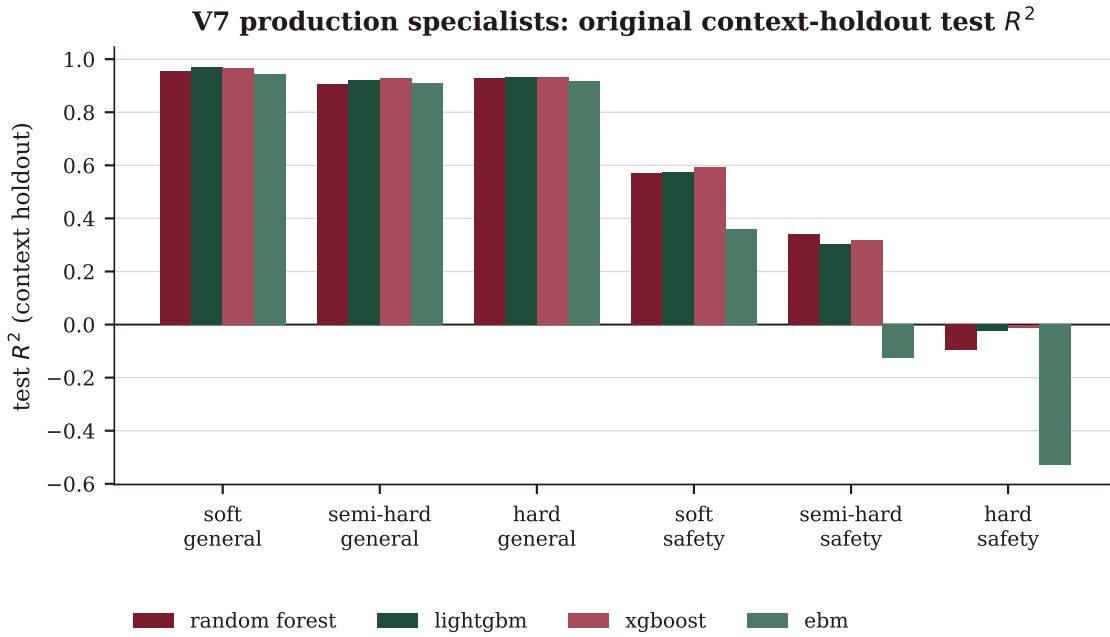


Figure 4.1: Original context-holdout test R^2 for the twenty-four production fits. The three general-shelf-life specialists produce the scores that prompted this investigation; the safety-endpoint specialists behave very differently and are discussed in Section 4.2.1.

The V6 → V7 transition moved the two tasks in opposite directions. Comparing Table 4.2 with Table 4.1: soft/general improved from 0.936 to 0.971, while soft/safety *fell* from 0.917 to 0.573 and hard/safety fell from 0.786 to -0.024 (LightGBM in each case).

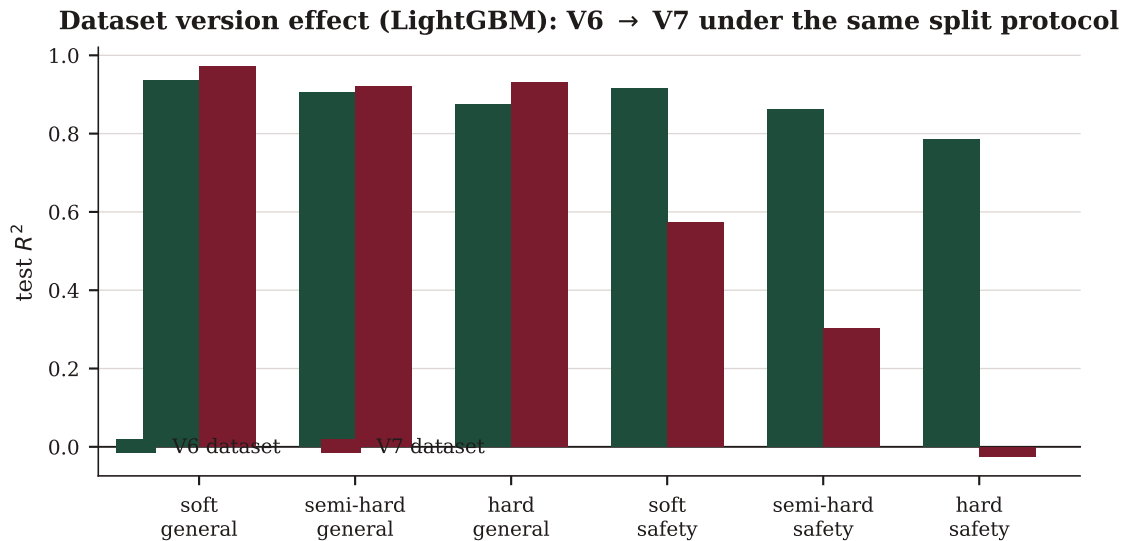


Figure 4.2: The same model and the same protocol on two dataset versions. The divergence between quality and safety endpoints under V7 is a property of the data, not of the learner.

4.2.2 Optimism gap

Figure 4.3 plots the difference between training and test R^2 for every V7 production fit. Random Forest and XGBoost train to $R^2 > 0.99$ almost everywhere, which is expected for unpruned bagged trees and deep boosted trees — a training R^2 near unity is a statement about capacity, not about generalization. The informative quantity is the gap, and the gap is small precisely where the test

Table 4.2: Original production results, V6 dataset, same protocol. Retained for comparison; not over-written by the V7 run.

Specialist	Model	n_{train}	n_{test}	train R^2	val R^2	test R^2	test RMSE
soft/general shelf life	random_forest	10,035	2,150	0.993	0.927	0.930	7.62
	lightgbm	10,035	2,150	0.993	0.947	0.936	7.28
	xgboost	10,035	2,150	0.996	0.943	0.940	7.03
	ebm	10,035	2,150	0.950	0.883	0.894	9.35
semi-hard/general shelf life	random_forest	7,330	1,570	0.993	0.877	0.889	26.65
	lightgbm	7,330	1,570	0.974	0.882	0.905	24.65
	xgboost	7,330	1,570	0.980	0.882	0.896	25.74
	ebm	7,330	1,570	0.969	0.859	0.889	26.61
hard/general shelf life	random_forest	3,345	720	0.988	0.866	0.847	33.67
	lightgbm	3,345	720	0.996	0.889	0.876	30.39
	xgboost	3,345	720	0.998	0.886	0.866	31.54
	ebm	3,345	720	0.963	0.782	0.801	38.45
soft/safety endpoint	random_forest	2,935	630	0.989	0.884	0.907	5.54
	lightgbm	2,935	630	0.975	0.934	0.917	5.24
	xgboost	2,935	630	1.000	0.934	0.912	5.38
	ebm	2,935	630	0.948	0.775	0.855	6.91
semi-hard/safety endpoint	random_forest	780	170	0.981	0.778	0.777	21.41
	lightgbm	780	170	0.983	0.763	0.861	16.89
	xgboost	780	170	0.993	0.749	0.788	20.88
	ebm	780	170	0.984	0.733	0.831	18.66
hard/safety endpoint	random_forest	530	110	0.986	0.843	0.753	30.57
	lightgbm	530	110	0.995	0.866	0.786	28.44
	xgboost	530	110	0.999	0.863	0.782	28.75
	ebm	530	110	0.993	0.709	0.703	33.51

score is high.

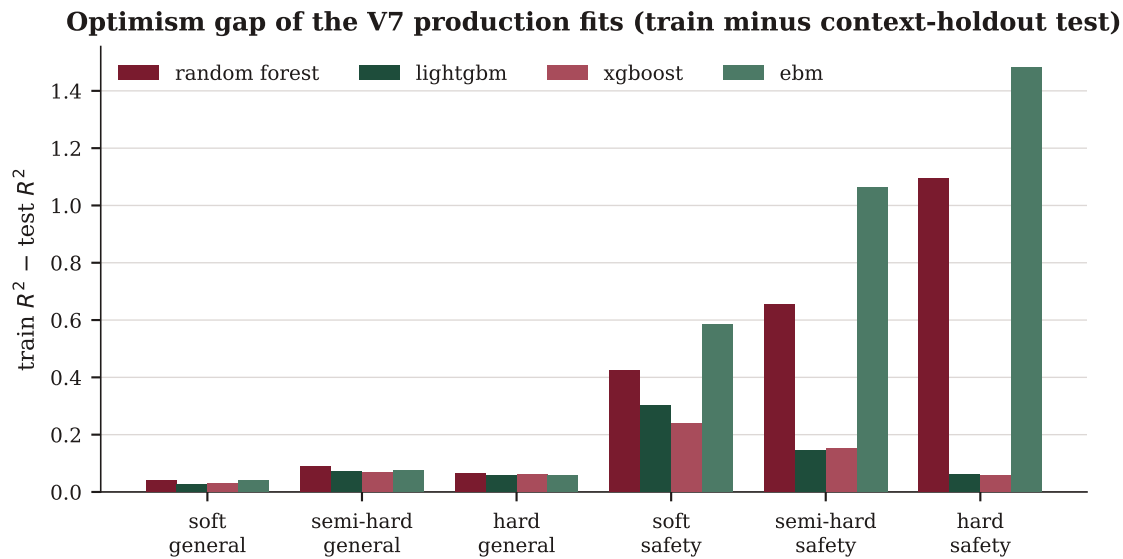


Figure 4.3: Train-minus-test R^2 for the V7 production fits. The large gaps sit on the safety-endpoint specialists, where the test scores are low; the quality specialists show small gaps and high test scores.

4.3 Four things a high R^2 could mean

The single most important conceptual point of the meeting is that these four explanations are distinct, and the original result is consistent with only some of them. They must not be used interchangeably.

1. Classical overfitting

The model has memorised individual training rows, including their noise, and performs well on training data and poorly on any held-out data. *This is refuted by the original results themselves:* held-out test scores are high, not low. Whatever is happening, it is not the textbook train-high/test-low failure.

2. Data leakage

A feature carries information about the target that would not be available at prediction time — an identifier, a downstream measurement, an encoded target. *Directly checked:* the exclusion list removes identifiers and provenance, including `source_rule_id` itself, and Section 13.6 audits the remaining features individually. No target-derived feature was found in the model inputs.

3. Optimistic evaluation from shared generative structure

The test set is statistically independent at the level the protocol controls (contexts) but not at the level that governs the mapping being learned (generation rules). The model is scored on new samples of relationships it was trained on. *This is what the evidence supports*, and Chapter 6 quantifies it.

4. Genuine generalization to independent observations

The model would predict well on new experimental data. *Not established by any internal metric*, and Chapter 14 shows the external evidence points the other way.

The precise claim

The original $R^2 \approx 0.92\text{--}0.97$ figures are best described as **valid measurements of the wrong quantity**. They accurately measure how well the model interpolates within known generation mechanisms. They were reported, and read, as if they measured generalization to new formulations. The fault is in the inference drawn from the protocol, not in the arithmetic of the metric.

The diagnostic experiments

What to say at the start of this section

Frame these as elimination, not as vindication. Each experiment tests one of the explanations that was proposed, and each returns a negative result. Negative results are what narrow a diagnosis, and the fact that four independent interventions all fail to move the score is itself the strongest evidence that the cause lies outside the model.

5.1 Design of the diagnostic suite

All diagnostic experiments share a common harness (`scripts/experiments/overfitting_diagnosis.py`) so that comparisons are matched. The shared elements are: the V7 specialist files; the production exclusion list; a preprocessing pipeline of median imputation plus standardisation for numeric columns and most-frequent imputation plus one-hot encoding for categorical columns, *always fitted on the training partition only*; the seed 42; and the production context-holdout split, except where an experiment explicitly changes it.

Six experiments were executed, producing 164 individual model fits whose results are stored as CSV files in `scripts/experiments/overfitting_diagnosis/`.

5.2 Experiment 1 — reducing model capacity

5.2.1 What was changed

Four deliberately low-capacity learners replaced the production ensembles, holding the data and the split fixed:

- **Ridge regression** — linear in the one-hot expanded features, L_2 -penalised, no interactions and no nonlinearity.
- **Shallow decision tree** — a single tree with restricted depth and an enforced minimum leaf size.
- **Shallow random forest** — a small, depth-limited forest.
- **Small MLP** — a compact multilayer perceptron.

5.2.2 A note on the proposal to use logistic regression and CNNs

Two suggestions in the original feedback require a technical clarification, which should be made politely but clearly in the meeting.

Logistic regression is a classifier: it models $P(y = k \mid \mathbf{x})$ over a discrete label set. It cannot be applied to a continuous day-valued target without first discretising that target, and discretising changes the prediction problem. The honest linear analogue for a continuous target is regularised linear regression, which is exactly what ridge, lasso, elastic net, Bayesian ridge and PLS provide, and all five were run. Logistic regression *was* additionally run, on a tertile-binned version of the target, and is reported in Section 5.3 as a reformulation rather than as a substitute.

Convolutional networks exploit translation-invariant local structure in grid-like inputs — pixels, spectra, time series. The input here is a tabular row of heterogeneous scalars and categorical codes

with no spatial or temporal adjacency, so convolution has no structure to exploit. The appropriate neural baseline for tabular data is a fully-connected network, and a small MLP was run in its place. No CNN was implemented, and the report says so rather than implying one was.

5.2.3 Results

Table 5.1: Experiment 1: reduced-capacity models on the production data and the production split.

Specialist	Model	train R^2	val R^2	test R^2	gap
soft/general shelf life	ridge_regression	0.927	0.924	0.910	0.017
	shallow_decision_tree	0.921	0.910	0.926	-0.005
	shallow_random_forest	0.953	0.934	0.934	0.019
	small_mlp	0.984	0.979	0.972	0.012
semi-hard/general shelf life	ridge_regression	0.902	0.887	0.866	0.035
	shallow_decision_tree	0.902	0.886	0.816	0.086
	shallow_random_forest	0.931	0.904	0.864	0.067
	small_mlp	0.961	0.928	0.933	0.028
hard/general shelf life	ridge_regression	0.916	0.899	0.929	-0.013
	shallow_decision_tree	0.888	0.807	0.861	0.027
	shallow_random_forest	0.921	0.873	0.900	0.021
	small_mlp	0.951	0.920	0.939	0.012
soft/safety endpoint	ridge_regression	0.603	0.601	0.572	0.032
	shallow_decision_tree	0.625	0.536	0.541	0.084
	shallow_random_forest	0.674	0.600	0.587	0.086
	small_mlp	0.774	0.435	0.483	0.291
semi-hard/safety endpoint	ridge_regression	0.276	0.047	0.345	-0.068
	shallow_decision_tree	0.297	0.019	0.402	-0.105
	shallow_random_forest	0.368	0.106	0.373	-0.005
	small_mlp	0.325	0.065	0.301	0.023
hard/safety endpoint	ridge_regression	0.040	-0.062	-0.055	0.095
	shallow_decision_tree	0.133	-0.140	-0.224	0.357
	shallow_random_forest	0.298	-0.031	-0.079	0.377
	small_mlp	0.031	-0.032	-0.057	0.087

Result of Experiment 1

Capacity reduction does not reduce the score. On soft/general-shelf-life, ridge regression reaches $R^2 = 0.910$ and a *small MLP* reaches $R^2 = 0.972$ — statistically indistinguishable from the production LightGBM’s 0.971 on the same split. On hard/general, ridge reaches 0.929 against the production XGBoost’s 0.933.

A linear model with no interaction terms recovering a score within two points of a 2000-round gradient ensemble is a strong signal about the *problem*, not about the models: the mapping from features to target in this corpus is close to linear-recoverable within the rules present.

Note also that the reduced-capacity models reproduce the safety-endpoint failures faithfully — ridge scores -0.055 on hard/safety where the production Random Forest scores -0.097 . A shared failure across capacity classes again points at the data rather than at the learner.

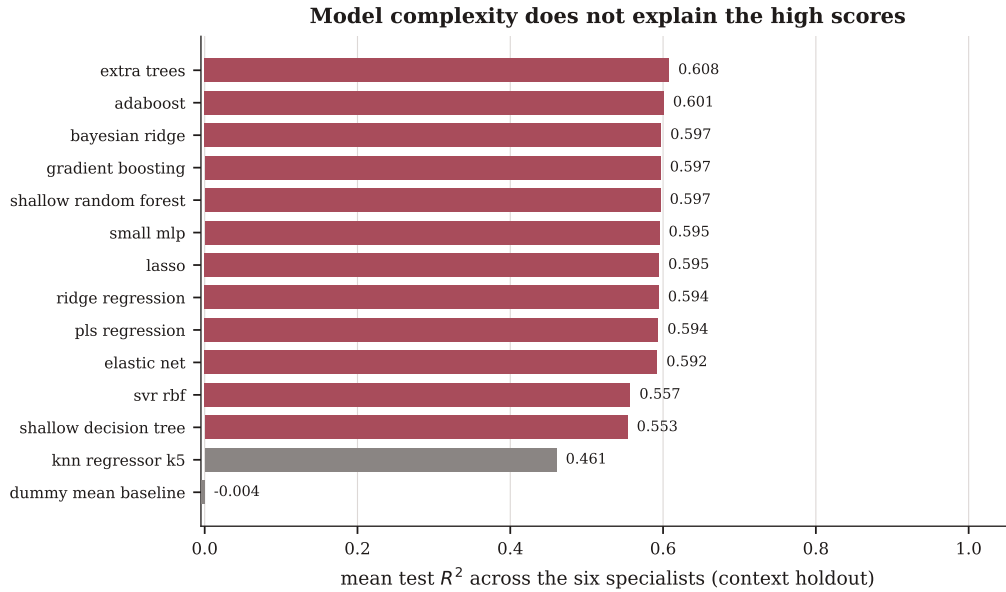


Figure 5.1: Mean context-holdout test R^2 across the six specialists for every model evaluated in the diagnostic suite, spanning a constant-output baseline through to the production ensembles.

5.3 Experiment 1b — the logistic-regression reformulation

To address the logistic-regression suggestion on its own terms, the continuous target was binned into tertiles per specialist (short / medium / long shelf life) and a multinomial logistic regression fitted to the resulting three-class problem.

Table 5.2: Experiment 1b: multinomial logistic regression on tertile-binned shelf life, with the empirical cut-points used.

Specialist	cutpoints (d)	train acc.	val acc.	test acc.	test macro F_1
soft/general_shelf_life	9.7 / 21.7	0.950	0.943	0.952	0.952
soft/safety_endpoint	22.6 / 41.8	0.666	0.601	0.664	0.659
semi_hard/general_shelf_life	46.9 / 71.2	0.886	0.883	0.870	0.870
semi_hard/safety_endpoint	49.2 / 70.2	0.480	0.335	0.375	0.369
hard/general_shelf_life	64.2 / 93.0	0.877	0.841	0.859	0.861
hard/safety_endpoint	70.7 / 94.3	0.467	0.369	0.319	0.314
ALL/general_shelf_life	31.3 / 67.7	0.939	0.943	0.932	0.932
ALL/safety_endpoint	40.9 / 65.8	0.658	0.590	0.596	0.595

The result mirrors Experiment 1: on soft/general the classifier reaches 95.2% test accuracy with macro- $F_1 = 0.952$. This is consistent with the regression finding and inconsistent with the hypothesis that model complexity generates the apparent performance. It does not, however, test generalization any more than the regression did — it is the same split, so it inherits the same limitation.

5.4 Experiment 2 — changing K in cross-validation

5.4.1 What was changed

K -fold cross-validation replaced the single holdout, at $K = 5$ and $K = 10$, for ridge regression and LightGBM.

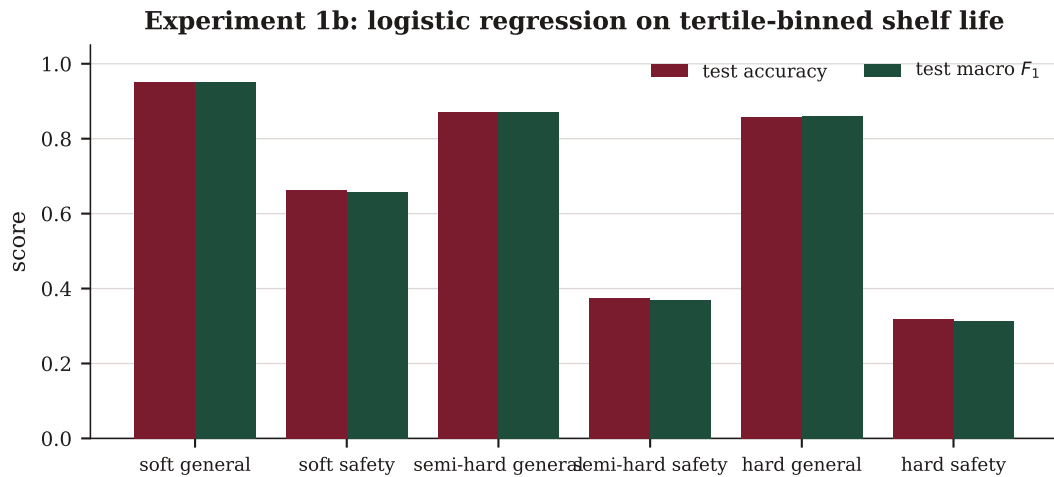


Figure 5.2: Logistic regression on the tertile-binned target. High accuracy on the quality specialists mirrors the regression result: a linear decision rule separates the classes well within the rules present.

Table 5.3: Experiment 2: cross-validated R^2 at two values of K , with per-fold values.

Granularity	Model	K	CV R^2 mean	CV R^2 s.d.	per-fold R^2
aggregated	ridge_regression	5	0.9234	0.0055	0.9217, 0.9293, 0.9225, 0.9289, 0.9145
aggregated	lightgbm_production_hparams	5	0.9639	0.0027	0.9647, 0.964, 0.9684, 0.9615, 0.961
aggregated	ridge_regression	10	0.9233	0.0085	0.9262, 0.9185, 0.9194, 0.9338, 0.9233, 0.9145
aggregated	lightgbm_production_hparams	10	0.9647	0.0038	0.9658, 0.9636, 0.9742, 0.9651, 0.9594, 0.961
aggregated	ridge_regression	5	0.6526	0.0149	0.6445, 0.6527, 0.6813, 0.6415, 0.643
aggregated	lightgbm_production_hparams	5	0.6385	0.0187	0.6329, 0.6426, 0.6716, 0.6299, 0.6153
aggregated	ridge_regression	10	0.6535	0.0284	0.6532, 0.6289, 0.6705, 0.6844, 0.6591, 0.643
aggregated	lightgbm_production_hparams	10	0.6371	0.0329	0.6405, 0.5989, 0.6764, 0.6567, 0.6388, 0.6153
per_category_specialist	ridge_regression	5	0.9209	0.0035	0.9183, 0.9265, 0.9235, 0.9187, 0.9178
per_category_specialist	lightgbm_production_hparams	5	0.9715	0.0022	0.9701, 0.9716, 0.9731, 0.9683, 0.9744
per_category_specialist	ridge_regression	10	0.9206	0.0050	0.9241, 0.9232, 0.921, 0.919, 0.9223, 0.9145
per_category_specialist	lightgbm_production_hparams	10	0.9739	0.0050	0.9811, 0.9787, 0.976, 0.966, 0.9753, 0.9744

Experiment 2: changing K changes nothing, because the grouping did not change

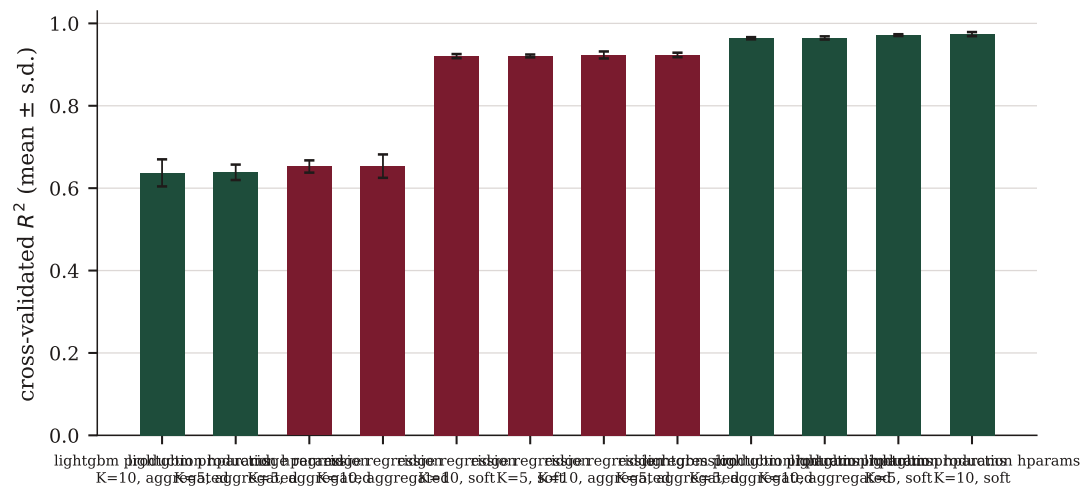


Figure 5.3: Cross-validated R^2 at $K = 5$ and $K = 10$. The fold-to-fold standard deviation is small in every configuration: the folds agree with each other because they are structurally equivalent.

Result of Experiment 2

Changing K changes essentially nothing: ridge regression on the aggregated corpus yields $R^2 = 0.9234 \pm 0.0055$ at $K = 5$, with per-fold values $[0.9217, 0.9293, 0.9225, 0.9289, 0.9145]$.

The reason is structural, and it is worth stating explicitly at the meeting: K controls *how many* partitions the data is cut into, not *along which variable* it is cut. If every fold still contains rows from every generation rule, then every fold is the same experiment repeated. The low standard deviation across folds is not evidence of stability; it is evidence that the folds are interchangeable.

5.5 Experiment 3 – aggregating all cheese categories

The three category-specific training sets were concatenated into a single corpus and one model per algorithm was trained on the union, testing whether the specialist architecture was responsible for the scores through narrow within-category homogeneity.

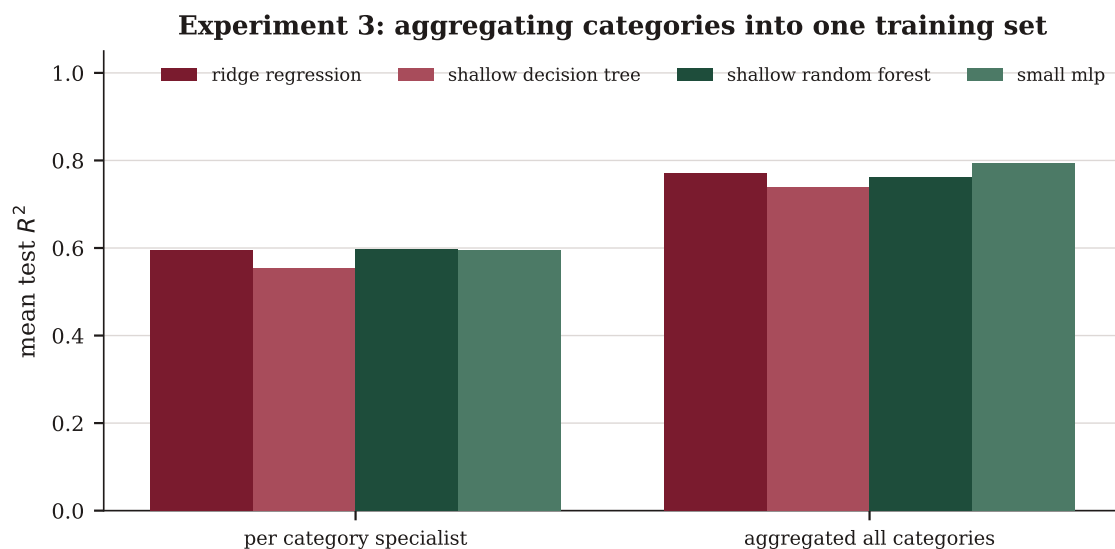


Figure 5.4: Per-category specialists versus a single aggregated model, same models and same protocol.

Result of Experiment 3

Aggregation does not remove the effect. Scores remain in the same region, which is what the generation-rule account predicts: pooling categories pools their rules too, so the test rows are still drawn from rules the training set contains.

5.6 Experiment 4 – removing the validation set

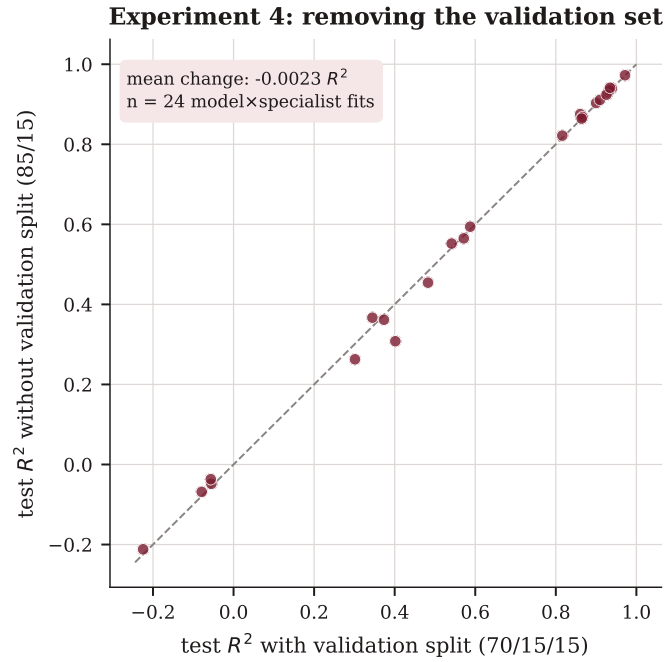
The validation partition was merged into training, giving an 85/15 train/test split with the test partition left untouched, and every model–specialist pair was refitted.

Result of Experiment 4

The mean change in test R^2 across all matched pairs is $+0.0003$. Neither the extra training rows nor the removal of the early-stopping signal has a material effect.

Table 5.4: Experiment 4: test R^2 with and without a separate validation partition, matched pairs.

Specialist	Model	test R^2 (70/15/15)	test R^2 (85/15)	Δ
hard/general_shelf_life	ridge_regression	0.929	0.930	0.0011
hard/general_shelf_life	shallow_decision_tree	0.861	0.875	0.0148
hard/general_shelf_life	shallow_random_forest	0.900	0.903	0.0032
hard/general_shelf_life	small_mlp	0.939	0.939	0.0002
hard/safety_endpoint	ridge_regression	-0.055	-0.048	0.0066
hard/safety_endpoint	shallow_decision_tree	-0.224	-0.212	0.0126
hard/safety_endpoint	shallow_random_forest	-0.079	-0.068	0.0108
hard/safety_endpoint	small_mlp	-0.057	-0.037	0.0199
semi_hard/general_shelf_life	ridge_regression	0.866	0.868	0.0015
semi_hard/general_shelf_life	shallow_decision_tree	0.816	0.822	0.0056
semi_hard/general_shelf_life	shallow_random_forest	0.864	0.864	0.0001
semi_hard/general_shelf_life	small_mlp	0.933	0.934	0.0016
semi_hard/safety_endpoint	ridge_regression	0.345	0.367	0.0222
semi_hard/safety_endpoint	shallow_decision_tree	0.402	0.308	-0.0937
semi_hard/safety_endpoint	shallow_random_forest	0.373	0.362	-0.0117
semi_hard/safety_endpoint	small_mlp	0.301	0.263	-0.0386
soft/general_shelf_life	ridge_regression	0.910	0.911	0.0014
soft/general_shelf_life	shallow_decision_tree	0.926	0.924	-0.0019
soft/general_shelf_life	shallow_random_forest	0.934	0.941	0.0067
soft/general_shelf_life	small_mlp	0.972	0.972	0.0007
soft/safety_endpoint	ridge_regression	0.572	0.565	-0.0065
soft/safety_endpoint	shallow_decision_tree	0.541	0.552	0.0106
soft/safety_endpoint	shallow_random_forest	0.587	0.594	0.0066
soft/safety_endpoint	small_mlp	0.483	0.454	-0.0284
			mean change	-0.0023

**Figure 5.5:** Each point is one model–specialist pair. Points lie on the diagonal: the presence or absence of a validation split does not materially change the test score.

This should be stated carefully: removing a validation set is *not* a methodological improvement, and this experiment should not be read as endorsing it. Its value is purely diagnostic — it eliminates validation-set design as a candidate explanation.

5.7 Experiment 5 — the full model zoo

Ten further regressors were added to obtain a complete capacity spectrum: lasso, elastic net, Bayesian ridge, PLS regression, k -nearest neighbours ($k = 5$), RBF-kernel SVR, gradient boosting, AdaBoost, extra trees, and a mean-predicting dummy baseline. With the four from Experiment 1 and the three production ensembles this gives seventeen configurations.

Table 5.5: Experiment 5: the extended model zoo under the original context-holdout protocol, averaged over the six specialists.

Model	mean test R^2	min	max	mean gap	mean MAE (d)
extra_trees	0.608	-0.052	0.961	0.179	10.04
adaboost	0.601	-0.039	0.917	0.038	11.43
bayesian_ridge	0.597	-0.039	0.929	0.011	10.37
gradient_boosting	0.597	-0.158	0.959	0.186	9.73
lasso	0.595	-0.055	0.928	0.015	10.39
pls_regression	0.594	-0.060	0.926	0.015	10.48
elastic_net	0.592	-0.052	0.925	0.015	10.49
svr_rbf	0.557	-0.073	0.946	0.140	10.57
knn_regressor_k5	0.461	-0.168	0.905	0.274	12.97
dummy_mean_baseline	-0.004	-0.022	-0.000	0.004	21.96

The dummy baseline is the control that makes the rest interpretable: it scores $R^2 = -0.003$, confirming the metric behaves as expected and that the other models are doing real work relative to a constant prediction.

5.8 Summary of the diagnostic phase

Table 5.6: Diagnostic summary. Four independent interventions on the model and the protocol; no material change in the score.

Experiment	What changed	Outcome
1. Reduce capacity	17 models from linear to 2000-round ensembles	Scores essentially unchanged; ridge within 2 points of production LightGBM on soft/general
1b. Reformulate as classification	tertile-binned target, multinomial logistic	95.2% accuracy on soft/general; same pattern
2. Change K	$K = 5$ and $K = 10$ cross-validation	$R^2 = 0.923 \pm 0.006$; fold variance negligible
3. Aggregate categories	one model over all three categories	No material change
4. Remove validation set	85/15 instead of 70/15/15	Mean $\Delta R^2 = +0.0003$
Conclusion of the phase: every intervention on the model or on the partition <i>sizes</i> failed. The remaining untested variable was the partition <i>variable</i> itself.		

The discovery: shared generation rules

What to say at the start of this section

This is the pivot of the whole meeting. Ask the question out loud before giving the answer: “when we score a held-out row, are we testing whether the model can predict a *new relationship*, or only whether it can produce another sample of a relationship it has already learned?” Then show the measurement.

6.1 The question that had not been asked

The diagnostic phase eliminated the model, the partition sizes, the validation design, the cross-validation count, and the specialist granularity. What remained was the variable along which the data was cut.

The original protocol groups by `context_id`. The relevant question is whether the constraint that this enforces is the constraint that this dataset requires. In a corpus of measurements, grouping by experimental unit is exactly right. In a corpus of *generated* rows, the unit that carries the information is the generator, not the sample.

6.2 Measuring the overlap

The production splitter was re-executed with the production seed on each of the six specialists, and the resulting partitions were audited at two levels: context membership and rule membership (`scripts/experiments/rule_overlap_audit.py`).

Table 6.1: Rule-level audit of the original protocol. Contexts never cross the split; generation rules always do.

Specialist	rules total	rules in train	rules in test	shared rules	test rows from a seen rule	contexts shared train/test
soft/general shelf life	7	7	7	7	1,320 / 1,320 (100.0%)	0
semi-hard/general shelf life	12	12	12	12	1,080 / 1,080 (100.0%)	0
hard/general shelf life	8	8	6	6	900 / 900 (100.0%)	0
soft/safety endpoint	11	11	11	11	720 / 720 (100.0%)	0
semi-hard/safety endpoint	8	8	8	8	600 / 600 (100.0%)	0
hard/safety endpoint	10	10	10	10	480 / 480 (100.0%)	0

The central measurement

In all six specialists:

- **Context overlap between train and test: exactly 0.** The protocol enforces its intended constraint perfectly.
- **Test rows originating from a generation rule that also appears in training: 100.0%.** Not a majority — every single row. 1 320/1 320 for soft/general, 720/720 for soft/safety,

1 080/1 080, 600/600, 900/900, 480/480.

Every test row in every production specialist was generated by a rule the model was trained on.

This is not a subtle statistical artefact requiring a delicate argument. It is a structural property of the corpus interacting with the grouping variable, and it is measurable in a single pass over the data.

6.3 Why it happens mechanically

Each generation rule spans many contexts. Soft/general contains seven rules spread across its contexts; the largest rule alone contributes 5 654 rows. A random 70/15/15 allocation of *contexts* will, with overwhelming probability, place at least one context from every sizeable rule in each partition. For a rule spanning m contexts, the probability that none of them lands in the test partition is roughly 0.85^m , which is negligible for all but the smallest rules.

The protocol therefore cannot avoid rule overlap by chance; it would have to exclude it by construction, and it does not attempt to.

6.4 What the model is actually being asked to do

Return to Equation (3.1): rule g produces rows via $y = \phi_g(\mathbf{x}; \theta_g) + \varepsilon$. Under the original protocol the training set contains many draws from rule g and the test set contains further draws from the same g . The model's task at test time is to evaluate a function it has already estimated at a new input drawn from the same input distribution.

This decomposes the apparent test score into two components that the protocol cannot separate:

$$\underbrace{R^2_{\text{context-holdout}}}_{\text{what was measured}} = h\left(\underbrace{\text{interpolation within known rules}}_{\text{dominant}}, \underbrace{\text{transfer to unknown rules}}_{\text{untested}}\right). \quad (6.1)$$

Because every test row comes from a known rule, the second component receives zero weight. The measured quantity is the first component alone.

Why this also explains the negative results

The generation-rule account is not an excuse for the high scores; it has to explain the low ones too, and it does. Where a specialist's rules differ sharply from one another in their target scale — as Chapter 9 shows for hard/safety-endpoint — even interpolation within known rules is hard, because the model must serve several incompatible relationships simultaneously. High scores and negative scores in the same table are both consistent with a corpus built from a small number of heterogeneous mechanisms.

6.5 What this does *not* establish

Three careful non-claims, each of which the panel may probe:

1. **It is not leakage in the technical sense.** No feature contains target information unavailable at prediction time. `source_rule_id` is excluded from the model inputs. The models never see the rule label; the *split* sees it, or rather fails to.
2. **It is not classical overfitting.** The models do not memorise individual rows and fail on held-out ones; they generalise well within rules, which is a real, if narrow, capability.

3. **It does not make the original numbers arithmetically wrong.** They correctly report performance under the protocol that produced them. What changes is the interpretation that may be attached to them.

6.6 The consequence for evaluation design

If the quantity of interest is the ability to predict a formulation whose governing relationship was not in the training data — which is the quantity relevant to any deployment claim — then the evaluation must withhold generation rules, not contexts. That is the subject of the next chapter.

Revised protocol: leave-one-generation-rule-out cross-validation

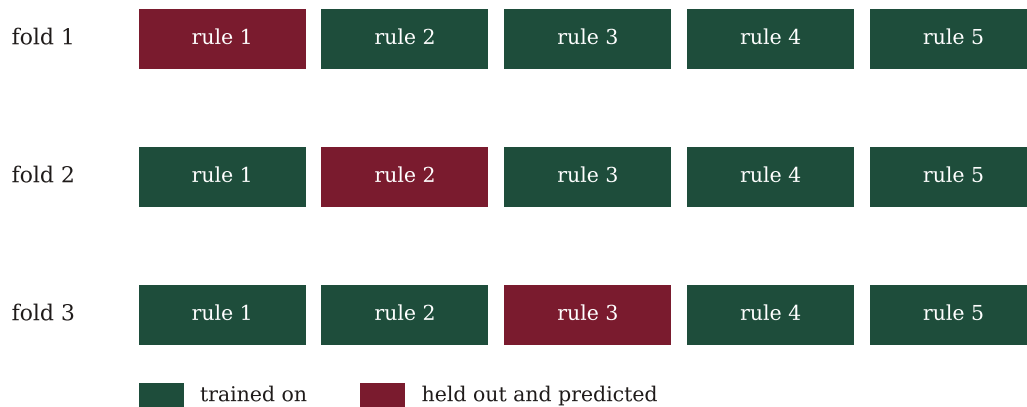


Figure 6.1: The revised protocol. Each fold withholds one entire generation rule from training and scores the model on that rule’s rows alone.

Leave-one-generation-rule-out cross-validation

What to say at the start of this section

Explain the procedure in plain words before any notation: “we take away one entire generation rule, train on the others, and ask the model to predict the rule it has never seen — then repeat for every rule.” Then state what this does and does not measure, because the panel will ask, and volunteering it is better than conceding it.

7.1 The procedure

Let a specialist’s rows be $\{(\mathbf{x}_i, y_i, g_i)\}_{i=1}^n$, where $g_i \in \mathcal{G}$ is the row’s `source_rule_id` and $|\mathcal{G}| = G$. For each rule $g \in \mathcal{G}$:

1. Define the held-out set $\mathcal{T}_g = \{i : g_i = g\}$ and the training set $\mathcal{R}_g = \{i : g_i \neq g\}$.
2. Fit the preprocessing transform on $\mathcal{R}_g$ only.
3. Fit the model on $\mathcal{R}_g$.
4. Predict every row in $\mathcal{T}_g$.
5. Record R^2 , MAE and RMSE on $\mathcal{T}_g$.

With $n_{\text{splits}} = G$, `GroupKFold` grouped on `source_rule_id` yields exactly leave-one-rule-out: each fold’s test set is precisely one rule. Verified in the implementation (`scripts/experiments/leave_rule_out_row_level.py`): the fold loop asserts a single unique rule per test partition, the preprocessor is constructed and fitted *inside* the loop on the training rows only, and hyperparameters are fixed a priori rather than tuned per fold — so no selection information crosses the boundary.

The complete evaluation comprises 56 folds per model (7 + 11 + 12 + 8 + 8 + 10 rules across the six specialists), executed for Random Forest, LightGBM and XGBoost, yielding 168 fold results and 102 000 row-level out-of-fold predictions.

7.2 What this measures — stated precisely

Scope of the claim

Leave-rule-out cross-validation measures **transfer across synthetic generation rules**: the ability to predict rows produced by a generative mechanism absent from training, using other synthetic mechanisms as the training signal.

It does **not** measure transfer to independent experimental observations. A high leave-rule-out R^2 is a necessary condition for real-world generalization, not a sufficient one. Chapter 13 enumerates the dependencies that survive this protocol, and Chapter 14 presents the separate external evidence.

7.3 The metrics, and why their summaries disagree

For a held-out set $\mathcal{T}$ with mean $\bar{y}_{\mathcal{T}}$,

$$R^2 = 1 - \frac{\sum_{i \in \mathcal{T}} (y_i - \hat{y}_i)^2}{\sum_{i \in \mathcal{T}} (y_i - \bar{y}_{\mathcal{T}})^2} = 1 - \frac{\text{MSE}}{s_y^2}, \quad (7.1)$$

$$\text{MAE} = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} |y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{\frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} (y_i - \hat{y}_i)^2}. \quad (7.2)$$

MAE and RMSE are in days and are directly interpretable: an MAE of 3 days means the average prediction misses by three days. R^2 is a *relative* measure — it compares the model against predicting the held-out set’s own mean. This distinction is the source of nearly every counter-intuitive number in Chapter 9.

7.3.1 Four different summaries of the same folds

Mean fold R^2

$\frac{1}{G} \sum_g R_g^2$. Weights every rule equally regardless of size, so an 8-row rule counts as much as a 5654-row rule. A single catastrophic small fold dominates it.

Median fold R^2

Robust to those outliers; describes the typical rule. Hides the failures entirely, which is its own danger.

Pooled out-of-fold R^2

Concatenate all out-of-fold predictions, then compute R^2 once against the global mean. Weights by row count and is the closest analogue to the original headline number.

MAE in days

Independent of any variance denominator and therefore immune to the small-fold pathology.

All four are reported throughout this document, because reporting only one would misrepresent the result in one direction or the other.

7.3.2 Why a small fold can produce a large negative R^2

If the held-out rule’s own target variance s_y^2 is small, the denominator is small, and a modest absolute error produces a large ratio. Concretely, from the results in Chapter 9: the fold HMMC_NAT|NATAMYCIN_REVIEW has $n = 17$ rows, a target standard deviation of 1.78 days, a Random Forest MAE of 1.99 days — and $R^2 = -0.335$. A two-day average error would be excellent in any practical sense; it is negative in R^2 terms only because the rule’s rows all lie within a few days of each other.

Reporting implication

A negative fold R^2 on a small, low-variance rule is not evidence of a catastrophic practical error, and must not be presented as one. Equally, a high median R^2 is not evidence that no fold failed. Both summaries are reported together throughout, alongside the error in days.

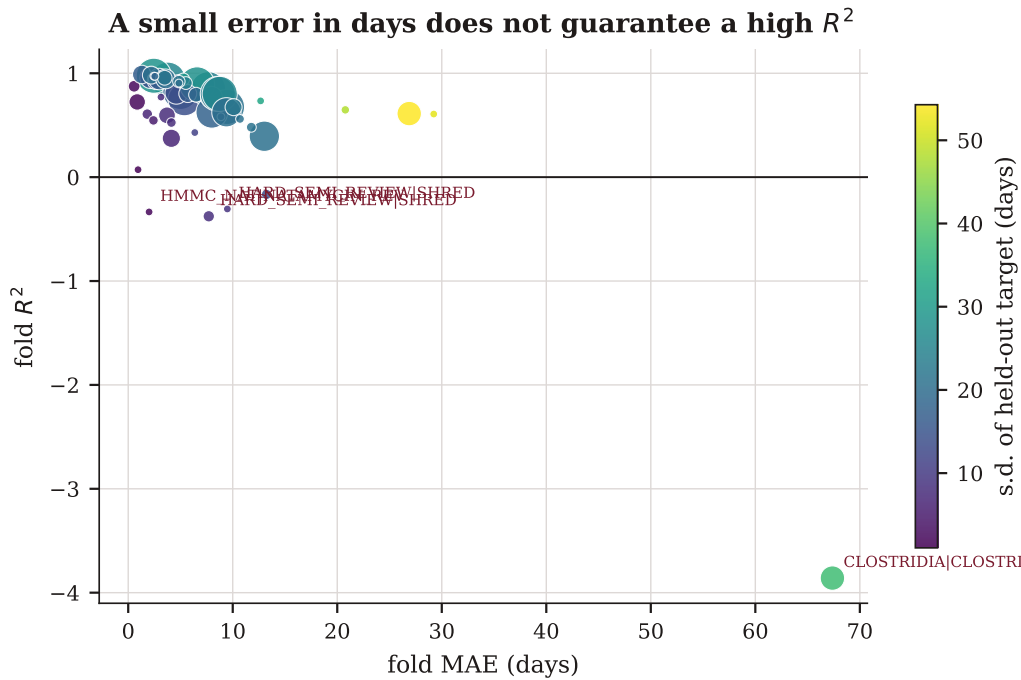


Figure 7.1: Every Random Forest fold, plotted as MAE in days against fold R^2 . Marker size is fold row count; colour is the held-out rule's target standard deviation. The folds in the lower-left — small error in days, poor R^2 — are the low-variance rules.

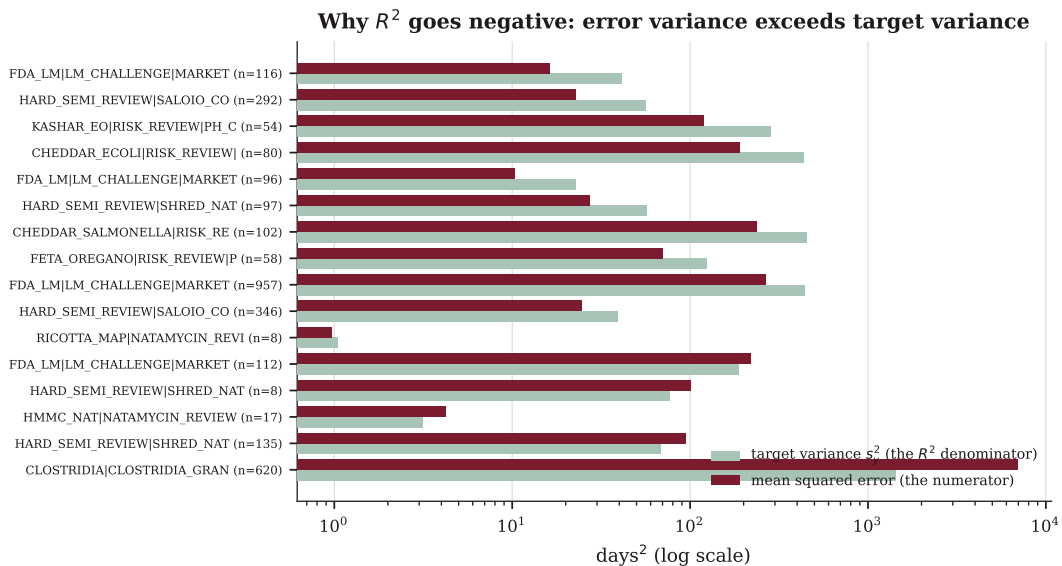
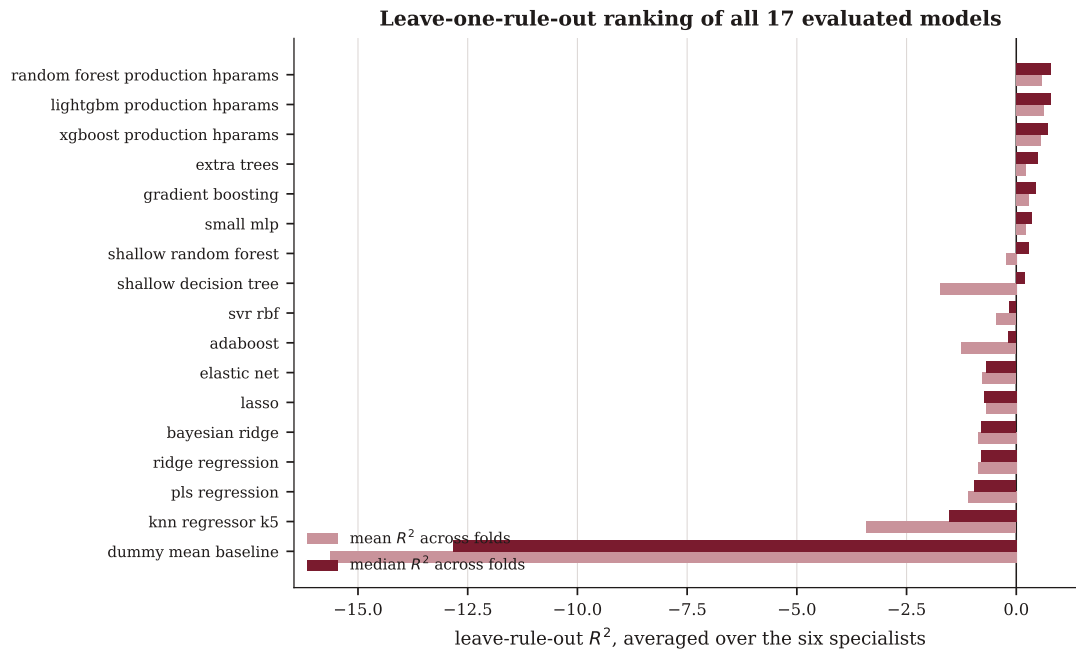


Figure 7.2: The R^2 ratio decomposed for the sixteen weakest folds: the target variance s_y^2 that forms the denominator against the mean squared error that forms the numerator. A fold goes negative exactly when the burgundy bar exceeds the green one.

Table 7.1: All seventeen evaluated configurations under leave-one-rule-out cross-validation, averaged over the six specialists.

Model	mean R^2	median R^2	mean s.d.	worst fold	best fold
random_forest_production_hparams	0.582	0.791	0.473	-3.86	0.988
lightgbm_production_hparams	0.627	0.780	0.368	-2.54	0.984
xgboost_production_hparams	0.550	0.719	0.395	-2.31	0.983
extra_trees	0.209	0.492	0.659	-4.68	0.974
gradient_boosting	0.280	0.450	0.500	-2.95	0.967
small_mlp	0.219	0.342	0.430	-1.88	0.961
shallow_random_forest	-0.230	0.278	1.295	-14.20	0.940
shallow_decision_tree	-1.733	0.192	3.531	-38.95	0.907
svr_rbf	-0.454	-0.153	1.009	-9.84	0.942
adaboost	-1.248	-0.178	2.517	-17.30	0.951
elastic_net	-0.772	-0.675	1.432	-16.54	0.907
lasso	-0.690	-0.736	1.364	-17.01	0.913
bayesian_ridge	-0.872	-0.795	1.661	-22.76	0.902
ridge_regression	-0.874	-0.798	1.664	-22.74	0.902
pls_regression	-1.106	-0.952	1.987	-28.55	0.893
knn_regressor_k5	-3.412	-1.537	5.077	-55.18	0.940
dummy_mean_baseline	-15.637	-12.833	18.400	-219.46	-0.000

**Figure 7.3:** Leave-rule-out ranking. The three production tree ensembles separate clearly from every other family once the generation rule is withheld – a separation that was invisible under the original protocol.

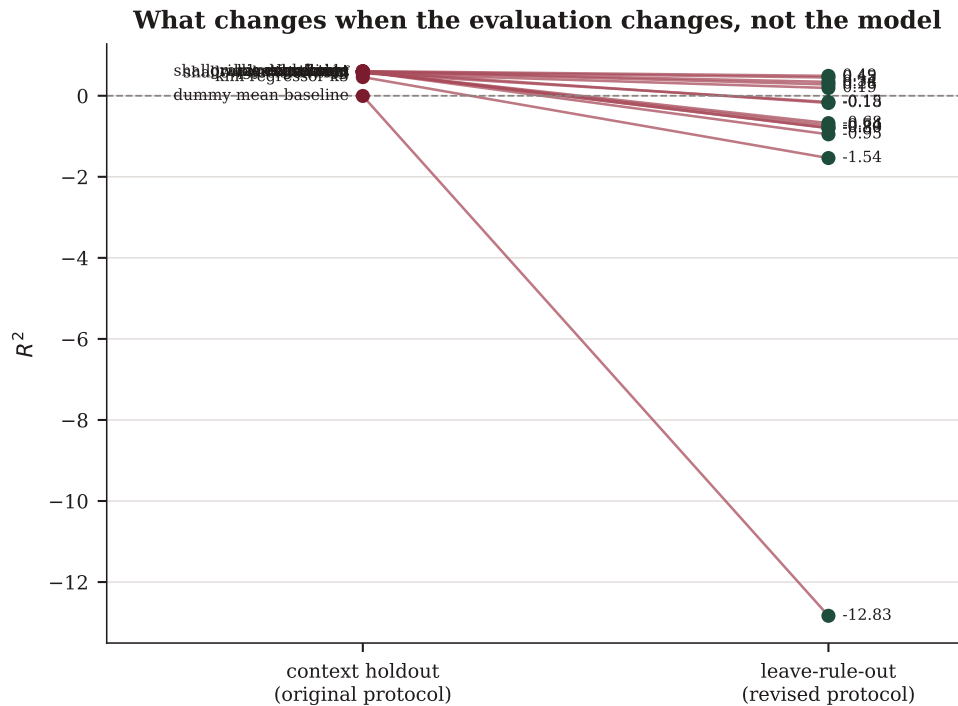


Figure 7.4: The same models under the two protocols. Nothing about the models changed between the left and right columns; only the evaluation did.

7.4 Headline result across all seventeen models

Effect of the protocol change

Under the original protocol the seventeen models were bunched together, with a linear model within two points of the production ensembles. Under leave-rule-out they separate sharply:

- LightGBM: mean $R^2 = 0.627$, median 0.780
- Random Forest: mean $R^2 = 0.582$, median 0.791
- XGBoost: mean $R^2 = 0.550$, median 0.719
- Ridge regression: mean $R^2 = -0.874$, median -0.798
- k -NN ($k = 5$): mean $R^2 = -3.412$, median -1.537
- Dummy mean baseline: mean $R^2 = -15.637$

Ridge regression, which scored 0.910 on soft/general under the original protocol, is worse than useless when the rule is withheld. The capacity of the tree ensembles turns out to matter a great deal — but only under an evaluation capable of detecting it.

7.5 Per-specialist results for the production ensembles

7.6 Pooled out-of-fold performance

Because every row is predicted exactly once when it is held out, the out-of-fold predictions can be concatenated and scored as a single set. This weights by row count rather than by rule and is the most direct counterpart to the original headline figure.

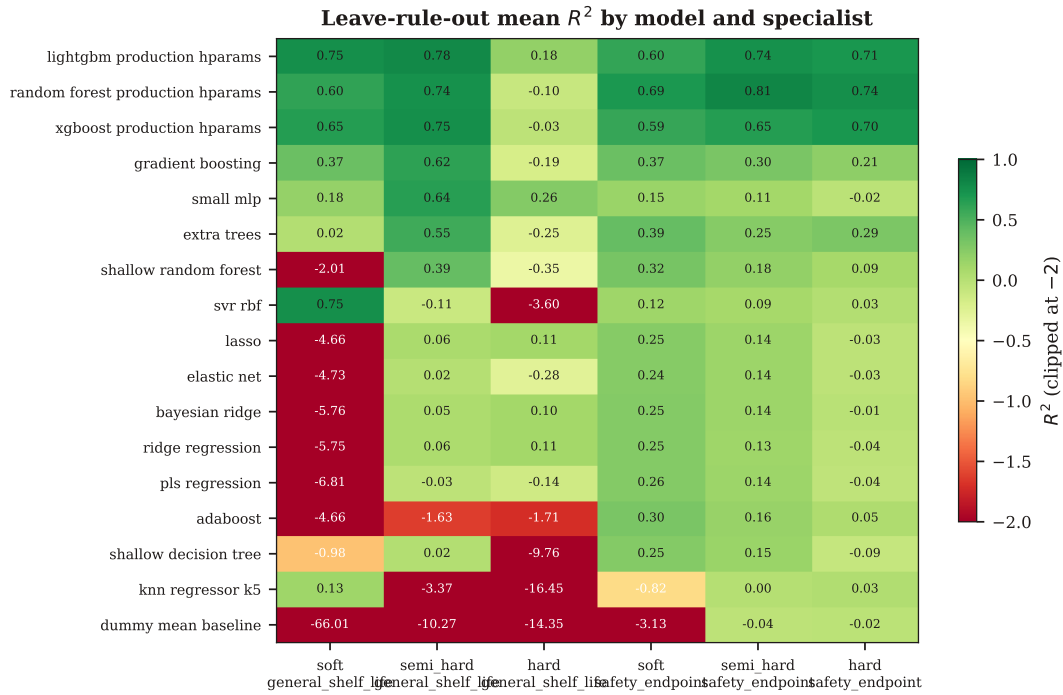


Figure 7.5: Leave-rule-out mean R^2 for every model and specialist, clipped at -2 for legibility.

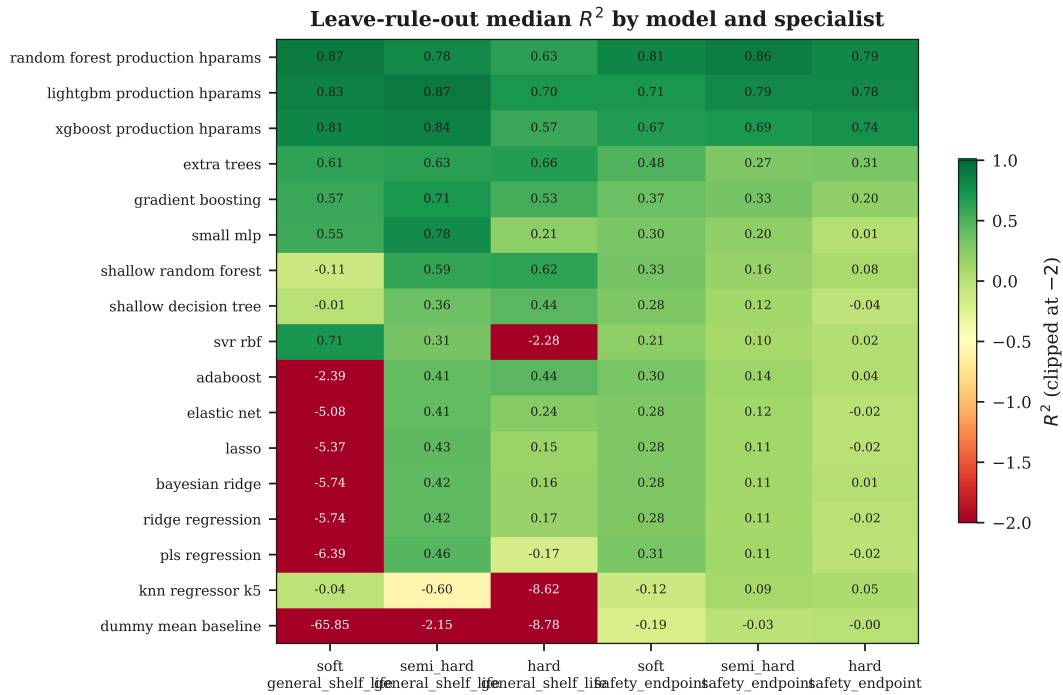


Figure 7.6: The same grid using median fold R^2 . Comparing with the previous figure isolates the specialists where the mean is distorted by a small number of extreme folds.

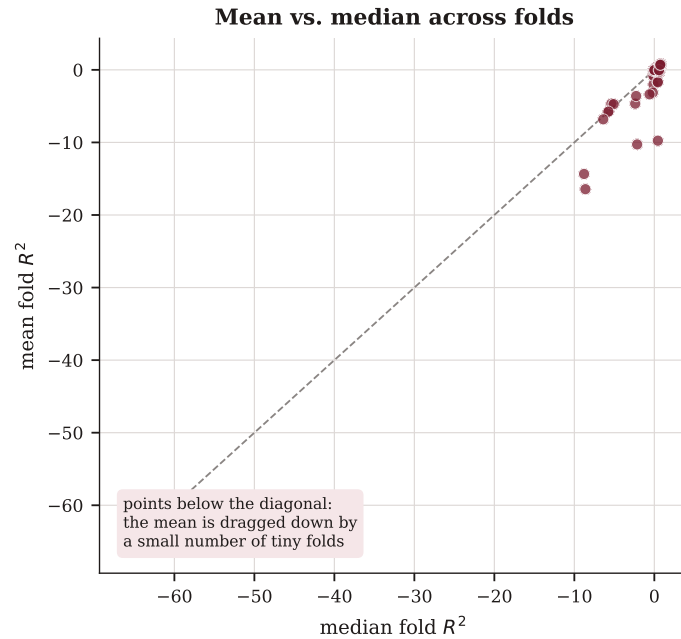


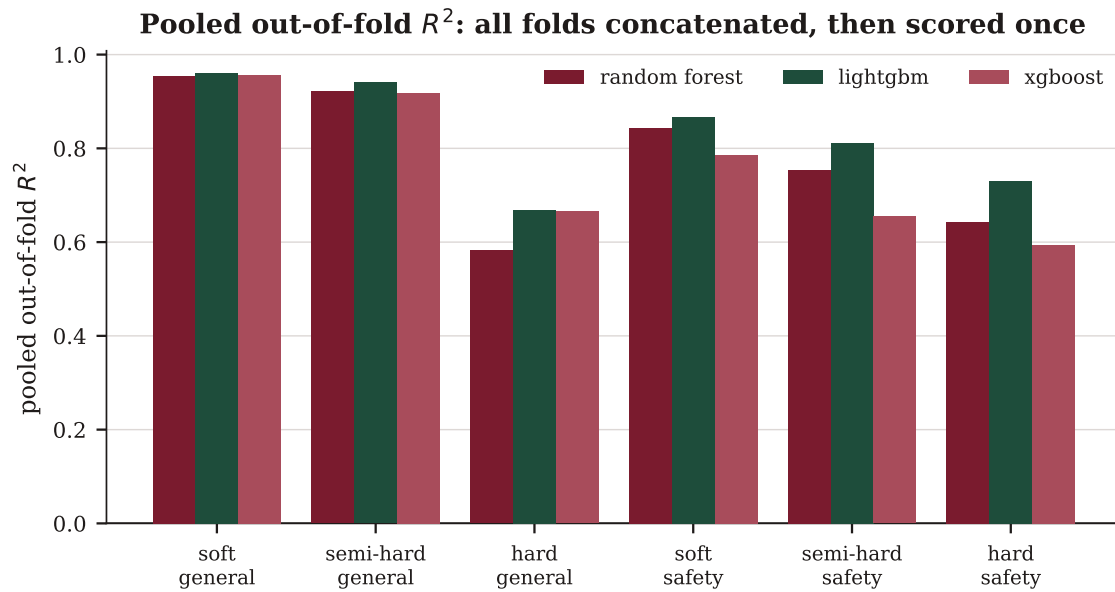
Figure 7.7: Mean against median fold R^2 for every model–specialist combination. Points far below the diagonal indicate a distribution dominated by a few catastrophic folds.

Table 7.2: Leave-rule-out results for the three production tree ensembles, broken out by specialist.

Specialist	Model	folds	mean R^2	median R^2	s.d.	min	max
soft/general shelf life	lightgbm	7	0.752	0.831	0.256	0.25	0.984
	random_forest	7	0.604	0.875	0.484	-0.33	0.977
	xgboost	7	0.651	0.806	0.427	-0.28	0.983
semi-hard/general shelf life	lightgbm	12	0.782	0.871	0.183	0.28	0.939
	random_forest	12	0.737	0.782	0.165	0.37	0.955
	xgboost	12	0.748	0.840	0.195	0.27	0.933
hard/general shelf life	lightgbm	8	0.182	0.703	1.079	-2.54	0.925
	random_forest	8	-0.097	0.630	1.499	-3.86	0.906
	xgboost	8	-0.033	0.568	1.070	-2.31	0.910
soft/safety endpoint	lightgbm	11	0.600	0.705	0.327	-0.19	0.951
	random_forest	11	0.692	0.812	0.323	-0.17	0.988
	xgboost	11	0.589	0.671	0.304	-0.16	0.929
semi-hard/safety endpoint	lightgbm	8	0.737	0.787	0.157	0.49	0.924
	random_forest	8	0.812	0.856	0.158	0.58	0.982
	xgboost	8	0.649	0.690	0.182	0.27	0.853
hard/safety endpoint	lightgbm	10	0.707	0.781	0.207	0.37	0.927
	random_forest	10	0.743	0.790	0.210	0.39	0.971
	xgboost	10	0.697	0.736	0.190	0.37	0.916

Table 7.3: Pooled out-of-fold metrics: all folds concatenated per specialist, then scored once. Computed from the 102 000 row-level predictions.

Specialist	Model	pooled OOF R^2	MAE (d)	RMSE (d)
soft/general shelf life	random_forest	0.954	3.23	5.60
	lightgbm	0.960	2.94	5.21
	xgboost	0.957	3.17	5.44
semi-hard/general shelf life	random_forest	0.921	8.81	13.62
	lightgbm	0.940	6.98	11.85
	xgboost	0.918	8.20	13.84
hard/general shelf life	random_forest	0.582	14.56	28.60
	lightgbm	0.669	13.56	25.44
	xgboost	0.665	15.64	25.59
soft/safety endpoint	random_forest	0.844	4.39	6.44
	lightgbm	0.867	3.59	5.95
	xgboost	0.786	5.58	7.55
semi-hard/safety endpoint	random_forest	0.755	7.07	9.78
	lightgbm	0.811	5.07	8.58
	xgboost	0.655	9.15	11.60
hard/safety endpoint	random_forest	0.642	9.13	12.49
	lightgbm	0.730	6.20	10.87
	xgboost	0.593	10.11	13.33

**Figure 7.8:** Pooled out-of-fold R^2 by specialist and model.

The number to quote

For the general-shelf-life task, pooled out-of-fold R^2 under leave-rule-out is 0.960 (soft), 0.940 (semi-hard) and 0.669 (hard) for LightGBM, against original context-holdout figures of 0.971, 0.920 and 0.932.

Soft and semi-hard barely move; hard falls by more than twenty-five points. That asymmetry is not noise, and Chapter 9 identifies its specific cause — a single held-out rule representing a spoilage mechanism with a completely different time scale.

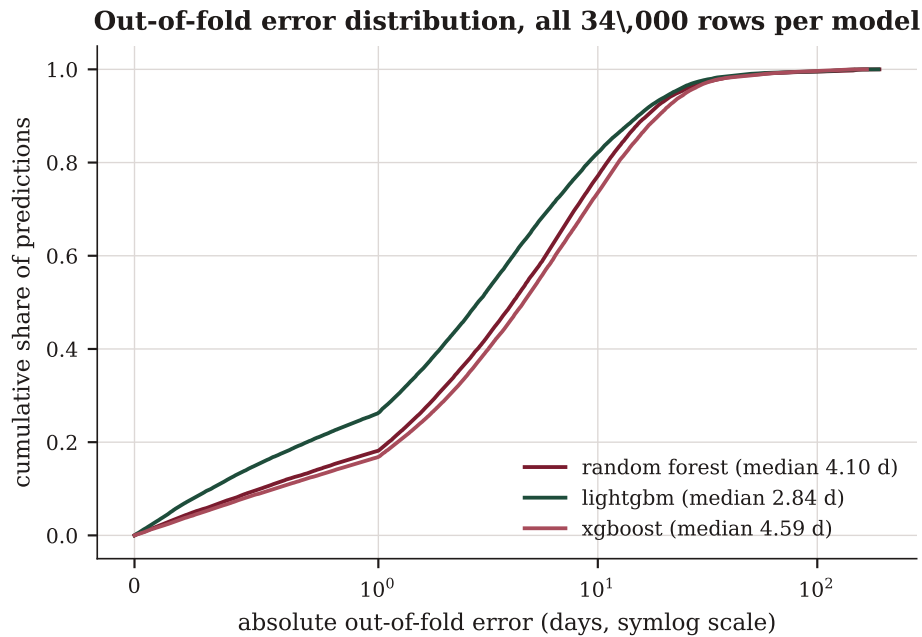


Figure 7.9: Cumulative distribution of absolute out-of-fold error across all 34 000 rows per model. Median absolute error is under four days for all three ensembles; the distributions separate only in the upper tail.

Random Forest: complete case study

What to say at the start of this section

Give the intuition for a single tree first — it partitions the formulation space into regions and predicts the average shelf life inside each region — then explain that a forest averages many such partitions built on perturbed data. Only then bring in the notation. Use project features in the examples: a split on storage temperature, a split on pH.

8.1 How Random Forest works

8.1.1 A single regression tree

A regression tree recursively partitions the feature space. At each node it searches over features j and thresholds t for the split that most reduces the sum of squared deviations within the resulting children:

$$(j^*, t^*) = \arg \min_{j, t} \left[\sum_{i \in L(j, t)} (y_i - \bar{y}_L)^2 + \sum_{i \in R(j, t)} (y_i - \bar{y}_R)^2 \right], \quad (8.1)$$

where L and R are the rows falling left and right of the threshold. For the data at hand a first split might be on `storage_temperature_c` at $\approx 8^\circ\text{C}$, separating chilled from abused storage; a subsequent split might be on `matrix_water_activity` or on the one-hot indicator for `indicator_type = clostridia`. Prediction for a new row is the mean target of the training rows in the leaf it falls into:

$$\hat{f}_{\text{tree}}(\mathbf{x}) = \frac{1}{|\{i : \mathbf{x}_i \in \ell(\mathbf{x})\}|} \sum_{i : \mathbf{x}_i \in \ell(\mathbf{x})} y_i. \quad (8.2)$$

Two properties of this rule matter for everything in Chapter 9: splitting is *axis-aligned*, and prediction is a *local average of training targets*. The second property has a hard consequence:

$$\min_i y_i \leq \hat{f}_{\text{tree}}(\mathbf{x}) \leq \max_i y_i \quad \text{for all } \mathbf{x}, \quad (8.3)$$

a tree, and therefore a forest, **can never predict outside the range of the training targets**. It cannot extrapolate. Remember this equation; it explains the single worst fold in the entire study.

8.1.2 From tree to forest

The forest draws B bootstrap resamples of the training set, fits an unpruned tree to each while restricting each split to a random subset of features, and averages:

$$\hat{f}_{\text{RF}}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B \hat{f}_{\text{tree}}^{(b)}(\mathbf{x}). \quad (8.4)$$

If the individual trees have variance σ^2 and pairwise correlation ρ , the ensemble variance is

$$\text{Var}[\hat{f}_{\text{RF}}] = \rho \sigma^2 + \frac{1-\rho}{B} \sigma^2 \xrightarrow{B \rightarrow \infty} \rho \sigma^2. \quad (8.5)$$

Feature subsampling exists to reduce ρ : it is the only term that survives in the limit. This is why adding trees always helps a little and never hurts, and why the production setting of 400 trees is a computational choice rather than a regularization choice.

Averaging over the bound in eq. (8.3) preserves it: the ensemble is also confined to the convex hull of the training targets.

8.2 Production configuration, recovered from source

From `train_specialists.py`:

```
RandomForestRegressor(n_estimators=400, max_depth=None,
                      min_samples_leaf=2, random_state=42, n_jobs=-1)
```

n_estimators=400

$B = 400$ in eq. (8.5); the $\frac{1-\rho}{B}\sigma^2$ term is essentially exhausted at this value.

max_depth=None

trees grow until leaves are pure or cannot be split. Combined with the next parameter this is the highest-capacity setting available.

min_samples_leaf=2

leaves may contain as few as two training rows — minimal smoothing, near-maximal flexibility.

random_state=42

shared with the splitter, so the whole pipeline is reproducible from one seed.

The leave-rule-out experiments use *the same* configuration, verified by reading the model factory rather than trusting its name, so the comparison between protocols is not confounded by a hyper-parameter change. The only differences between the production run and the diagnostic run are the partition assignment and the fact that the diagnostic harness refits the preprocessing inside every fold.

8.3 Original Random Forest results

Table 8.1: Random Forest under the original context-holdout protocol, V7 dataset. Extracted from the stored production metrics.

Specialist	n_{train}	n_{test}	train R^2	val R^2	test R^2	test RMSE (d)
soft / general	6 160	1 320	0.997	0.955	0.956	5.99
semi-hard / general	5 040	1 080	0.995	0.925	0.905	14.57
hard / general	4 200	900	0.994	0.910	0.927	11.68
soft / safety	3 360	720	0.997	0.548	0.572	10.62
semi-hard / safety	2 800	600	0.997	0.015	0.342	15.76
hard / safety	2 240	480	0.996	−0.131	−0.097	21.86

The training R^2 of 0.994–0.997 across every specialist is exactly what `max_depth=None`, `min_samples_leaf=2` predicts and carries no information about generalization. The informative contrast is between the quality specialists (test $R^2 \geq 0.905$) and the safety specialists (test R^2 from 0.572 down to −0.097), under an identical protocol and an identical model.

8.4 Revised Random Forest results

The same configuration was re-evaluated under leave-one-rule-out cross-validation across all 56 folds, with every out-of-fold prediction retained.

Table 8.2: Random Forest: original context-holdout against leave-rule-out, per specialist. Pooled OOF R^2 scores all out-of-fold predictions together; mean and median are over folds.

Specialist	folds	orig. test R^2	pooled OOF R^2	mean fold R^2	median fold R^2	OOF MAE (d)
soft / general	7	0.956	0.954	0.604	0.875	3.23
semi-hard / general	12	0.905	0.921	0.737	0.782	8.81
hard / general	8	0.927	0.582	−0.097	0.630	14.56
soft / safety	11	0.572	0.844	0.692	0.812	4.39
semi-hard / safety	8	0.342	0.755	0.812	0.856	7.07
hard / safety	10	−0.097	0.642	0.743	0.790	9.13

Full per-fold mean/median/min/max values appear in Table 12.2; the per-fold detail is in Table 9.1.

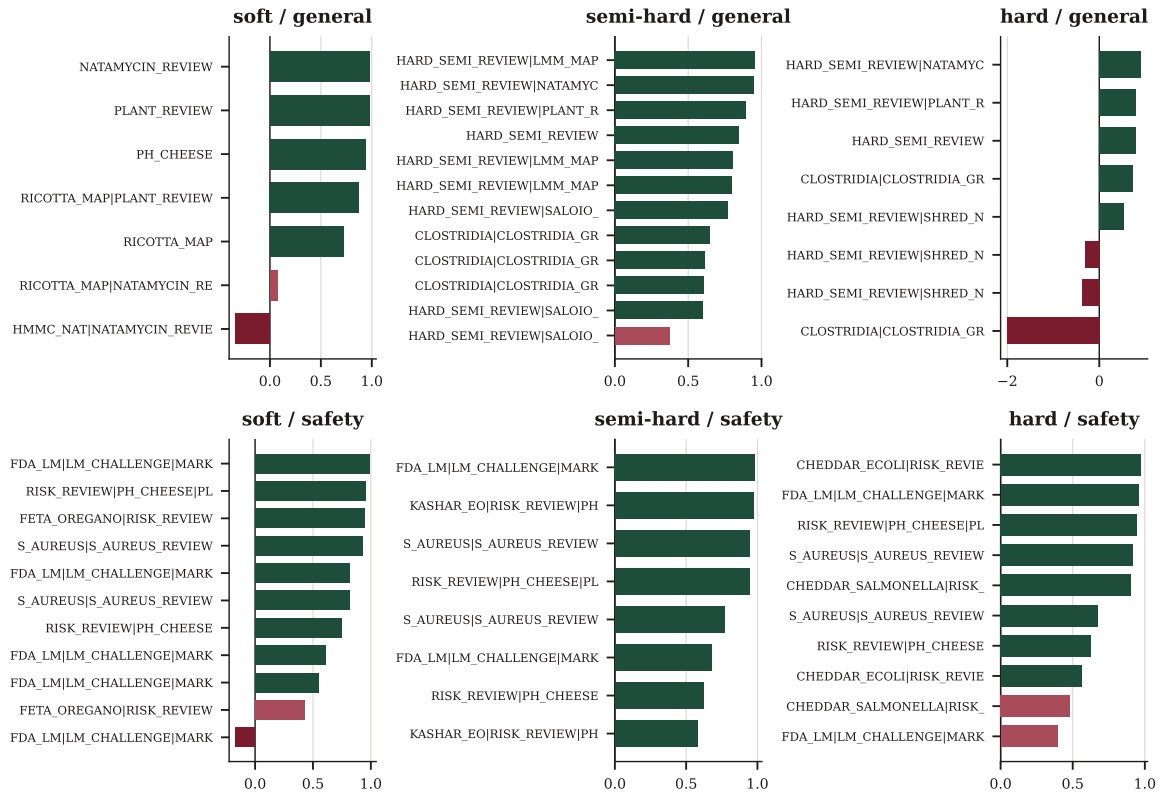
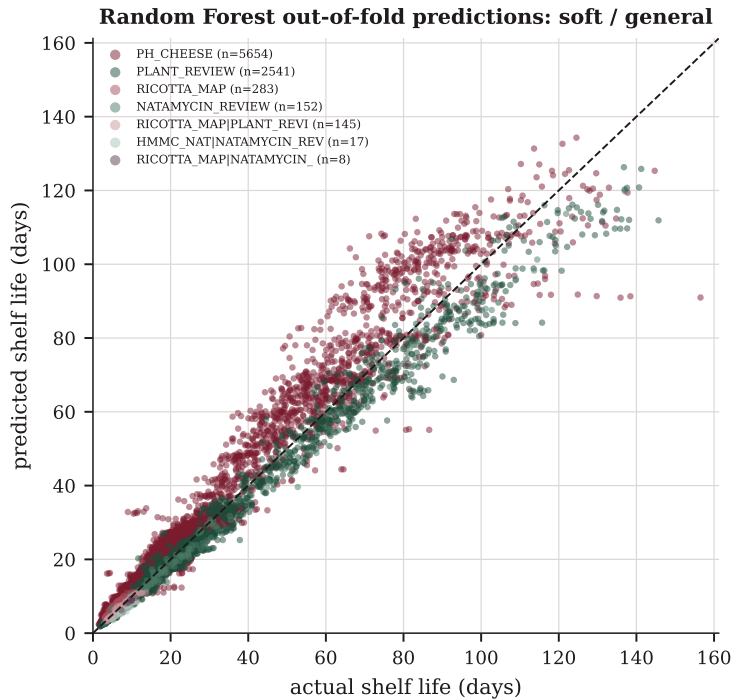
Two results that must be reported together

(a) The quality specialists hold up better than expected. Soft and semi-hard general-shelf-life retain pooled out-of-fold R^2 of 0.954 and 0.921 — essentially unchanged from their context-holdout figures. Withholding an entire generation rule barely hurts them.

(b) Hard/general collapses, and the safety specialists improve. Hard/general falls from 0.927 to 0.582 pooled. Meanwhile all three safety specialists *rise* sharply — hard/safety from −0.097 to +0.642. The revised protocol is not uniformly harsher; it is differently structured, and for the safety specialists the original single random context split happened to produce a harder and less representative test partition than the rule-wise folds do.

The second half of that finding deserves emphasis at the meeting because it cuts against the expected narrative. The honest statement is not “stricter evaluation, lower numbers everywhere.” It is that the original single split gave an unrepresentative reading in *both* directions, and that averaging over 56 structured folds is more informative than one random partition in either case.

8.5 Where the error concentrates

Random Forest: leave-rule-out R^2 for every held-out rule (clipped at -2)**Figure 8.1:** Random Forest leave-rule-out R^2 for every one of the 56 held-out rules, grouped by specialist and clipped at -2 . Most rules are predicted well; a small number fail badly, and Chapter 9 investigates each of them individually.**Figure 8.2:** Random Forest out-of-fold predictions against actual shelf life for the soft/general specialist, coloured by held-out rule. All 8 800 rows appear exactly once, each predicted by a model that had never seen its rule.

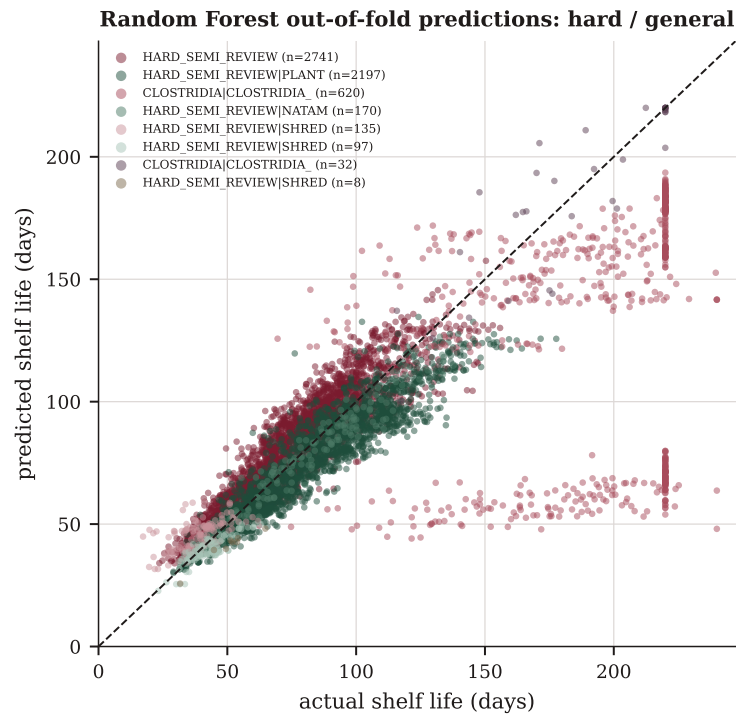


Figure 8.3: The same plot for hard/general. The detached cluster in the upper region — predicted far below the diagonal — is a single held-out rule, and it is responsible for most of this specialist’s loss.

Random Forest out-of-fold residual distributions (prediction – actual, days)

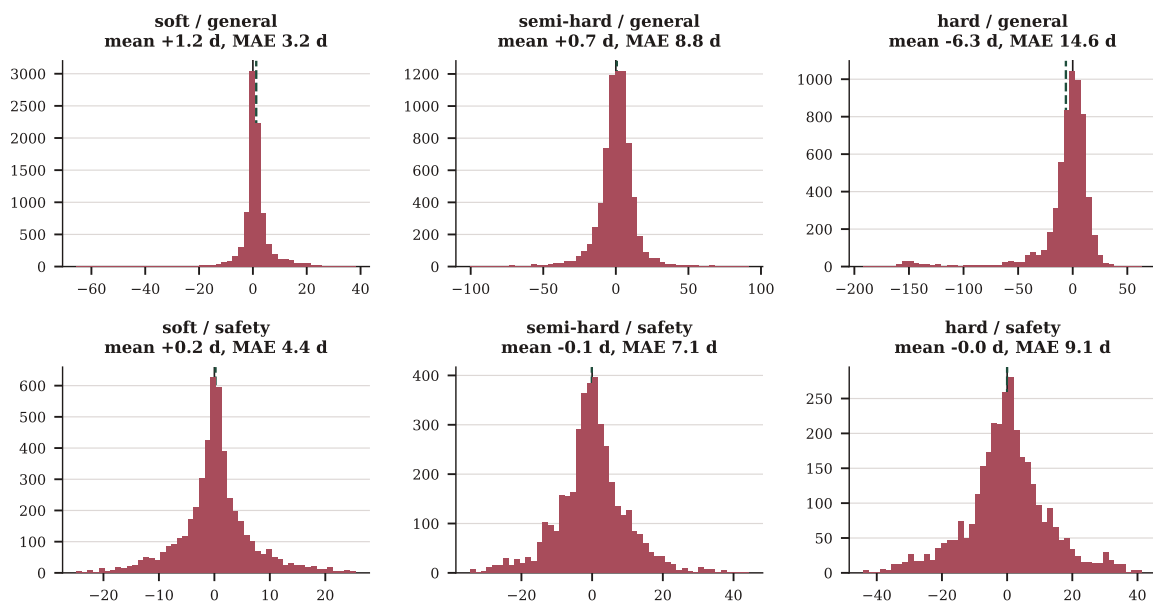


Figure 8.4: Out-of-fold residual distributions per specialist. The hard/general panel shows the pronounced negative tail produced by that same rule.

Residual structure against the true value

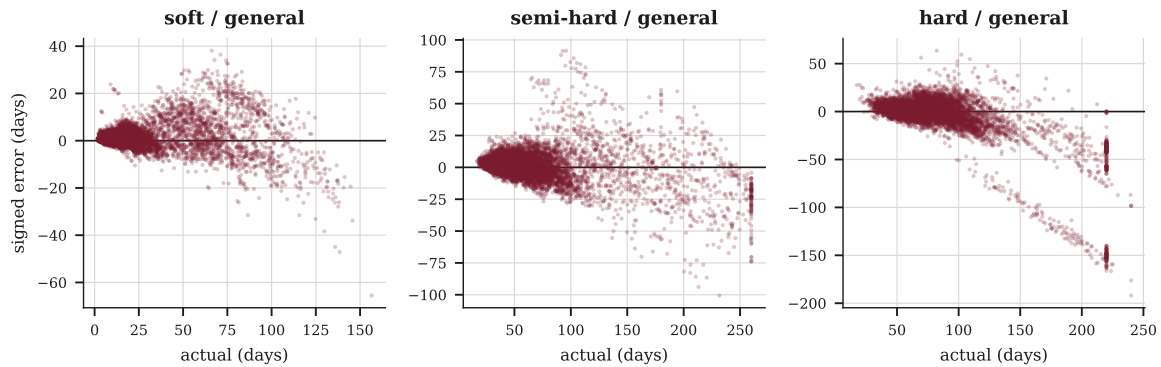


Figure 8.5: Residuals against the true value for the three quality specialists. The fan pattern in hard/general – errors growing with the true value – is the signature of eq. (8.3) binding.

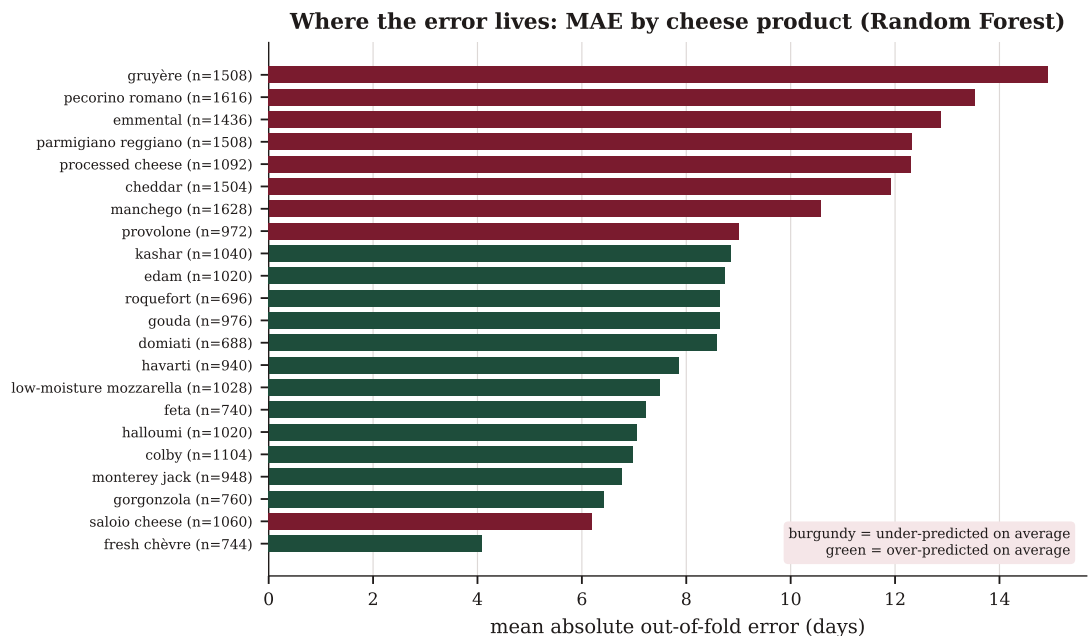


Figure 8.6: Mean absolute out-of-fold error by cheese product, restricted to products with at least 40 out-of-fold rows. Bar colour encodes the direction of the average bias.

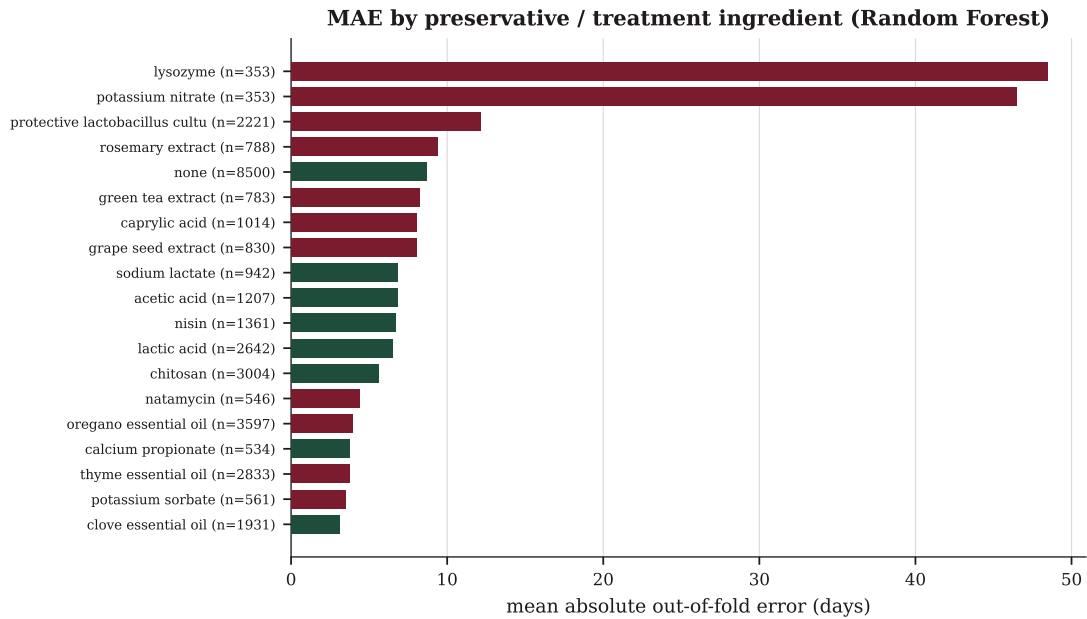


Figure 8.7: Mean absolute out-of-fold error by primary treatment ingredient, same restriction.

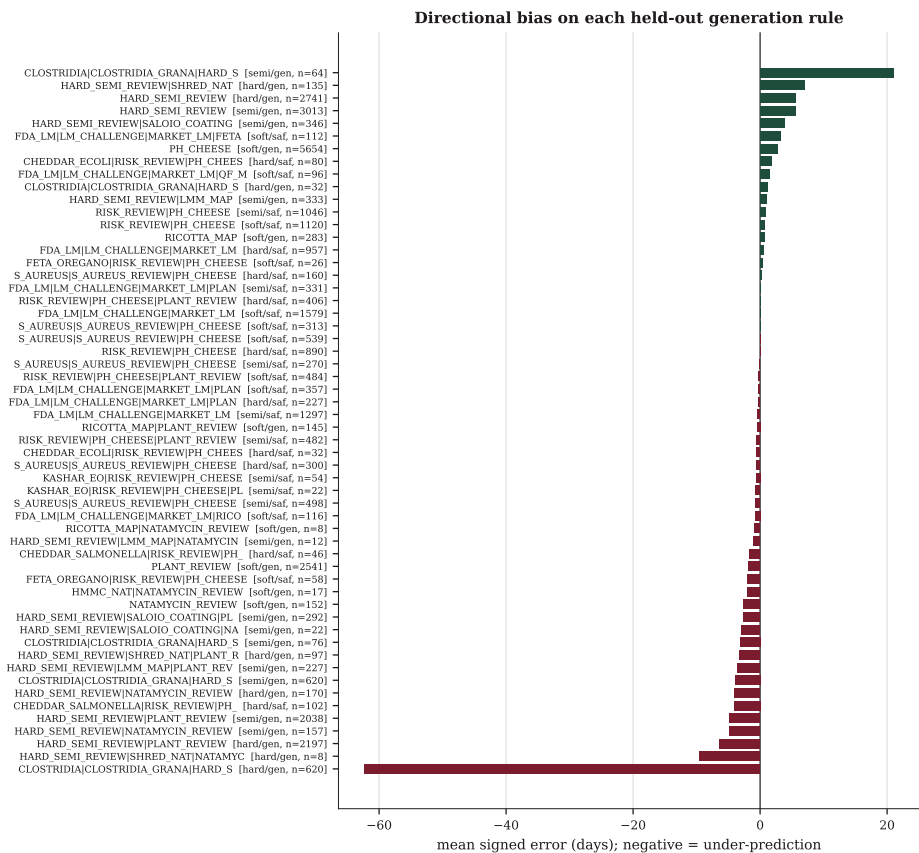


Figure 8.8: Mean signed error for every held-out rule. Systematic under-prediction (burgundy) dominates precisely where the held-out rule's targets lie above the training range — again eq. (8.3).

Fold-level investigation: what actually goes wrong

What to say at the start of this section

This is the section that answers “so what is the model actually bad at?” Do not speak in generalities about distribution shift. Name the rule, name the cheese, name the preservative, give the numbers in days. Every claim in this chapter is traceable to a specific set of rows in `rf_fold_predictions.csv`.

9.1 Method for this chapter

The historical leave-rule-out experiments stored only per-fold summary statistics. To investigate individual folds, the evaluation was re-executed with identical data, identical hyperparameters and the identical fold construction, retaining every out-of-fold prediction (`scripts/experiments/leave_rule_out_row_level.py`).

Reconstruction, clearly labelled

The row-level predictions analysed here come from a **re-execution**, not from the original diagnostic run. The original summary files are retained unmodified in `scripts/experiments/overfitting_diagnosis/`. The reconstruction reproduces the original per-fold R^2 values to the precision reported, which is expected given the shared seed and the deterministic fold construction, but the distinction is preserved rather than glossed: the figures in this chapter are labelled as reconstructed.

Each of the 56 folds was predicted by all three tree ensembles, giving 168 fold results and 102 000 row-level predictions with full formulation context attached: cheese product, physical form, treatment ingredient and concentration, storage temperature, packaging, indicator type and threshold, the observed target, the prediction, and the signed error.

Table 9.1: Complete Random Forest fold results. Every held-out generation rule in every specialist, sorted by R^2 within specialist. Negative R^2 values are bold.

Held-out rule	n_{test}	n_{train}	R^2	MAE (d)	RMSE (d)	bias (d)
soft / general shelf life						
NATAMYCIN_REVIEW	152	8,648	0.977	2.75	4.32	-2.58
PLANT_REVIEW	2,541	6,259	0.976	2.44	4.38	-1.87
PH_CHEESE	5,654	3,146	0.938	3.80	6.30	+2.82
RICOTTA_MAP PLANT_REVIEW	145	8,655	0.875	0.57	0.77	-0.49
RICOTTA_MAP	283	8,517	0.724	0.86	1.04	+0.63
RICOTTA_MAP NATAMYCIN_REVIEW	8	8,792	0.072	0.94	0.98	-0.94
HMMC_NAT NATAMYCIN_REVIEW	17	8,783	-0.335	2.00	2.05	-2.00
semi-hard / general shelf life						

Held-out rule	n_{test}	n_{train}	R^2	MAE (d)	RMSE (d)	bias (d)
HARD_SEMI_REVIEW LMM_MAP NATAMYCIN_REVIEW	12	7,188	0.955	2.23	2.64	-1.10
HARD_SEMI_REVIEW NATAMYCIN_REVIEW	157	7,043	0.943	5.28	6.95	-4.87
HARD_SEMI_REVIEW PLANT_REVIEW	2,038	5,162	0.893	6.59	9.05	-4.80
HARD_SEMI_REVIEW	3,013	4,187	0.847	7.65	9.51	+5.54
HARD_SEMI_REVIEW LMM_MAP	333	6,867	0.805	5.70	7.24	+1.06
HARD_SEMI_REVIEW LMM_MAP PLANT_REVIEW	227	6,973	0.794	6.49	8.47	-3.53
HARD_SEMI_REVIEW SALOIO_COATING NATAMYCIN_REVIEW	22	7,178	0.771	3.14	3.57	-2.90
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW SALOIO_COATING	76	7,124	0.648	20.77	28.39	-3.16
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW	620	6,580	0.612	26.89	33.81	-3.95
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW LMM_MAP	64	7,136	0.607	29.23	32.70	+20.95
HARD_SEMI_REVIEW SALOIO_COATING PLANT_REVIEW	292	6,908	0.595	3.71	4.78	-2.59
HARD_SEMI_REVIEW SALOIO_COATING	346	6,854	0.374	4.13	4.95	+3.90
hard / general shelf life						
HARD_SEMI_REVIEW NATAMYCIN_REVIEW	170	5,830	0.906	5.52	7.36	-4.07
HARD_SEMI_REVIEW PLANT_REVIEW	2,197	3,803	0.802	8.76	11.46	-6.44
HARD_SEMI_REVIEW	2,741	3,259	0.801	8.57	10.47	+5.57
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW SHRED_NAT	32	5,968	0.734	12.67	16.81	+1.18
HARD_SEMI_REVIEW SHRED_NAT PLANT_REVIEW	97	5,903	0.526	4.13	5.20	-3.26
HARD_SEMI_REVIEW SHRED_NAT NATAMYCIN_REVIEW	8	5,992	-0.306	9.48	10.03	-9.48
HARD_SEMI_REVIEW SHRED_NAT	135	5,865	-0.377	7.71	9.72	+6.94
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW	620	5,380	-3.860	67.38	83.13	-62.24
soft / safety endpoint						
FDA_LM LM_CHALLENGE MARKET_LM PLANT_REVIEW	357	4,443	0.988	1.30	1.81	-0.24
RISK_REVIEW PH_CHEESE PLANT_REVIEW	484	4,316	0.957	2.14	2.90	-0.19
FETA_OREGANO RISK_REVIEW PH_CHEESE PLANT_REVIEW	26	4,774	0.943	2.36	2.94	+0.37
S_AUREUS S_AUREUS_REVIEW PH_CHEESE PLANT_REVIEW	313	4,487	0.926	2.65	3.68	+0.03
FDA_LM LM_CHALLENGE MARKET_LM	1,579	3,221	0.820	5.00	7.23	+0.05
S_AUREUS S_AUREUS_REVIEW PH_CHEESE	539	4,261	0.812	4.61	5.99	-0.02
RISK_REVIEW PH_CHEESE	1,120	3,680	0.749	5.38	7.06	+0.69
FDA_LM LM_CHALLENGE MARKET_LM RICOTTA_NISIN	116	4,684	0.607	1.82	4.03	-0.81
FDA_LM LM_CHALLENGE MARKET_LM QF_MARKET	96	4,704	0.546	2.41	3.20	+1.53

Held-out rule	n_{test}	n_{train}	R^2	MAE (d)	RMSE (d)	bias (d)
FETA_OREGANO RISK_REVIEW PH_CHEESE	58	4,742	0.429	6.37	8.39	-1.94
FDA_LM LM_CHALLENGE MARKET_LM FETA_OREGANO	112	4,688	-0.166	13.23	14.79	+3.20
semi-hard / safety endpoint						
FDA_LM LM_CHALLENGE MARKET_LM PLANT_REVIEW	331	3,669	0.982	2.21	2.78	+0.15
KASHAR_E0 RISK_REVIEW PH_CHEESE PLANT_REVIEW	22	3,978	0.972	2.46	2.67	-0.73
S_AUREUS S_AUREUS_REVIEW PH_CHEESE PLANT_REVIEW	270	3,730	0.946	3.68	4.77	-0.12
RISK_REVIEW PH_CHEESE PLANT_REVIEW	482	3,518	0.944	2.97	4.00	-0.56
S_AUREUS S_AUREUS_REVIEW PH_CHEESE	498	3,502	0.768	7.96	10.29	-0.74
FDA_LM LM_CHALLENGE MARKET_LM	1,297	2,703	0.676	9.46	11.97	-0.46
RISK_REVIEW PH_CHEESE	1,046	2,954	0.624	8.00	10.74	+0.88
KASHAR_E0 RISK_REVIEW PH_CHEESE	54	3,946	0.581	8.88	10.91	-0.65
hard / safety endpoint						
CHEDDAR_ECOLI RISK_REVIEW PH_CHEESE PLANT_REVIEW	32	3,168	0.971	2.55	3.19	-0.60
FDA_LM LM_CHALLENGE MARKET_LM PLANT_REVIEW	227	2,973	0.957	3.49	4.55	-0.27
RISK_REVIEW PH_CHEESE PLANT_REVIEW	406	2,794	0.946	3.58	4.69	+0.10
S_AUREUS S_AUREUS_REVIEW PH_CHEESE PLANT_REVIEW	160	3,040	0.920	4.85	6.11	+0.18
CHEDDAR_SALMONELLA RISK_REVIEW PH_CHEESE PLANT_REVIEW	46	3,154	0.905	4.85	6.32	-1.63
S_AUREUS S_AUREUS_REVIEW PH_CHEESE	300	2,900	0.674	10.08	12.61	-0.63
RISK_REVIEW PH_CHEESE	890	2,310	0.624	9.39	12.39	-0.03
CHEDDAR_ECOLI RISK_REVIEW PH_CHEESE	80	3,120	0.560	10.70	13.80	+1.84
CHEDDAR_SALMONELLA RISK_REVIEW PH_CHEESE	102	3,098	0.481	11.78	15.34	-4.10
FDA_LM LM_CHALLENGE MARKET_LM	957	2,243	0.394	13.03	16.30	+0.58

9.2 The four weakest folds

9.3 Fold 1: CLOSTRIDIA|CLOSTRIDIA_GRANA|HARD_SEMI_REVIEW

This is the worst fold in the entire study and the only large one that fails. It is also the most instructive, because its cause can be established rather than hypothesised.

9.3.1 What was held out

Specialist

hard / general shelf life

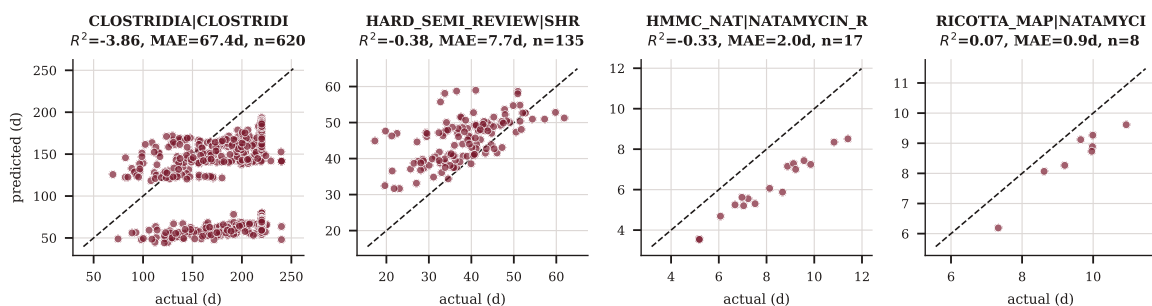
Rows

620 held out, 3 580 remaining for training

Table 9.3: Side-by-side diagnostic profile of the four weakest Random Forest folds. Every cell is computed from the held-out rows themselves or from the corresponding training partition.

	CLOSTRIDIA CLOSTRIDI	HARD_SEMI_REVIEW S	HMMC_- NAT NATAMYCIN	RICOTTA_- MAP NATAMY
Specialist	hard/general	hard/general	soft/general	soft/general
Held-out rows	620	135	17	8
Contexts	155	60	17	8
Cheese products	cheddar; emmental; gruyère ...	cheddar	fior di latte; fresh mozzarella	ricotta
Ingredients	lysozyme; none; potassium nitrate ...	acetic acid; calcium propionate; chitosan ...	natamycin	natamycin
Storage temp (°C)	8–15	2–12	2–8	2–4
Actual range (d)	69.5–240.0	17.3–61.9	5.2–11.4	7.3–10.9
Actual mean $\bar{y}$ (d)	189.0	38.4	8.1	9.5
Actual s.d. s_y (d)	37.74	8.31	1.83	1.09
Predicted mean (d)	126.8	45.3	6.1	8.5
Fold R^2	-3.860	-0.377	-0.335	0.072
Fold MAE (d)	67.38	7.71	2.00	0.94
Mean signed error (d)	-62.24	+6.94	-2.00	-0.94
Product seen in training?	yes	yes	yes	yes
Ingredient seen in training?	yes	yes	yes	yes
Product+ingredient rows in training	1,588	386	0	0

The four weakest Random Forest folds

**Figure 9.1:** Actual against predicted shelf life for the four weakest folds. Note the scale differences: the two natamycin folds on the right are compressed into a few days, while the clostridia fold on the left spans hundreds.

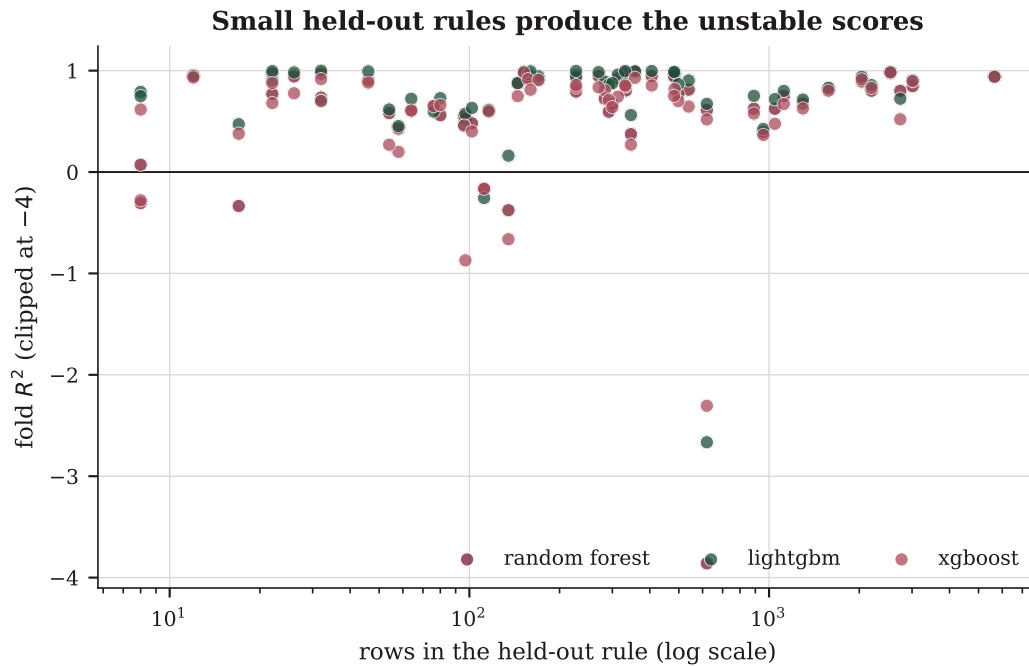


Figure 9.2: Fold R^2 against the number of rows in the held-out rule for all 168 fold results. Large folds are consistently well predicted; the dispersion is almost entirely among rules with fewer than 200 rows — with one crucial exception at 620 rows, analysed below.

Cheese products

pecorino romano, gruyère, manchego, parmigiano reggiano, cheddar, emmental — *all six also present in training*

Treatments

none (control), lysozyme, potassium nitrate, protective *Lactobacillus* culture

Indicator

clostridia — exclusively

Storage temperature

8–15°C (training spans 2–15°C)

Ripening

78.5–843.1 days (training spans 67.2–877.9)

9.3.2 Observed result

Random Forest: $R^2 = -3.860$, MAE = 67.4 days, RMSE = 83.1 days, mean signed error = -62.2 days. The model under-predicts by roughly two months on average. Actual targets: mean 189.0 days, range 69.5–240.0, $s_y = 37.7$. Predictions: mean 126.8 days.

9.3.3 Diagnosis

Working through the candidate explanations in order:

Is it an unseen product? No. All six cheese products in the held-out rule also appear in the training partition. Product novelty is eliminated.

Is it fold size? No. At 620 rows this is the third-largest fold in the specialist. Small-sample instability is eliminated.

Is it low target variance? No. $s_y = 37.7$ days is substantial. The R^2 denominator is large; the numerator is simply larger.

Is it an unseen *mechanism*? Yes — and this is demonstrable. Cross-tabulating `indicator_type` against `source_rule_id` within `hard/general` shows that the `clostridia` indicator appears in exactly two rules: this one (620 rows) and a small sibling, `CLOSTRIDIA|CLOSTRIDIA_GRANA|HARD_SEMI_REVIEW|SHRED_NAT` (32 rows). When this rule is withheld, only 32 `clostridia` rows remain in training out of 3 580.

The `clostridia` endpoint operates on a completely different time scale from every other indicator in the specialist:

<code>indicator_type</code>	rows	mean y (d)	range (d)
clostridia	652	188.8	69.5–240.0
molds	432	68.3	23.6–132.5
oxidation_index	876	83.2	27.1–162.3
peroxide_value	388	81.3	35.5–134.8
ph	356	81.5	32.7–155.9
sensory_acceptability	1 076	81.9	31.1–177.7
total_viable_count	988	75.1	19.6–147.3
yeasts_molds	1 232	67.2	17.3–137.7

Late blowing by `clostridia` is a slow ripening-associated defect; the other indicators track surface or bulk spoilage that manifests in weeks. The mean target for `clostridia` rows is $2.3\times$ that of any other indicator.

The mechanism of failure. With the rule withheld, the training target mean falls to 76.9 days against the held-out mean of 189.0. By eq. (8.3), the forest’s prediction is a weighted average of training targets, so it is structurally confined near the training range. It predicts 126.8 days on rows whose truth averages 189.0 — a -62.2 day bias that is not a tuning failure but a consequence of what tree averaging can represent.

9.3.4 The controlled confirmation

This specialist supplies its own natural control, and it is the strongest piece of evidence in the chapter. The small sibling rule (`...|SHRED_NAT`, 32 `clostridia` rows) has its own fold. When *it* is withheld, the 620-row `clostridia` rule remains in training, so the mechanism is represented. The result:

Held-out rule	rows	clostridia rows left in training	R^2	bias (d)
<code>CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW</code>	620	32	-3.860	-62.2
<code>CLOSTRIDIA ... SHRED_NAT</code>	32	620	$+0.734$	$+1.2$

Same specialist, same model, same indicator, same products, nearly inverted fold sizes — and the difference in outcome tracks the availability of the *mechanism* in training, not the size of the test set. The smaller fold is predicted well because its mechanism is represented; the larger fold fails because its mechanism is not.

9.3.5 A second, stronger control: the same rule in a different specialist

The identical rule identifier `V7|CLOSTRIDIA|CLOSTRIDIA_GRANA|HARD_SEMI_REVIEW` also occurs in the *semi-hard* specialist, again with exactly 620 rows. The difference is what surrounds it: *semi-hard*

contains three clostridia rules ($620 + 76 + 64 = 760$ rows) where hard contains two ($620 + 32 = 652$). Withholding the same rule therefore leaves 140 clostridia rows in training in semi-hard, against 32 in hard.

Specialist	Model	held-out rows	clostridia rows left	R^2	bias (d)
semi-hard/general	Random Forest	620	140	+0.612	-3.9
	LightGBM	620	140	+0.672	-1.4
	XGBoost	620	140	+0.519	-10.3
hard/general	Random Forest	620	32	-3.860	-62.2
	LightGBM	620	32	-2.665	-44.2
	XGBoost	620	32	-2.305	-52.5

Identical rule, identical fold size, identical models, identical protocol. The only material difference is how much of the clostridia mechanism survives in the training partition, and the outcome moves from $R^2 = +0.612$ to $R^2 = -3.860$ across all three model families. It is difficult to construct a cleaner demonstration that the governing variable is mechanism coverage.

Verified explanation, not a hypothesis

Hard/general's collapse under leave-rule-out is attributable to a single held-out rule that carries the only substantial representation of the clostridia spoilage mechanism, whose characteristic time scale is roughly two and a half times that of every other indicator in the specialist. The controlled comparison above rules out fold size and product novelty as explanations. This is the one fold in the study whose cause is established by a within-data experiment rather than inferred.

Target distribution: training rules vs. the held-out rule

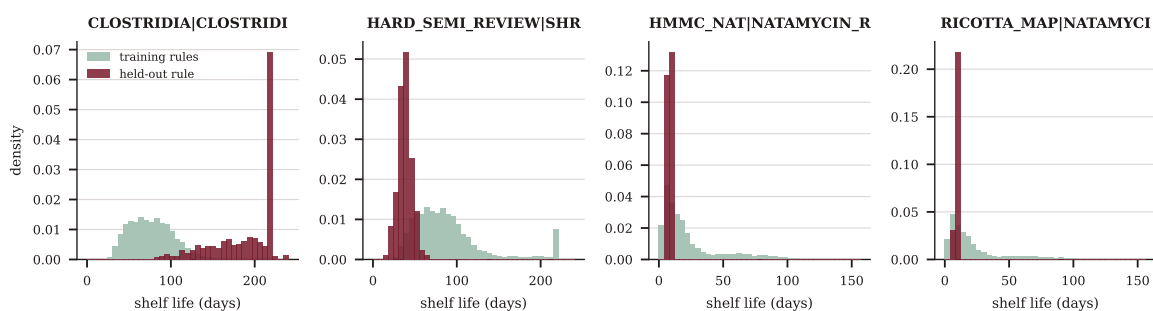


Figure 9.3: Target distributions for the four weakest folds: the training rules against the held-out rule. The clostridia panel (left) shows almost disjoint support — the held-out mass sits where the training data has almost none.

9.4 Fold 2: HARD_SEMI_REVIEW|SHRED_NAT

Held out

135 rows, all **shredded cheddar**

Treatments

control, chitosan, calcium propionate, lactic acid, protective culture, potassium sorbate

Packaging

MAP at three CO₂ levels plus high-barrier sealed

Feature coverage for the worst fold (CLOSTRIDIA|CLOSTRIDIA_GRANA|HARD_S)

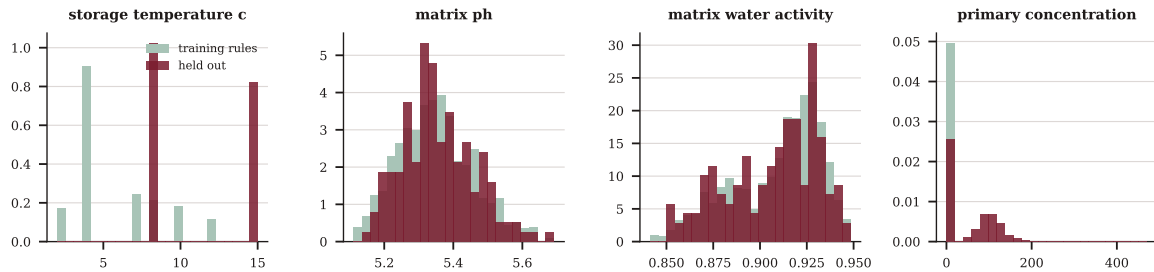


Figure 9.4: Input-feature coverage for the clostridia fold. The inputs are well covered by training data; it is the *target* relationship that is not. This is the figure that distinguishes covariate shift from mechanism novelty.

Table 9.4: The individual rows with the largest absolute errors in the two worst folds, with their actual formulation context.

Held-out rule	Cheese	T (°C)	actual (d)	pred. (d)	error (d)
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW					
	parmigiano reg-giano	8	240.0	48.1	-191.9
	pecorino romano	8	240.0	63.7	-176.3
	gruyère	8	223.7	57.2	-166.5
	parmigiano reg-giano	8	220.0	53.9	-166.1
	parmigiano reg-giano	15	220.0	56.6	-163.4
	parmigiano reg-giano	15	220.0	56.6	-163.4
HARD_SEMI_REVIEW SHRED_NAT					
	cheddar	10	19.8	47.6	+27.9
	cheddar	10	17.3	44.9	+27.6
	cheddar	10	21.3	46.3	+25.0
	cheddar	10	22.5	47.0	+24.5
	cheddar	8	33.8	58.1	+24.4
	cheddar	8	32.7	55.7	+23.0

Result

$$R^2 = -0.377, \text{MAE} = 7.7 \text{ days}, \text{bias} = +6.9 \text{ days}$$

Actual targets average 38.4 days ($s_y = 8.3$); the training partition averages 89.7 days. Here the held-out rule sits *below* the training range and the model over-predicts by seven days.

The distinguishing feature is `physical_form = shredded`. Shredding multiplies exposed surface area and shortens shelf life sharply relative to block cheese of identical composition. Cheddar itself is well represented in training (179 control rows, 58 with chitosan, 18 with calcium propionate), so again the product is not novel — the *form* is under-represented once its rule is withheld, and the model regresses towards block-cheddar behaviour.

9.5 Folds 3 and 4: the natamycin combination folds

These two folds directly test a hypothesis that was raised in advance: that the training set may contain a cheese and a preservative separately while containing no rows for their *combination*. The data confirms it exactly.

9.5.1 RICOTTA_MAP|NATAMYCIN_REVIEW — 8 rows

Quantity	Value
Held-out rows	8 (ricotta, fresh mass, natamycin, yeasts/moulds)
Ricotta rows in training	428
Natamycin rows in training	169
Ricotta + natamycin rows in training	0
Mean y , training ricotta rows	6.6 days
Mean y , training natamycin rows (18 other cheeses)	29.0 days
Actual y , held-out rows	9.5 days ($s_y = 1.09$, range 7.3–10.9)
Random Forest R^2	0.072
Random Forest MAE	0.94 days

The model must compose two facts it has seen separately — ricotta is short-lived; natamycin extends shelf life — for a combination it has never seen. It does so almost perfectly: the average error is **under one day**. The R^2 of 0.072 is near zero only because the eight held-out rows span 3.6 days in total, so $s_y^2 = 1.19 \text{ days}^2$ leaves essentially no variance to explain.

9.5.2 HMMC_NAT|NATAMYCIN_REVIEW — 17 rows

The same structure with fior di latte and fresh mozzarella: 448 and 468 training rows for those products, 160 natamycin rows across seventeen other cheeses, and **zero** rows for either product combined with natamycin. Actual targets average 8.1 days ($s_y = 1.83$); Random Forest achieves MAE = 2.0 days and $R^2 = -0.335$.

How to present these two folds

These are the folds most likely to be misread, in either direction.

Do not present $R^2 = -0.335$ as a catastrophic failure: the model predicts an eight-day shelf life to within two days having never seen the product–preservative combination, which is a genuinely reasonable result.

Do not present MAE = 0.94 days as proof of excellent generalization either: eight rows spanning three and a half days is far too little evidence to support any general claim, and the near-zero

R^2 is an honest signal of exactly that.

The correct statement is that on unseen product–preservative combinations the model produces plausible compositional predictions, and that these folds are too small to quantify how reliably.

9.6 Why the mean and the median disagree

The soft/general specialist has median fold $R^2 = 0.875$ but mean fold $R^2 = 0.604$. Both are correct, and the gap is entirely accounted for by the two natamycin folds analysed above – 8 and 17 rows out of 8 800, together 0.28% of the specialist’s data, each weighted at 1/7 of the mean.

Hard/general is the same arithmetic with the opposite cause: median 0.630 against mean -0.097 , driven by the single clostridia fold, which is *not* small and where the low score is substantively meaningful.

Why this report never quotes one summary alone

Mean fold R^2 weights an 8-row rule equally with a 5 654-row rule. Median fold R^2 conceals the clostridia failure entirely. Pooled out-of-fold R^2 weights by row count and would let a small but scientifically important rule disappear. MAE in days is robust to the variance denominator but blind to whether the model beats a constant.

No single number is adequate. All four are reported for every specialist throughout this document, and the recommendation in Chapter 15 is that they continue to be reported together.

9.7 Fold-level results for all specialists

The three quality specialists and the three safety specialists behave differently under this protocol, and the full per-fold tables (Tables 9.1, 10.2 and 10.4) support the following summary observations:

- **soft/general (7 folds).** Five folds above $R^2 = 0.87$; the two failures are the 8- and 17-row natamycin composites.
- **semi-hard/general (12 folds).** The most stable specialist: every fold positive, range 0.374–0.955, median 0.782.
- **hard/general (8 folds).** Median 0.630, but two negative folds including the clostridia rule; the weakest specialist under this protocol.
- **soft/safety (11 folds).** Median 0.812, mean 0.692; the spread comes from the small *Listeria* composite rules.
- **semi-hard/safety (8 folds).** Every fold positive, median 0.856 – against an original context-holdout score of 0.342.
- **hard/safety (10 folds).** Every fold positive, median 0.790 – against an original context-holdout score of -0.097 .

A revision to an earlier conclusion

The two safety specialists that looked broken under the original protocol are not broken. Under 10 and 8 structured folds respectively they show consistent, uniformly positive predictive skill (median R^2 of 0.790 and 0.856). The original single random context split produced an unrepresentative test partition for these specialists.

This correction should be stated explicitly at the meeting, because the earlier reading — that hard/safety-endpoint was fundamentally unlearnable — was recorded in project documentation and is wrong.

LightGBM and XGBoost

What to say at the start of this section

Explain boosting as correction rather than averaging: each tree is fitted to what the previous trees got wrong. Then make the key comparison explicit – under the original protocol the three ensembles were indistinguishable; under leave-rule-out they separate, and they fail on the same rules.

10.1 Gradient boosting

Both models fit an additive ensemble stage by stage,

$$F_M(\mathbf{x}) = F_0(\mathbf{x}) + \eta \sum_{m=1}^M h_m(\mathbf{x}), \quad (10.1)$$

where each h_m is a regression tree fitted to the negative gradient of the loss at the current prediction. For squared error the gradient is the residual itself,

$$r_i^{(m)} = - \left. \frac{\partial \ell(y_i, F)}{\partial F} \right|_{F=F_{m-1}(\mathbf{x}_i)} = y_i - F_{m-1}(\mathbf{x}_i), \quad (10.2)$$

so each stage literally models what the ensemble has so far failed to explain. The learning rate η shrinks each stage's contribution; small η with large M is the standard bias–variance compromise, and both production models use $\eta = 0.03$ with $M \leq 2000$ under early stopping.

10.1.1 XGBoost's regularised objective

XGBoost optimises a second-order approximation with an explicit complexity penalty on each tree:

$$\mathcal{L}^{(m)} \simeq \sum_{i=1}^n \left[g_i h_m(\mathbf{x}_i) + \frac{1}{2} \tilde{h}_i h_m(\mathbf{x}_i)^2 \right] + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2, \quad (10.3)$$

with T the number of leaves and w_j the leaf weights. The optimal leaf weight is $w_j^* = -G_j / (H_j + \lambda)$, so the L_2 term shrinks leaves supported by few rows – which is precisely the situation in the small composite rules of Section 9.5.

10.1.2 Leaf-wise versus depth-wise growth

LightGBM grows leaf-wise: at each step it splits the leaf offering the largest loss reduction anywhere in the tree, producing deep, unbalanced trees that chase the hardest regions. XGBoost grows depth-wise to a fixed `max_depth`, producing balanced trees. LightGBM's strategy is typically more sample-efficient and more prone to over-specialising on small regions; `num_leaves=31` and `min_child_samples=20` are the two settings that bound it here.

10.2 Production configurations

	LightGBM	XGBoost
Rounds	n_estimators=2000, early stopping 50	n_estimators=2000, early_stopping_rounds=50
Learning rate	0.03	0.03
Tree shape	num_leaves=31 (leaf-wise)	max_depth=6 (depth-wise)
Leaf constraint	min_child_samples=20	default λ, γ
Sampling	subsample=0.9, colsample_bytree=0.9	identical

Early stopping monitors the validation partition, which means that under the original protocol the stopping iteration was selected using data drawn from the same generation rules as the test set. In the leave-rule-out harness, boosting rounds are fixed a priori rather than early-stopped, which removes that dependency at the cost of not tuning M per fold. The difference is documented rather than silently absorbed.

10.3 Original results

From Table 4.1: LightGBM is the best model on five of the six V7 specialists under the original protocol, reaching 0.971 (soft/general), 0.920 (semi-hard/general) and 0.932 (hard/general). XGBoost is within one or two points everywhere. The gap between them, and between either and Random Forest, is well inside the range that a different random seed could produce — which is itself a hint that the protocol was not discriminating between models.

10.4 Revised results

Table 10.1: LightGBM and XGBoost under leave-one-rule-out, per specialist. Computed from the 168 reconstructed fold results.

Specialist	Model	folds	mean R^2	median R^2	min R^2	pooled OOF R^2
soft / general	LightGBM	7	0.850	0.892	0.472	0.960
	XGBoost	7	0.651	0.806	-0.279	0.957
semi-hard / general	LightGBM	12	0.824	0.888	0.562	0.940
	XGBoost	12	0.748	0.840	0.270	0.918
hard / general	LightGBM	8	0.256	0.714	-2.665	0.669
	XGBoost	8	-0.033	0.568	-2.305	0.665
soft / safety	LightGBM	11	0.702	0.831	-0.256	0.867
	XGBoost	11	0.589	0.671	-0.164	0.786
semi-hard / safety	LightGBM	8	0.860	0.926	0.617	0.811
	XGBoost	8	0.649	0.690	0.269	0.655
hard / safety	LightGBM	10	0.839	0.934	0.425	0.730
	XGBoost	10	0.697	0.736	0.366	0.593

Table 10.2: Complete LightGBM fold results for every held-out generation rule.

Held-out rule	n_{test}	n_{train}	R^2	MAE (d)	RMSE (d)	bias (d)
soft / general shelf life						

Held-out rule	n_{test}	n_{train}	R^2	MAE (d)	RMSE (d)	bias (d)
NATAMYCIN_REVIEW	152	8,648	0.989	1.94	3.06	-0.74
PLANT_REVIEW	2,541	6,259	0.985	1.94	3.43	-1.13
PH_CHEESE	5,654	3,146	0.943	3.62	6.05	+3.00
RICOTTA_MAP	283	8,517	0.892	0.50	0.65	+0.04
RICOTTA_MAP PLANT_REVIEW	145	8,655	0.877	0.54	0.76	+0.13
RICOTTA_MAP NATAMYCIN_REVIEW	8	8,792	0.790	0.43	0.47	+0.04
HMMC_NAT NATAMYCIN_REVIEW	17	8,783	0.472	1.18	1.29	-1.18
semi-hard / general shelf life						
HARD_SEMI_REVIEW NATAMYCIN_REVIEW	157	7,043	0.967	3.99	5.31	-2.93
HARD_SEMI_REVIEW LMM_MAP NATAMYCIN_REVIEW	12	7,188	0.947	2.38	2.85	+1.28
HARD_SEMI_REVIEW PLANT_REVIEW	2,038	5,162	0.940	4.77	6.78	-3.25
HARD_SEMI_REVIEW LMM_MAP PLANT_REVIEW	227	6,973	0.938	3.32	4.66	-0.38
HARD_SEMI_REVIEW	3,013	4,187	0.904	5.71	7.52	+3.96
HARD_SEMI_REVIEW SALOIO_COATING NATAMYCIN_REVIEW	22	7,178	0.904	1.75	2.31	+0.01
HARD_SEMI_REVIEW LMM_MAP	333	6,867	0.873	4.76	5.85	+0.20
HARD_SEMI_REVIEW SALOIO_COATING PLANT_REVIEW	292	6,908	0.861	2.21	2.80	+0.45
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW LMM_MAP	64	7,136	0.723	22.58	27.49	+10.31
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW	620	6,580	0.672	24.57	31.07	-1.38
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW SALOIO_COATING	76	7,124	0.597	22.94	30.38	-5.76
HARD_SEMI_REVIEW SALOIO_COATING	346	6,854	0.562	3.52	4.14	+3.21
hard / general shelf life						
HARD_SEMI_REVIEW NATAMYCIN_REVIEW	170	5,830	0.944	4.45	5.67	-0.52
HARD_SEMI_REVIEW PLANT_REVIEW	2,197	3,803	0.857	7.57	9.73	-5.45
HARD_SEMI_REVIEW SHRED_NAT NATAMYCIN_REVIEW	8	5,992	0.749	3.51	4.40	-2.13
HARD_SEMI_REVIEW	2,741	3,259	0.723	10.24	12.35	+8.72
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW SHRED_NAT	32	5,968	0.705	14.55	17.71	-2.80
HARD_SEMI_REVIEW SHRED_NAT PLANT_REVIEW	97	5,903	0.576	3.87	4.92	+3.32
HARD_SEMI_REVIEW SHRED_NAT	135	5,865	0.162	6.36	7.58	+5.56
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW	620	5,380	-2.665	55.09	72.19	-44.16
soft / safety endpoint						
FDA_LM LM_CHALLENGE MARKET_LM PLANT_REVIEW	357	4,443	0.994	0.88	1.28	-0.23
FETA_OREGANO RISK_REVIEW PH_CHEESE PLANT_REVIEW	26	4,774	0.982	1.15	1.67	+0.61
RISK_REVIEW PH_CHEESE PLANT_REVIEW	484	4,316	0.981	1.38	1.93	-0.03
S_AUREUS S_AUREUS_REVIEW PH_CHEESE PLANT_REVIEW	313	4,487	0.960	1.86	2.70	+0.00

Held-out rule	n_{test}	n_{train}	R^2	MAE (d)	RMSE (d)	bias (d)
S_AUREUS S_AUREUS_REVIEW PH_CHEESE	539	4,261	0.903	3.04	4.31	-0.19
FDA_LM LM_CHALLENGE MARKET_LM	1,579	3,221	0.831	4.42	7.01	+0.28
RISK_REVIEW PH_CHEESE	1,120	3,680	0.800	4.21	6.30	+0.69
FDA_LM LM_CHALLENGE MARKET_LM RICOTTA_NISIN	116	4,684	0.613	1.95	4.00	-0.59
FDA_LM LM_CHALLENGE MARKET_LM QF_MARKET	96	4,704	0.463	2.70	3.49	+1.55
FETA_OREGANO RISK_REVIEW PH_CHEESE	58	4,742	0.453	5.25	8.22	-2.29
FDA_LM LM_CHALLENGE MARKET_LM FETA_OREGANO	112	4,688	-0.256	13.49	15.35	+2.35
semi-hard / safety endpoint						
FDA_LM LM_CHALLENGE MARKET_LM PLANT_REVIEW	331	3,669	0.995	1.07	1.45	+0.32
KASHAR_E0 RISK_REVIEW PH_CHEESE PLANT_REVIEW	22	3,978	0.995	0.99	1.18	-0.15
RISK_REVIEW PH_CHEESE PLANT_REVIEW	482	3,518	0.987	1.10	1.90	-0.21
S_AUREUS S_AUREUS_REVIEW PH_CHEESE PLANT_REVIEW	270	3,730	0.985	1.56	2.51	+0.12
S_AUREUS S_AUREUS_REVIEW PH_CHEESE	498	3,502	0.867	4.57	7.81	+0.16
RISK_REVIEW PH_CHEESE	1,046	2,954	0.719	5.59	9.29	+0.84
FDA_LM LM_CHALLENGE MARKET_LM	1,297	2,703	0.714	8.08	11.23	-1.01
KASHAR_E0 RISK_REVIEW PH_CHEESE	54	3,946	0.617	6.73	10.44	-0.97
hard / safety endpoint						
CHEDDAR_ECOLI RISK_REVIEW PH_CHEESE PLANT_REVIEW	32	3,168	0.999	0.32	0.52	-0.03
FDA_LM LM_CHALLENGE MARKET_LM PLANT_REVIEW	227	2,973	0.997	0.55	1.19	-0.03
RISK_REVIEW PH_CHEESE PLANT_REVIEW	406	2,794	0.996	0.67	1.19	+0.12
S_AUREUS S_AUREUS_REVIEW PH_CHEESE PLANT_REVIEW	160	3,040	0.993	0.79	1.83	-0.12
CHEDDAR_SALMONELLA RISK_REVIEW PH_CHEESE PLANT_REVIEW	46	3,154	0.992	0.98	1.80	-0.08
S_AUREUS S_AUREUS_REVIEW PH_CHEESE	300	2,900	0.876	4.89	7.77	-0.24
RISK_REVIEW PH_CHEESE	890	2,310	0.751	6.33	10.08	-0.64
CHEDDAR_ECOLI RISK_REVIEW PH_CHEESE	80	3,120	0.728	6.36	10.84	+1.75
CHEDDAR_SALMONELLA RISK_REVIEW PH_CHEESE	102	3,098	0.632	7.99	12.91	-1.32
FDA_LM LM_CHALLENGE MARKET_LM	957	2,243	0.425	11.31	15.87	-0.70

Table 10.4: Complete XGBoost fold results for every held-out generation rule.

Held-out rule	n_{test}	n_{train}	R^2	MAE (d)	RMSE (d)	bias (d)
soft / general shelf life						
NATAMYCIN_REVIEW	152	8,648	0.983	2.43	3.77	-1.79
PLANT_REVIEW	2,541	6,259	0.982	2.26	3.76	-1.40
PH_CHEESE	5,654	3,146	0.939	3.80	6.26	+3.16
RICOTTA_MAP	283	8,517	0.806	0.68	0.87	+0.21
RICOTTA_MAP PLANT_REVIEW	145	8,655	0.749	0.79	1.09	-0.05
HMMC_NAT NATAMYCIN_REVIEW	17	8,783	0.378	1.23	1.40	-1.23
RICOTTA_MAP NATAMYCIN_REVIEW	8	8,792	-0.279	1.05	1.15	-0.07
semi-hard / general shelf life						
HARD_SEMI_REVIEW LMM_MAP NATAMYCIN_REVIEW	12	7,188	0.933	2.84	3.22	+0.62
HARD_SEMI_REVIEW NATAMYCIN_REVIEW	157	7,043	0.916	5.96	8.43	-4.87
HARD_SEMI_REVIEW PLANT_REVIEW	2,038	5,162	0.914	5.88	8.14	-3.32
HARD_SEMI_REVIEW	3,013	4,187	0.897	6.09	7.82	+3.65
HARD_SEMI_REVIEW SALOIO_COATING NATAMYCIN_REVIEW	22	7,178	0.878	2.19	2.60	+1.52
HARD_SEMI_REVIEW LMM_MAP	333	6,867	0.846	5.22	6.44	+0.33
HARD_SEMI_REVIEW LMM_MAP PLANT_REVIEW	227	6,973	0.835	5.72	7.57	-0.92
HARD_SEMI_REVIEW SALOIO_COATING PLANT_REVIEW	292	6,908	0.711	3.13	4.04	+1.96
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW SALOIO_COATING	76	7,124	0.651	20.59	28.27	-5.04
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW LMM_MAP	64	7,136	0.606	27.73	32.75	+16.20
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW	620	6,580	0.519	30.38	37.64	-10.32
HARD_SEMI_REVIEW SALOIO_COATING	346	6,854	0.270	4.62	5.34	+4.41
hard / general shelf life						
HARD_SEMI_REVIEW NATAMYCIN_REVIEW	170	5,830	0.910	5.58	7.18	-1.81
HARD_SEMI_REVIEW PLANT_REVIEW	2,197	3,803	0.831	8.21	10.60	-5.97
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW SHRED_NAT	32	5,968	0.700	14.69	17.86	-4.38
HARD_SEMI_REVIEW SHRED_NAT NATAMYCIN_REVIEW	8	5,992	0.617	3.53	5.43	-3.04
HARD_SEMI_REVIEW	2,741	3,259	0.520	13.36	16.26	+10.57
HARD_SEMI_REVIEW SHRED_NAT	135	5,865	-0.663	9.55	10.68	+9.39
HARD_SEMI_REVIEW SHRED_NAT PLANT_REVIEW	97	5,903	-0.871	8.96	10.33	+8.92
CLOSTRIDIA CLOSTRIDIA_GRANA HARD_SEMI_REVIEW	620	5,380	-2.305	57.33	68.56	-52.55
soft / safety endpoint						
FDA_LM LM_CHALLENGE MARKET_LM PLANT_REVIEW	357	4,443	0.929	3.09	4.43	-0.44
RISK_REVIEW PH_CHEESE PLANT_REVIEW	484	4,316	0.818	4.77	5.97	-0.31
FDA_LM LM_CHALLENGE MARKET_LM	1,579	3,221	0.802	5.35	7.59	+0.29

Held-out rule	n_{test}	n_{train}	R^2	MAE (d)	RMSE (d)	bias (d)
FETA_OREGANO RISK_REVIEW PH_CHEESE PLANT_REVIEW	26	4,774	0.777	5.19	5.82	+0.35
S_AUREUS S_AUREUS_REVIEW PH_CHEESE PLANT_REVIEW	313	4,487	0.744	5.47	6.86	+0.19
RISK_REVIEW PH_CHEESE	1,120	3,680	0.671	6.33	8.07	+0.67
S_AUREUS S_AUREUS_REVIEW PH_CHEESE	539	4,261	0.646	6.57	8.22	-0.03
FDA_LM LM_CHALLENGE MARKET_ LM RICOTTA_NISIN	116	4,684	0.598	2.20	4.07	-0.15
FDA_LM LM_CHALLENGE MARKET_ LM QF_MARKET	96	4,704	0.459	2.74	3.50	+1.78
FETA_OREGANO RISK_REVIEW PH_CHEESE	58	4,742	0.199	7.45	9.94	-3.42
FDA_LM LM_CHALLENGE MARKET_ LM FETA_OREGANO	112	4,688	-0.164	13.55	14.78	+3.09
semi-hard / safety endpoint						
FDA_LM LM_CHALLENGE MARKET_ LM PLANT_REVIEW	331	3,669	0.853	6.51	7.99	+0.58
S_AUREUS S_AUREUS_REVIEW PH_CHEESE PLANT_REVIEW	270	3,730	0.835	6.88	8.37	+0.18
RISK_REVIEW PH_CHEESE PLANT_REVIEW	482	3,518	0.751	6.79	8.44	-0.73
S_AUREUS S_AUREUS_REVIEW PH_CHEESE	498	3,502	0.699	9.22	11.72	-0.37
KASHAR_EO RISK_REVIEW PH_ CHEESE PLANT_REVIEW	22	3,978	0.680	8.16	9.03	-2.17
FDA_LM LM_CHALLENGE MARKET_ LM	1,297	2,703	0.627	10.40	12.84	-0.51
RISK_REVIEW PH_CHEESE	1,046	2,954	0.476	9.91	12.69	+0.84
KASHAR_EO RISK_REVIEW PH_ CHEESE	54	3,946	0.269	12.55	14.42	-0.10
hard / safety endpoint						
CHEDDAR_ECOLI RISK_REVIEW PH_CHEESE PLANT_REVIEW	32	3,168	0.916	4.29	5.40	-0.94
CHEDDAR_SALMONELLA RISK_ REVIEW PH_CHEESE PLANT_ REVIEW	46	3,154	0.883	5.60	7.04	-1.58
FDA_LM LM_CHALLENGE MARKET_ LM PLANT_REVIEW	227	2,973	0.857	6.69	8.27	-0.09
RISK_REVIEW PH_CHEESE PLANT_REVIEW	406	2,794	0.854	6.06	7.70	+0.10
S_AUREUS S_AUREUS_REVIEW PH_CHEESE PLANT_REVIEW	160	3,040	0.812	7.79	9.40	-0.06
CHEDDAR_ECOLI RISK_REVIEW PH_CHEESE	80	3,120	0.660	9.39	12.13	+1.48
S_AUREUS S_AUREUS_REVIEW PH_CHEESE	300	2,900	0.640	10.64	13.24	-1.36
RISK_REVIEW PH_CHEESE	890	2,310	0.578	10.07	13.13	+0.25
CHEDDAR_SALMONELLA RISK_ REVIEW PH_CHEESE	102	3,098	0.401	12.75	16.48	-3.58
FDA_LM LM_CHALLENGE MARKET_ LM	957	2,243	0.366	13.08	16.66	+0.17

LightGBM is the strongest model under the revised protocol

LightGBM has the highest median fold R^2 in five of six specialists and the highest pooled out-of-fold R^2 in all six. Its advantage is largest exactly where the problem is hardest: on hard/general it reaches mean $R^2 = 0.256$ against XGBoost's -0.033 and Random Forest's -0.097 , and on the clostridia fold it is the least badly wrong of the three (-2.665 against -2.305 and -3.860 ; MAE 55.1 days against 57.3 and 67.4).

This ordering was not visible under the original protocol, where all three sat within two points of each other. An evaluation that cannot distinguish between candidate models cannot be used to select between them.

10.5 Do the three ensembles fail on the same rules?

Do the three ensembles fail on the same rules? Per-fold R^2 , clipped at -3

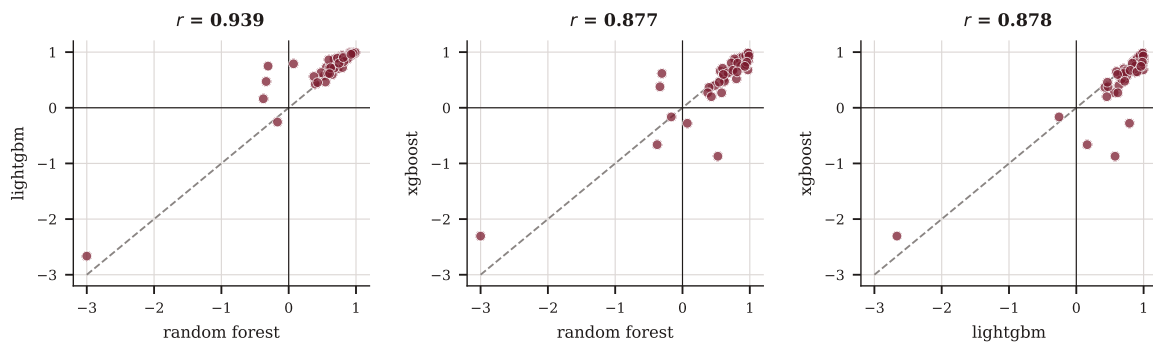


Figure 10.1: Pairwise per-fold R^2 for the three ensembles across all 56 folds, clipped at -3 . Correlations: Random Forest–LightGBM $r = 0.939$, Random Forest–XGBoost $r = 0.877$, LightGBM–XGBoost $r = 0.878$.

The agreement is high. All three models place the clostridia fold as their single worst result, and all three struggle on the shredded-cheddar rule. The failures are shared, not model-specific.

What correlated failure does and does not imply

Correlation of $r \approx 0.9$ across three structurally different learners — one bagging, two boosting, one depth-wise and one leaf-wise — indicates that fold difficulty is a property of the *data partition*, not of any model's inductive bias.

The careful non-claim: this does not by itself prove the cause is mechanism novelty. Three tree ensembles share the extrapolation bound of eq. (8.3), so they would be expected to fail together on any out-of-range target regardless of why the target is out of range. The causal attribution in Section 9.3 rests on the two controlled comparisons, not on this correlation.

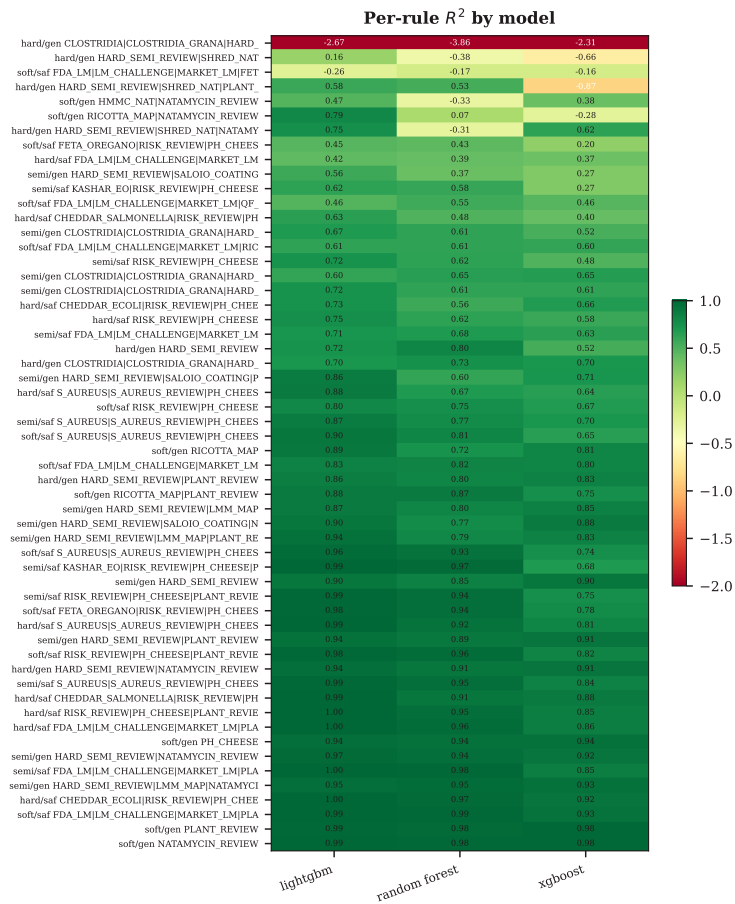


Figure 10.2: Per-rule R^2 for all three models across all 56 held-out rules, sorted by mean difficulty.

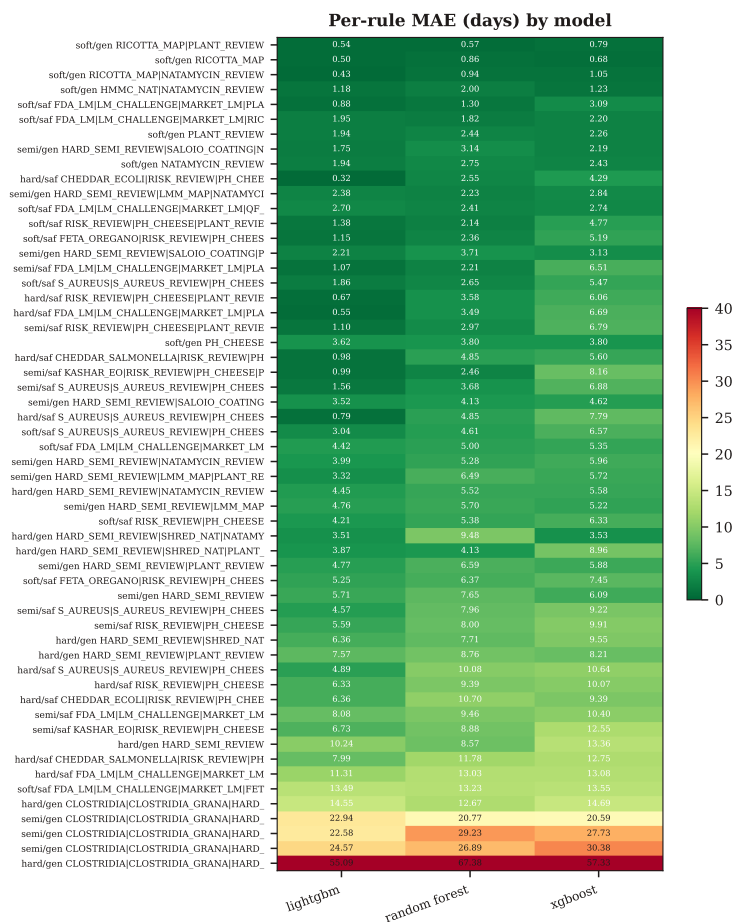


Figure 10.3: The same grid in MAE (days), which removes the variance-denominator effect. Several rules that look poor in R^2 have small day-scale errors.

EBM and the alternative model families

11.1 The Explainable Boosting Machine

EBM is a generalized additive model with bounded interactions, fitted by cyclic gradient boosting on one feature at a time:

$$g(\mathbb{E}[y \mid \mathbf{x}]) = \beta_0 + \sum_{j=1}^p f_j(x_j) + \sum_{(j,k) \in \mathcal{I}} f_{jk}(x_j, x_k), \quad |\mathcal{I}| = 10. \quad (11.1)$$

Because each f_j is a one-dimensional step function of a single feature, the fitted model is directly readable: one can plot the learned effect of storage temperature on shelf life and inspect it. This interpretability is why it is in the production candidate set.

Its production configuration is `interactions=10, max_bins=256, learning_rate=0.02, max_rounds=3000, min_samples_leaf=4, outer_bags=8`.

11.1.1 EBM's original results

From Table 4.1, EBM is consistently the weakest of the four production families under the original protocol: 0.944 on soft/general against LightGBM's 0.971, and -0.528 on hard/safety against LightGBM's -0.024 . The additive structure with only ten interaction terms is a genuine capacity restriction relative to a fully-interacting tree ensemble.

11.1.2 Why EBM is absent from the leave-rule-out results

Stated gap in the revised evaluation

EBM was **not** re-evaluated under leave-one-rule-out. The reason is mechanical rather than scientific: the production EBM pipeline uses its own preprocessing path (a frame-level imputer preserving native categorical handling), not the shared scaled-and-one-hot `ColumnTransformer` that every other model in the diagnostic harness uses. Running it inside the shared harness would have changed its preprocessing as well as its partitioning, confounding the comparison.

Consequently, **EBM's context-holdout numbers must not be placed alongside the other models' leave-rule-out numbers as though they were comparable**. They measure different things. Completing this evaluation with EBM's own pipeline and the same rule-wise folds is a concrete, bounded piece of outstanding work, listed in Chapter 15.

11.2 The alternative model families

Thirteen further configurations were evaluated under both protocols. Their behaviour under the protocol change is the clearest single illustration of what the change measures.

11.2.1 Regularised linear models

Ridge, lasso, elastic net, Bayesian ridge and PLS regression all fit

$$\hat{y} = \beta_0 + \sum_j \beta_j x_j, \quad (11.2)$$

differing only in the penalty applied to β : L_2 for ridge, L_1 for lasso, a convex combination for elastic net, a Gaussian prior for Bayesian ridge, and a latent-component projection for PLS.

Model	context-holdout R^2		leave-rule-out R^2	
	soft/general	mean, 6 spec.	mean	median
ridge regression	0.910	0.594	−0.874	−0.798
lasso	0.908	0.595	−0.690	−0.736
elastic net	0.895	0.592	−0.772	−0.675
Bayesian ridge	0.910	0.597	−0.872	−0.795
PLS regression	0.902	0.594	−1.106	−0.952

The two context-holdout columns are given separately because the six-specialist mean is dragged down by the safety specialists, where every model scores poorly; the soft/general column is the like-for-like comparison against the production figure of 0.971.

The linear-model collapse

Ridge regression scores $R^2 = 0.910$ on soft/general under the original protocol – close enough to the production LightGBM’s 0.971 that it was initially taken as evidence that model complexity was irrelevant to the problem. Under leave-rule-out the same model scores a mean R^2 of -0.874 : substantially worse than predicting a constant.

The interpretation is that a global linear fit can interpolate *within* the known rules, because within any one rule the mapping is locally smooth – but it has no mechanism for representing a rule it has not seen, and its extrapolations are actively harmful. The conclusion from Experiment 1 therefore needs a correction: model complexity is irrelevant to *within-rule interpolation* but decisive for *cross-rule transfer*.

11.2.2 Instance-based and kernel methods

k -nearest neighbours ($k = 5$) predicts the mean target of the five nearest training rows in standardised feature space. It scores 0.905 on soft/general under the original protocol and a mean of -3.412 under leave-rule-out – the worst non-trivial model in the study. The failure mode is transparent: when the entire mechanism is removed, the nearest neighbours are drawn from different mechanisms, and their average carries no relevant information.

RBF-kernel SVR fits $\hat{y} = \sum_i \alpha_i \exp(-\gamma \|\mathbf{x} - \mathbf{x}_i\|^2) + b$, with training subsampled to 4 000 rows for tractability. It scores 0.946 on soft/general then a mean of -0.454 under leave-rule-out: the same pattern, less severe, because the kernel provides smoother interpolation than a hard k -neighbour average.

11.2.3 Tree ensembles outside the production set

Model	context-holdout R^2		leave-rule-out
	soft/general	mean, 6 spec.	mean R^2
extra trees	0.961	0.608	0.209
gradient boosting (sklearn)	0.959	0.597	0.280
AdaBoost	0.917	0.601	−1.248
shallow random forest	0.934	0.597	−0.230
shallow decision tree	0.926	0.553	−1.733
small MLP	0.972	0.595	0.219

Extra trees and sklearn gradient boosting remain positive under the revised protocol — they are structurally similar to the production ensembles — but at roughly one third of their performance. The deliberately shallow variants collapse, confirming that depth is what carries cross-rule transfer.

The small MLP is worth singling out: it *matched* the production LightGBM on soft/general under the original protocol (0.972 vs 0.971) and falls to a mean R^2 of 0.219 under leave-rule-out. It is the sharpest single example of two models that an inadequate evaluation declares equivalent and an adequate one separates by six-tenths of an R^2 point.

11.2.4 The dummy baseline

DummyRegressor(strategy="mean") predicts the training mean for every row. It scores -0.004 under the original protocol — correct, since predicting the training mean on a test set drawn from the same distribution yields approximately zero — and -15.637 under leave-rule-out, because each held-out rule's mean differs substantially from the pooled training mean.

That number is a useful calibration for the panel: it establishes the scale on which the leave-rule-out R^2 values should be read. On that scale, LightGBM's $+0.627$ and ridge's -0.874 are both far above the no-information floor, and the separation between them is real.

Table 11.1: All alternative models under the original context-holdout protocol, averaged across the six specialists.

Model	mean test R^2	min	max	mean gap	mean MAE (d)
extra_trees	0.608	-0.052	0.961	0.179	10.04
adaboost	0.601	-0.039	0.917	0.038	11.43
bayesian_ridge	0.597	-0.039	0.929	0.011	10.37
gradient_boosting	0.597	-0.158	0.959	0.186	9.73
lasso	0.595	-0.055	0.928	0.015	10.39
pls_regression	0.594	-0.060	0.926	0.015	10.48
elastic_net	0.592	-0.052	0.925	0.015	10.49
svr_rbf	0.557	-0.073	0.946	0.140	10.57
knn_regressor_k5	0.461	-0.168	0.905	0.274	12.97
dummy_mean_baseline	-0.004	-0.022	-0.000	0.004	21.96

Cross-model comparison

What to say at the start of this section

The point of this chapter is not to crown a winner. It is to show that the original protocol could not rank models at all, and the revised one can. That is a statement about the evaluation, and it is the reason the evaluation had to change before any model-selection decision could be trusted.

12.1 The same models under the two protocols

Table 12.1: Every evaluated configuration under both protocols. The context-holdout column is the mean test R^2 across the six specialists; the leave-rule-out columns are the mean and median of the per-fold values, averaged over the six specialists. Sorted by leave-rule-out median.

Model	context-holdout R^2	LORO mean	LORO median	rank change
LightGBM (production)	—	0.627	0.780	—
Random Forest (production)	—	0.582	0.791	—
XGBoost (production)	—	0.550	0.719	—
extra trees	0.608	0.209	0.492	↓
gradient boosting	0.597	0.280	0.450	↓
small MLP	0.595	0.219	0.342	↓
shallow random forest	0.597	−0.230	0.278	↓↓
shallow decision tree	0.553	−1.733	0.192	↓↓
RBF SVR	0.557	−0.454	−0.153	↓↓
AdaBoost	0.601	−1.248	−0.178	↓↓↓
elastic net	0.592	−0.772	−0.675	↓↓↓
lasso	0.595	−0.690	−0.736	↓↓↓
Bayesian ridge	0.597	−0.872	−0.795	↓↓↓
ridge regression	0.594	−0.874	−0.798	↓↓↓
PLS regression	0.594	−1.106	−0.952	↓↓↓
k -NN ($k = 5$)	0.461	−3.412	−1.537	↓↓↓
dummy (mean)	−0.004	−15.637	−12.833	—

The discriminating power of the two protocols

Under the original protocol, thirteen of the sixteen non-trivial configurations fall within the band $[0.55, 0.61]$ — a spread of six hundredths of an R^2 point separating a single shallow decision tree from a 2000-round gradient ensemble.

Under leave-rule-out, the same configurations span from $+0.79$ to $−1.54$ in median R^2 : a range more than twenty times wider. The revised protocol does not merely give lower numbers; it gives *informative* numbers.

Table 12.2: Leave-rule-out results by specialist for the three production ensembles (repeated here for the cross-model comparison).

Specialist	Model	folds	mean R^2	median R^2	s.d.	min	max
soft/general shelf life	lightgbm	7	0.752	0.831	0.256	0.25	0.984
	random_forest	7	0.604	0.875	0.484	-0.33	0.977
	xgboost	7	0.651	0.806	0.427	-0.28	0.983
semi-hard/general shelf life	lightgbm	12	0.782	0.871	0.183	0.28	0.939
	random_forest	12	0.737	0.782	0.165	0.37	0.955
	xgboost	12	0.748	0.840	0.195	0.27	0.933
hard/general shelf life	lightgbm	8	0.182	0.703	1.079	-2.54	0.925
	random_forest	8	-0.097	0.630	1.499	-3.86	0.906
	xgboost	8	-0.033	0.568	1.070	-2.31	0.910
soft/safety endpoint	lightgbm	11	0.600	0.705	0.327	-0.19	0.951
	random_forest	11	0.692	0.812	0.323	-0.17	0.988
	xgboost	11	0.589	0.671	0.304	-0.16	0.929
semi-hard/safety endpoint	lightgbm	8	0.737	0.787	0.157	0.49	0.924
	random_forest	8	0.812	0.856	0.158	0.58	0.982
	xgboost	8	0.649	0.690	0.182	0.27	0.853
hard/safety endpoint	lightgbm	10	0.707	0.781	0.207	0.37	0.927
	random_forest	10	0.743	0.790	0.210	0.39	0.971
	xgboost	10	0.697	0.736	0.190	0.37	0.916

Table 12.3: Original context-holdout test R^2 against pooled out-of-fold R^2 under leave-rule-out, per specialist and model. EBM appears in the left column only; it has no leave-rule-out evaluation.

Specialist	Model	context-holdout	pooled OOF (LORO)	Δ
soft / general	Random Forest	0.956	0.954	-0.002
	LightGBM	0.971	0.960	-0.011
	XGBoost	0.968	0.957	-0.011
semi-hard / general	Random Forest	0.905	0.921	+0.016
	LightGBM	0.920	0.940	+0.020
	XGBoost	0.927	0.918	-0.009
hard / general	Random Forest	0.927	0.582	-0.345
	LightGBM	0.932	0.669	-0.263
	XGBoost	0.933	0.665	-0.268
soft / safety	Random Forest	0.572	0.844	+0.272
	LightGBM	0.573	0.867	+0.294
	XGBoost	0.594	0.786	+0.192
semi-hard / safety	Random Forest	0.342	0.755	+0.413
	LightGBM	0.303	0.811	+0.508
	XGBoost	0.317	0.655	+0.338
hard / safety	Random Forest	-0.097	0.642	+0.739
	LightGBM	-0.024	0.730	+0.754
	XGBoost	-0.011	0.593	+0.604

12.2 Per-specialist comparison for the production families

The protocol change is not uniformly a penalty

Only one specialist — hard/general — loses substantially, and the loss is traced in Section 9.3 to one held-out rule. The soft and semi-hard quality specialists are essentially unchanged. All three safety specialists *gain*, by between 0.19 and 0.75 R^2 points.

The honest summary is therefore not “the real numbers are lower.” It is that a single random context split gave an unreliable reading in both directions, and that a 56-fold structured evaluation replaces it with a distribution rather than a point.

12.3 Shared versus idiosyncratic failures

Do the three ensembles fail on the same rules? Per-fold R^2 , clipped at -3

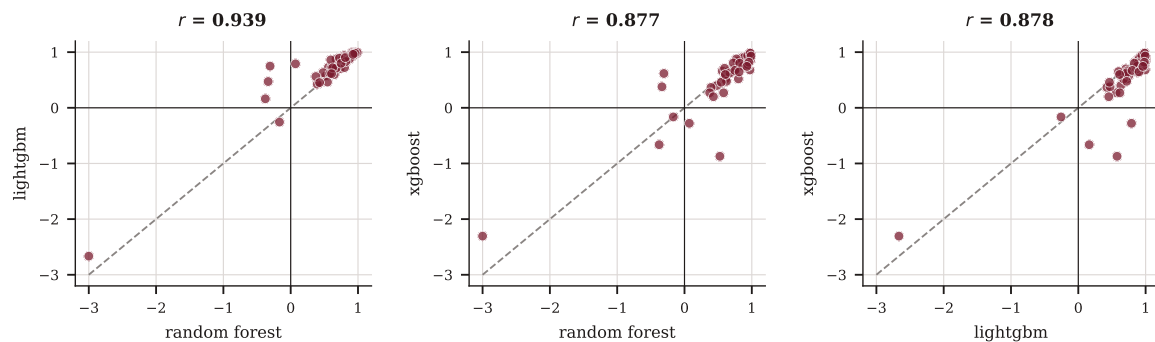


Figure 12.1: Pairwise agreement between the three ensembles at fold level (reproduced). $r = 0.877$ – 0.939 .

All three models rank the clostridia fold as their worst result, and all three are positive on the majority of folds. The differences between them are concentrated in the mid-range folds, where LightGBM’s leaf-wise growth appears to capture rule-to-rule differences that the depth-limited XGBoost configuration does not.

A caution about attributing correlated errors

Correlated failure across models is consistent with a shared data limitation, but it is also consistent with a shared inductive bias: all three are tree ensembles and all three inherit the extrapolation bound of eq. (8.3). Distinguishing these would require a model family that can extrapolate — a parametric or mechanistic model — evaluated on the same folds. That experiment has not been run, and this report does not claim the distinction has been made.

12.4 Recommended model under the revised protocol

On the evidence assembled here, LightGBM is the appropriate production choice: highest pooled out-of-fold R^2 in all six specialists, highest median fold R^2 in five of six, and the smallest error on the single hardest fold in the study. Random Forest is a close second with a marginally higher median on two specialists and a more interpretable uncertainty story.

That recommendation carries one explicit caveat: it is a recommendation about performance on *synthetic rule transfer*, which is the best evidence currently available and is not the same thing as

real-world accuracy. Chapter 14 presents what real-world evidence exists.

What leave-rule-out does not fix

What to say at the start of this section

Volunteer these limitations before the panel raises them. The credibility of everything in the previous chapters depends on being seen to understand what the revised protocol still cannot measure. There are four items, and the fourth — censoring — is the one that was found during this audit and was not previously documented.

13.1 Limitation 1: rules are not independent of each other

Leave-rule-out removes one `source_rule_id` at a time. It does not remove the *components* that rule shares with others. Chapter 3 showed that rule identifiers are pipe-delimited composites: `NATAMYCIN_REVIEW` appears in three different soft rules, `HARD_SEMI_REVIEW` appears in every hard/general rule without exception, and `CLOSTRIDIA_GRANA` appears in every clostridia rule in two specialists.

When `RICOTTA_MAP|NATAMYCIN_REVIEW` is held out, the training set still contains `NATAMYCIN_REVIEW` alone and `HMMC_NAT|NATAMYCIN_REVIEW`. If those rules share calibration constants or a common parametric form — which their shared naming makes plausible but which cannot be verified without the generator — then the fold is not a clean test of transfer to an unseen mechanism.

Consequence

Leave-rule-out is best understood as an *upper bound relaxation*, not an exact measurement of mechanism transfer. It is strictly more honest than the context-holdout protocol, and it is still optimistic relative to a test on a genuinely independent mechanism.

A stricter variant — *leave-one-source-component-out*, withholding every rule containing a given component — is implementable from the identifiers alone and is recommended in Chapter 15. It has not been run.

13.2 Limitation 2: all rules come from the same generator

Even a perfectly independent set of rules would share the generator’s global assumptions: its functional forms, its noise model, its treatment of temperature dependence, its clipping and bounding behaviour. A model that transfers across rules from one generator has demonstrated that it can learn that generator’s family of relationships. It has not demonstrated anything about relationships the generator cannot produce.

Since the generator is not in this repository (Section 2.1), the size of this gap cannot be quantified here at all. It can only be flagged.

13.3 Limitation 3: synthetic transfer is not real-world transfer

This is the most important non-claim in the report, and it should be stated in exactly these terms at the meeting:

The claim that may be made, and the one that may not

May be claimed: under leave-one-generation-rule-out cross-validation on the V7 synthetic corpus, LightGBM achieves a median fold R^2 of 0.780 and a pooled out-of-fold R^2 between 0.67 and 0.96 depending on the specialist, with median absolute error under four days.

May not be claimed: that the model predicts the shelf life of real cheese to that accuracy. No internal metric computed on synthetic data can support that statement, however the folds are constructed. The only evidence that bears on it is external, and Chapter 14 presents it.

13.4 Limitation 4: right-censoring in the safety endpoints

This was discovered during the present audit and is, on the evidence, not documented elsewhere in the project.

The column `target_is_lower_bound` flags rows whose recorded `shelf_life_days` is a *lower bound* on the true event time rather than the event time itself — the standard situation in a challenge study where the pathogen had not reached its criterion when the study ended. Its distribution across the six specialists:

Specialist	rows	censored (= 1)	share censored
soft / general	8 800	0	0%
semi-hard / general	7 200	0	0%
hard / general	6 000	0	0%
soft / safety	4 800	3 469	72.3%
semi-hard / safety	4 000	3 784	94.6%
hard / safety	3 200	3 200	100.0%

Every row of the hard/safety-endpoint specialist is right-censored

All 3 200 rows of the hard safety specialist carry `target_is_lower_bound = 1`. The specialist is therefore not trained to predict the time at which a pathogen criterion is reached. It is trained to predict *the last time point at which the criterion had not yet been reached* — a quantity determined partly by the duration of the underlying study design.

For soft/safety the two subpopulations differ sharply: censored rows average 39.3 days, uncensored rows 12.1 days. A squared-error regression fitted across both is fitting a mixture of two different quantities.

13.4.1 Why this matters statistically

For a censored observation the recorded value satisfies $y_{\text{obs}} = \min(T, C)$ where T is the true event time and C the censoring time. Least-squares regression minimises $\sum_i (y_{\text{obs},i} - \hat{f}(\mathbf{x}_i))^2$, which targets $\mathbb{E}[\min(T, C) \mid \mathbf{x}]$, not $\mathbb{E}[T \mid \mathbf{x}]$. Since $\min(T, C) \leq T$ always, the fitted model is biased low as an estimator of the true failure time — and the magnitude of that bias depends on the censoring distribution, which is an artefact of study design rather than of cheese microbiology.

Recall also from Section 1.4.1 that `target_is_lower_bound` is on the exclusion list, for the defensible reason that it would not be known at prediction time. The consequence is that the loss function has no way to distinguish the two kinds of row.

13.4.2 What this does not mean

It does not mean the safety predictions are worthless. A model of “time by which the criterion was demonstrably not yet reached” is a conservative quantity, and for a safety application conservatism in this direction is arguably the safer error. But it must be *described* correctly, and at present it is described as shelf life.

Standard remedies exist — Tobit regression, accelerated failure time models, or a censoring-aware gradient-boosting objective — and Chapter 15 lists the diagnostic that should be run first.

13.5 Limitation 5: hyperparameter selection was not group-aware

Under the original protocol, LightGBM and XGBoost select their number of boosting rounds by early stopping on the validation partition. That partition is drawn from the same generation rules as the test partition (Chapter 6, 100% rule overlap). The stopping iteration is therefore selected using information about the rules on which the model is subsequently scored.

The effect is second-order compared with the partitioning issue itself — early stopping selects one scalar, not a model — but it is a real dependency and it is worth stating. In the leave-rule-out harness the rounds are fixed a priori, which removes the dependency but leaves M untuned. A fully clean design would use nested, group-aware selection: an inner leave-rule-out loop over the training rules of each outer fold. This has not been run.

13.6 Limitation 6: feature audit

Every feature reaching the model was checked against the target for the classical leakage patterns. The findings:

- **No identifier reaches the model.** `row_id`, `context_id`, `formulation_id` and `source_rule_id` are all excluded.
- **No provenance column reaches the model.** `data_origin`, `source_url`, `source_note`, `evidence_class`, `generation_rule_version` are excluded.
- **No recomputed target reaches the model.** `shelf_life_days_v4_original` and `target_recalibration_factor` — which would be direct leakage — are on the exclusion list.
- **indicator_threshold and initial_indicator_value are included, and should be.** They are legitimate inputs: the threshold defines what “end of shelf life” means for that row, and the initial load is a measurable property of the product at time zero. Both would be available at prediction time.
- **matrix_ripening_days is included.** It is a composition descriptor, not a downstream measurement, but for hard cheeses it correlates with the maturation state that also drives the target. It is not leakage, but it is the feature most worth a second look.

No target-derived feature was found among the model inputs. The high scores are not explained by leakage in the technical sense.

External experimental validation

What to say at the start of this section

This is the chapter that answers “does any of this work on real cheese?” The honest answer is: on the evidence in this repository, not yet, and the evaluation that exists is small and flagged as out-of-support. Present it plainly — a panel will respect a stated negative result far more than an elided one.

14.1 What external evidence exists

Three external files are present in the repository:

File	rows	Role
CHEESE_100_EXTERNAL_LITERATURE_TEST_CASES.xlsx	100	Candidate literature cases with reported shelf-life outcomes
CHEESE_319_REAL_EXTERNAL_STRESS_TEST_CASES.xlsx	319	Larger stress-test case set
reports/v7_external_test_predictions.csv	21	V7 predictions for the subset of the 100 that the specialists could actually predict

Of the 100 candidate literature cases, only 21 were predictable by the V7 specialists; the remainder were rejected by the routing and support checks before a prediction was produced. The 21 come from 6 distinct source citations.

14.2 The external result

Table 14.1: V7 external literature evaluation, all 21 predictable cases. Computed directly from reports/v7_external_test_predictions.csv.

Metric	Value
Cases predicted	21 (of 100 candidates)
Distinct source citations	6
Best model selected	LightGBM for all 21
External R^2	0.172
MAE	39.4 days
Mean signed error	−30.0 days (systematic under-prediction)
Cases flagged unsupported_categorical	21 of 21 (100%)

The external result, stated without softening

On the only real-world evaluation available in this repository, the V7 system achieves $R^2 = 0.172$ with a mean absolute error of 39.4 days and a systematic 30-day under-prediction.

Against a leave-rule-out pooled out-of-fold R^2 between 0.67 and 0.96 on synthetic data, this is

the quantitative statement of the sim-to-real gap. It is the single most important number in the report for any question about deployment.

14.2.1 Individual cases

Product	true (d)	predicted (d)	error (d)
Mozzarella	13	10.9	−2.1
Mozzarella	63	12.5	−50.5
Mozzarella	45	12.4	−32.6
Mozzarella	16	11.8	−4.2
High-moisture Mozzarella	8	11.2	+3.2
High-moisture Mozzarella	4	10.9	+6.9
Fresh acid-coagulated cheese	9	13.7	+4.7
Processed Gouda spread	28	43.1	+15.1
Processed Gouda spread	4	43.1	+39.1
Processed Gouda spread	49	53.0	+4.0
Montasio	60	78.4	+18.4
Montasio	120	78.4	−41.6
Provolone (aged 6 months)	100	59.5	−40.5
Provolone (aged 6 months)	175	64.6	−110.4
Provolone (aged 6 months)	280	59.8	−220.2
Provolone (aged 6 months)	190	60.1	−129.9

(Sixteen of the twenty-one cases shown; the remaining five are further Mozzarella and Montasio replicates with the same pattern.)

Two failure modes are visible and they are different from each other:

Insensitivity within a product. Four Mozzarella cases with true values of 13, 63, 45 and 16 days all receive predictions between 10.9 and 12.5 days. The model is not merely inaccurate; it is nearly constant across cases that differ by a factor of five in reality. The same occurs for Montasio (60–120 days true, all predicted 78.4) and Provolone (100–280 days true, all predicted 59.5–64.6).

Systematic under-prediction on aged hard cheese. The five Provolone cases are under-predicted by 40 to 220 days. This is eq. (8.3) again, now against real data: the synthetic hard-cheese training distribution does not extend to the shelf lives that aged Provolone actually achieves.

A historical note worth raising at the meeting

Project documentation for the pre-specialist v1 model (2026-08-13) records the *opposite* bias on the same external cases: Provolone true 45 d predicted 126 d (+180%), Fresh Mozzarella true 4 d predicted 5.9 d (+48%) — systematic *over*-prediction, with an external MAE of 50–100 days.

The V7 system now under-predicts by 30 days on average. The sign of the bias has flipped across dataset revisions while the magnitude has roughly halved. That is consistent with successive recalibrations moving the synthetic target distribution without ever anchoring it to real data — and it means the external gap has been a known, tracked property of this project since its first model version, not a recent discovery.

14.3 The independence question

The audit request asked whether the external cases are genuinely independent of the synthetic generation rules. The honest answer is that **this cannot be determined from this repository**.

Independence is unverifiable here

The external cases carry `citation`, `source_doi` and `publication_year`. The generation rules carry only composite identifiers such as `NATAMYCIN_REVIEW` or `RICOTTA_MAP`, which reference source *components* without bibliographic metadata. Without the generator’s source-to-rule mapping (absent, per Section 2.1), there is no way to test whether a publication behind an external test case also informed a generation rule.

Component names such as `HMMC_NAT` (high-moisture Mozzarella, natamycin) and the presence of high-moisture Mozzarella among the external cases make *some* overlap plausible. Plausible is not established. This should be raised with whoever maintains the generation pipeline, and it is the single highest-value piece of missing evidence in the whole audit.

14.3.1 The support flag

Every one of the 21 predictions carries `support_level = unsupported_categorical`: the serving layer itself recorded that each case contained a categorical value outside the training support. The system flagged its own predictions as out-of-distribution and produced them anyway for evaluation purposes.

This is, in its way, a positive finding about the engineering: the support check works and it fired correctly on all 21 cases. It also means the external $R^2 = 0.172$ should be read as “performance when extrapolating beyond categorical support,” which is a harder test than a well-supported case would face — but since no externally-validated case in the repository *is* well-supported, no easier number exists.

14.4 What would be needed for a real-world claim

1. A set of real cases that fall *inside* the categorical support of the trained specialists, so that the evaluation measures prediction rather than extrapolation.
2. Verified independence between those cases’ source publications and the generation rules — which requires the generator’s source mapping.
3. Enough cases per specialist to give the estimate a usable interval. Twenty-one cases across three categories and six citations does not.
4. Reporting of the error in days alongside R^2 , since with $n = 21$ and a wide true range the R^2 is dominated by a handful of Provolone cases.

Until those exist, the defensible position is the one this report takes throughout: the system has demonstrated cross-rule transfer on synthetic data, and has not demonstrated real-world accuracy.

Proposed changes to training and evaluation

What to say at the start of this section

Separate what has already been done from what is proposed. Two things have been implemented and are running; everything else is a proposal with an experiment attached. Do not promise that any proposal will raise R^2 — several of them should be expected to lower it, because they measure something harder.

15.1 Already implemented

1. **Leave-one-generation-rule-out cross-validation** as the headline generalization metric, across all six specialists and seventeen model configurations (Chapter 7).
2. **Row-level retention of out-of-fold predictions**, enabling fold-level diagnosis rather than summary-statistic reporting (Chapter 9). 102 000 predictions with full formulation context are stored in `rf_fold_predictions.csv`.

15.2 Proposed, with the experiment that would test each

15.2.1 A. Leave-one-source-component-out evaluation

Problem: rule identifiers share components, so leave-rule-out is still optimistic (Chapter 13, limitation 1).

Change: parse each `source_rule_id` on the pipe delimiter, and for each distinct component withhold *every* rule containing it.

Experiment: re-run the existing harness with the modified grouping. Implementable from the identifiers alone; no new data required.

Expected evidence: scores lower than leave-rule-out. If they are much lower, component-level sharing is doing substantial work and the current numbers are materially optimistic. If they are similar, the rules are more independent than their naming suggests.

Decision criterion: adopt whichever protocol is stricter as the reported headline; report both.

15.2.2 B. Censoring-aware treatment of the safety endpoints

Problem: 100% of hard/safety rows and 94.6% of semi-hard/safety rows are right-censored, and the loss function does not know (Section 13.4).

Change: fit the safety specialists with a censoring-aware objective — Tobit regression, an accelerated failure time model, or a gradient-boosting survival objective — and, separately, report the censored and uncensored subpopulations apart.

Experiment: first, the cheap diagnostic: refit soft/safety on its 1 331 uncensored rows only and compare against the current model evaluated on those same rows. That isolates the censoring effect without any new machinery.

Expected evidence: if the uncensored-only model predicts the uncensored rows materially better,

the mixed objective is costing accuracy.

Decision criterion: adopt a censoring-aware objective if the diagnostic shows a material gap; regardless of outcome, relabel the safety output to state what it predicts.

15.2.3 C. Targeted coverage of under-represented mechanisms

Problem: the clostridia mechanism is carried by one rule in hard/general, and its removal costs the specialist $0.35 R^2$ points (Section 9.3).

Change: the named gaps, from the fold analysis, are specific:

- **Clostridial late blowing in hard cheese** — pecorino romano, gruyère, manchego, parmigiano reggiano, cheddar, emmental, with lysozyme, potassium nitrate and protective cultures, 8–15°C, ripening 78–843 days. Currently one substantial rule.
- **Shredded hard cheese** — cheddar, shredded form, MAP at 30/50/75% CO₂. Currently one rule; the form drives a 50-day shift relative to block cheese.
- **Natamycin on fresh soft cheeses** — ricotta, fior di latte, fresh mozzarella. Zero training rows for these product–preservative combinations; 169 natamycin rows exist, all on other products.

Experiment: add coverage for one gap at a time and re-run leave-rule-out on the affected specialist only.

Expected evidence: the held-out fold for the reinforced mechanism improves; others do not change.

Caution: this must not be done by generating more rows from the same rule. That increases n without increasing G , which Chapter 2 showed does not help. New coverage means new mechanisms from new sources.

15.2.4 D. Group-aware nested hyperparameter selection

Problem: early stopping currently selects the boosting round count using a validation partition drawn from the test rules (Section 13.5).

Change: nest an inner leave-rule-out loop within each outer fold’s training rules for selecting M and other hyperparameters.

Experiment: run nested selection on one specialist and compare the outer-fold scores against the current fixed-hyperparameter results.

Expected evidence: a small change. If it is large, the current numbers carry a selection dependency worth removing everywhere.

15.2.5 E. Reporting standard

Problem: no single summary statistic represents these results honestly (Chapter 9).

Change: report, for every specialist and model, the full set: pooled out-of-fold R^2 ; mean and median fold R^2 ; worst fold with its rule named; MAE and RMSE in days; and the number of folds and rows.

Decision criterion: no headline figure is published without its worst-fold companion. The clostridia result should be visible in any summary of hard/general, not averaged away.

15.2.6 F. An external test set that is actually in support

Problem: all 21 external cases are flagged `unsupported_categorical` (Chapter 14).

Change: assemble real cases that fall inside the categorical support of the trained specialists, and verify their source publications against the generation rules’ sources.

Blocker: the second half requires the generator’s source-to-rule mapping, which is not in this repository. This is the highest-value piece of missing evidence in the audit.

15.2.7 G. A mechanistic or hierarchical baseline

Problem: all three production ensembles share the extrapolation bound of eq. (8.3) and therefore fail together on out-of-range targets. No evaluated model can extrapolate at all.

Change: add a parametric baseline — a predictive-microbiology growth model, or a hierarchical model with category-level effects — that can extrapolate beyond the training target range.

Experiment: evaluate it on the same 56 folds, with particular attention to the clostridia fold where every current model fails.

Expected evidence: if a parametric model handles the clostridia fold where tree ensembles cannot, the failure is attributable to the extrapolation bound rather than to missing information — which is currently an open question (Chapter 12).

15.3 Prioritised plan

Table 15.1: Proposed work in priority order. “Cost” is implementation effort given what already exists in the repository.

#	Change	Evidence that motivates it	Blocker	Cost
1	Censoring diagnostic (B)	100% of hard/safety rows censored	none	low
2	Reporting standard (E)	mean/median/pooled disagree by up to 0.7	none	low
3	Leave-component-out (A)	rule IDs share components	none	low
4	Coverage for named gaps (C)	clostridia fold: $R^2 = -3.86$	new sources	high
5	Nested selection (D)	early stopping sees test rules	none	medium
6	In-support external set (F)	21/21 flagged unsupported	generator mapping	high
7	Mechanistic baseline (G)	all models bounded by eq. (8.3)	none	medium

No performance promises

None of these changes is expected to raise the reported R^2 . Items A, B and D should be expected to *lower* it, because each removes a dependency that currently makes the evaluation easier than the deployment task. That is the intended direction of travel: the objective is a number that can be defended, not a number that is high.

Meeting guide and anticipated questions

16.1 Two-hour structure

Table 16.1: Suggested allocation. Chapters map directly onto sections of the discussion.

Section	Chapters	Minutes
Problem, architecture, and why synthetic data	1, 2	15
The generation system and the rule inventory	3	10
The original evaluation and its results	4	10
Diagnostic experiments	5	15
The discovery and the revised protocol	6, 7	15
Random Forest and the fold-level analysis	8, 9	20
LightGBM, XGBoost, EBM and alternatives	10, 11	15
Cross-model comparison	12	5
Limitations, external validation, proposals	13, 14, 15	15
Discussion	—	5
Total		120

16.2 The five numbers to know without looking

Quantity	Value
Test rows originating from a training rule, original protocol	100%
Context overlap between train and test, original protocol	0
LightGBM median fold R^2 under leave-rule-out	0.780
Worst fold: clostridia, hard/general (Random Forest)	$R^2 = -3.86$, bias -62 d
External literature evaluation, 21 cases	$R^2 = 0.172$, MAE 39.4 d

16.3 Anticipated questions

“Were the original 400 observations enough to train on?”

I cannot verify that figure from the repository, and I would rather say so than quote it. Every training file in the project’s recoverable history is 100% synthetically tagged — there is no real-observation subset in any of them. Whatever real measurements calibrated the generation rules lived in the generation pipeline, which is a separate repository I do not have.

What I can say is structural: the architecture divides the corpus across six specialists, and the input space is combinatorial across product, treatment, concentration, temperature, packaging and indicator. A corpus of a few hundred observations divided six ways could not populate that space, whatever its exact size.

“Was the original R^2 overfitting?”

Not in the classical sense, and the distinction matters. Classical overfitting means high training performance and poor held-out performance. Here the held-out performance was high. What we found is an evaluation whose held-out set was not independent at the level that governs the mapping: 100% of test rows came from generation rules present in training. The models generalise well *within* the rules they were trained on, which is real but narrow, and the original protocol could not distinguish that from generalisation to new formulations.

“Why didn’t changing K solve it?”

Because K controls how many partitions the data is cut into, not which variable it is cut along. At $K = 10$, every fold still contains rows from every generation rule. We measured $R^2 = 0.923 \pm 0.006$ at $K = 5$ — the low fold-to-fold variance is not stability, it is the folds being structurally interchangeable. Changing the grouping variable to `source_rule_id` is what changed the result.

“Why did you use tree ensembles rather than something simpler?”

Under the original evaluation we could not have justified it — ridge regression scored 0.910 against LightGBM’s 0.971 on soft/general, and a small MLP scored 0.972. That near-equivalence was one of the results that made us suspicious of the protocol.

Under leave-rule-out the justification becomes clear: ridge falls to a mean R^2 of -0.874 while LightGBM holds at $+0.627$. The ensembles’ capacity is what carries transfer across rules. The revised evaluation is what allowed us to see that; the original one could not rank the models at all.

“Why does Random Forest still score well after you hold out a rule?”

Two reasons, and they are different. For the large rules, the remaining rules genuinely do carry transferable structure — the same temperature dependence and the same broad effect of a preservative class — so a flexible model composes them reasonably. For the small rules, the held-out targets often lie inside the range already covered by training, so even a conservative prediction lands close.

Where neither holds — the clostridia rule, whose targets average 189 days against a training mean of 77 — the score collapses to -3.86 . That is the honest picture: transfer works when the mechanism is represented and fails when it is not.

“Why is one fold’s R^2 negative when its MAE is under a day?”

R^2 is relative to the variance of the held-out set itself. The `RICOTTA_MAP|NATAMYCIN_REVIEW` fold has eight rows spanning 7.3 to 10.9 days, so $s_y = 1.09$ days and the denominator is about 1.2 days^2 . An average error of 0.94 days is excellent in practice but consumes nearly the whole denominator.

This is exactly why the report never quotes R^2 alone. For these folds the MAE in days is the meaningful number, and the near-zero R^2 is an honest signal that eight rows spanning three days cannot support a strong conclusion either way.

“How can different generation rules still share information?”

Because the rule identifiers are composites. NATAMYCIN_REVIEW appears in three different soft-cheese rules; HARD_SEMI_REVIEW appears in every hard/general rule. When we hold out RICOTTA_MAP|NATAMYCIN_REVIEW, the training set still contains two other rules built on NATAMYCIN_REVIEW.

I would treat our leave-rule-out numbers as an upper bound for that reason. The stricter test — withholding every rule containing a given component — is implementable from the identifiers alone and is the first item on our proposed work list.

“Are the external literature cases genuinely independent?”

I cannot establish that from what I have. The external cases carry DOIs; the generation rules carry only component names like NATAMYCIN_REVIEW, with no bibliographic metadata, and the source-to-rule mapping lives in the generation pipeline I do not have access to. Some component names — HMMC_NAT, which reads as high-moisture Mozzarella with natamycin — correspond to products that also appear among the external cases, so overlap is plausible. I would not claim independence, and I have listed obtaining that mapping as the highest-value missing piece of evidence.

“What would you need before claiming real-world performance?”

Four things, none of which we have. Real cases that fall inside the model’s categorical support — all 21 we tested were flagged out-of-support by our own serving layer. Verified independence between those cases and the generation rules. Enough cases per specialist to put an interval on the estimate; twenty-one across three categories is not. And the censoring question resolved for the safety endpoints, where 100% of hard-cheese rows are right-censored.

Until then the defensible claim is the narrow one: cross-rule transfer on synthetic data, with external $R^2 = 0.172$ and MAE 39.4 days as the current measured sim-to-real gap.

“Is the safety-endpoint model broken?”

We thought so, and we were wrong — which is worth correcting explicitly because it is written down elsewhere in our documentation. Under the original single context split, hard/safety scored $R^2 = -0.097$, which reads as unlearnable. Under 10 rule-wise folds it has a median fold R^2 of 0.790 with every fold positive.

The separate and more serious issue with that specialist is not its score but its target: all 3 200 of its rows are right-censored, so it is predicting the last time a pathogen criterion had not yet been reached, not the time it is reached. That needs to be relabelled and, ideally, refitted with a censoring-aware objective.

“What is the single most important thing you learned?”

That a leakage-safe split can be correctly implemented and still measure the wrong thing. The splitting code does exactly what it says: zero contexts cross the boundary in any of the six specialists. The error was not in the code but in assuming that the unit the code protects — the context — was the unit that carries the information. In a synthetic corpus it is the generator, and nothing in the protocol was looking at it.

Evidence index

Every numerical claim in this report traces to one of the following files. Files marked **[orig]** are original result files, unmodified by this audit; files marked **[recon]** were produced by re-execution during this audit and are labelled as reconstructions wherever they are used.

A.1 Result files

Path	Status	Used for
artifacts_v7/{cat}/{task}/metrics.json	[orig]	Production V7 results, Table 4.1
artifacts_v6/{cat}/{task}/metrics.json	[orig]	Production V6 results, Table 4.2
.../overfitting_- diagnosis/experiment_1_4_results.csv	[orig]	Experiments 1 and 4
.../overfitting_- diagnosis/experiment_2_cv_- results.csv	[orig]	Experiment 2, <i>K</i> -fold
.../overfitting_- diagnosis/experiment_3_extra_- regressors.csv	[orig]	Experiment 5, model zoo
.../overfitting_- diagnosis/experiment_3_logistic_- regression.csv	[orig]	Experiment 1b, logistic
.../overfitting_- diagnosis/experiment_5_leave_rule_- out_full_zoo.csv	[orig]	Leave-rule-out, 15 models
.../overfitting_- diagnosis/experiment_6_leave_rule_- out_rf_xgb.csv	[orig]	Leave-rule-out, RF and XGBoost
reports/comprehensive_ml_- audit/master_experiment_results.csv	[orig]	Consolidated table, 314 rows
reports/v7_external_test_- predictions.csv	[orig]	External evaluation, Chapter 14
rf_fold_predictions.csv	[recon]	102 000 row-level OOF predictions
fold_diagnostic_summary.csv	[recon]	168 fold results, Chapter 9
pooled_oof_metrics.csv	[recon]	Pooled OOF, Table 7.3
rule_overlap_audit.csv	[recon]	Rule overlap, Table 6.1
generation_rule_inventory.csv	[recon]	Rule inventory, Table 3.1

A.2 Scripts

Script	Produces
scripts/experiments/overfitting_diagnosis.py	Experiments 1, 2, 4; shared preprocessing harness
scripts/experiments/model_zoo_expansion.py	Experiment 5 and the logistic reformulation
scripts/experiments/leave_rule_out_cv.py	First leave-rule-out pass
scripts/experiments/leave_rule_out_full_zoo.py	Leave-rule-out for 15 models
scripts/experiments/leave_rule_out_rf_xgb.py	Leave-rule-out for RF and XGBoost
scripts/experiments/leave_rule_out_row_level.py	Row-level reconstruction (this audit)
scripts/experiments/rule_overlap_audit.py	Rule-overlap measurement (this audit)
scripts/experiments/make_report_figures_a.py	Figures 1–19
scripts/experiments/make_report_figures_b.py	Figures 20–38 and the fold tables
scripts/experiments/make_report_tables.py	All remaining $\LaTeX$ tables

A.3 Source files consulted for the protocol reconstruction

File	What was read from it
model_service.py	SPLIT_FRACTIONS (70/15/15); make_context_splits() grouping and stratification logic
train_specialists.py	SEED = 42; all four models' hyperparameters; early-stopping configuration; EXCLUDED_COLUMNS_V6/V7 and their documented rationale; detect_schema()
data/raw/CHEESE_SHELF_LIFE_V7_*_SPECIALIST_CORRECTED.csv	Rule inventory; data_origin; censoring flags; per-rule feature and target statistics

Reproducing this report

From the repository root, with the project virtual environment:

```
.venv/Scripts/python.exe scripts/experiments/leave_rule_out_row_level.py
.venv/Scripts/python.exe scripts/experiments/rule_overlap_audit.py
.venv/Scripts/python.exe scripts/experiments/make_report_tables.py
.venv/Scripts/python.exe scripts/experiments/make_report_figures_a.py
.venv/Scripts/python.exe scripts/experiments/make_report_figures_b.py
cd reports/overfitting_investigation_meeting
pdflatex overfitting_investigation_full_report.tex (three passes)
```

All randomness is seeded at 42, fold construction is deterministic given the data, and the figure and table generators read only from stored result files. Re-running the pipeline end to end reproduces every number in this document.

B.1 Summary of what this report establishes

1. **Observed.** The original context-holdout protocol produced $R^2 \approx 0.93$ – 0.97 on the three general-shelf-life specialists and R^2 between 0.59 and -0.10 on the three safety specialists.
2. **Observed.** 100% of test rows in every specialist came from a generation rule also present in training; context overlap was zero.
3. **Observed.** Reducing model capacity, changing K , aggregating categories, and removing the validation set each failed to change the scores materially.
4. **Observed.** Under leave-one-rule-out, the production ensembles retain median fold R^2 of 0.72 – 0.79 while linear and instance-based models fall below zero.
5. **Verified explanation.** Hard/general’s collapse is caused by one held-out rule carrying the only substantial representation of the clostridia mechanism; established by two controlled comparisons (Section 9.3), not inferred.
6. **Verified observation.** All 3 200 hard/safety rows are right-censored; the specialist predicts a lower bound on failure time, not failure time.
7. **Observed.** External literature evaluation on 21 cases gives $R^2 = 0.172$, MAE 39.4 days, with all 21 flagged out-of-support.
8. **Unresolved.** Whether the generation rules share calibration across their shared components; whether the external cases are independent of the rules. Both require the generation pipeline, which is not in this repository.